Open-FDD Documentation

Generated by scripts/build_docs_pdf.py

June 16, 2026

Generated June 16, 2026

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1 Home

2 Open-FDD

Open-FDD is an open-source **building edge platform** for BACnet integration, feather historian storage, Arrow-native fault detection, and operator dashboards — designed for **trusted LAN / OT edge** deployment behind Caddy, not direct public internet exposure.

2.1 Who it is for

Role	What you get
IT / controls engineer	Pull three GHCR images, log in, poll BACnet, view faults
Commissioning integrator	Discover points, bind BRICK model, pin FDD rules in Rule Lab

Role	What you get
Developer / maintainer	Extend the Operator Bridge, publish images, run CI and LAN security scans

2.2 Start here

Path	Go to
Try it	[Quick Start — Run with Docker]({{ “/quick-start/docker/”
Operate it	Updating the stack({{ “/quick-start/updating/”
Develop it	Developer Guide({{ “/developer/”

2.3 v3 edge stack

Three primary containers on a Linux edge host:

Container	Role
openfdd-bridge	Operator Bridge — API, dashboard, feather **ingest**
openfdd-commission	BACnet discover/ read/write + **poll loop**
openfdd-mcp-rag	Doc-search sidecar for agent tools

Host **Caddy** on `:80` or `:443` is the normal LAN front door. BACnet UDP `:47808` is bound by **commission** only — the bridge ingests `samples.csv` into the feather historian.

Details: Containers({{ “/architecture/containers/” | relative_url }}) · Deployment modes({{ “/architecture/deployment-modes/” | relative_url }})

> **LAN / OT edge only.** Do not expose the Operator Bridge to the public internet without TLS, strong auth, and site review. BACnet writes are disabled by default — see [BACnet write guard]({{ “/security/bacnet-writes/” | relative_url }}).

2.4 Distribution

Channel	Use when
GHCR `ghcr.io/bbartling/openfdd-*`	Production or trial edge deploy
This repository	Custom builds, commissioning, Rule Lab development
PyPI `open-fdd`	Arrow runtime lint/test without full UI

2.5 License

MIT — see repository `LICENSE`.

3 Open-FDD Central / Portfolio / RCx reports

4 Open-FDD Central / Portfolio / RCx reports

Planning doc for the next feature branch: `feature/openfdd-central-portfolio-rcx-reports`.

4.1 Boundary

Layer	Role
Open-FDD Edge	Lives at the building. BACnet, local FDD, model/ trends/faults, safe REST APIs.
Open-FDD Central	Multi-building desk over Tailscale/VPN. No direct BACnet. No equipment commands. Validation jobs, RCx reports, portfolio health.

Existing code to extend (do not rewrite):

- `portfolio/collector/` — edge login, rollup fetch, CSV history
- `portfolio/dash/` — multi-site Dash dashboard
- `scripts/portfolio_collect.py` — collector CLI
- Edge APIs: `/api/building/portfolio-rollup`, `/api/faults/status`, `/api/fdd/results`

4.2 Incremental build order

1. **Edge registry** — extend `portfolio/sites.json` schema (name, base URL, site_id, last check-in).
2. **Remote edge client** — reuse `portfolio/collector/edge_client.py`; add validation job endpoints.
3. **Portfolio page** — edge health, BACnet/model/fault status, last check-in.
4. **One-off validation** — trigger read-only harness against one edge (`acme_overnight_fdd_validate` pattern).
5. **24-hour validation planner** — dropdown intervals (2h / 6h / 12h), store cycle results.
6. **Fault-hour analytics** — elapsed fault hours from historian + FDD runs.
7. **Minimal DOCX generator** — `python-docx` + `matplotlib` (`BytesIO` charts).
8. **Report download endpoint** — query edges via REST, render `.docx`.
9. **Tests/fixtures** — deterministic edge mocks; no live BACnet in CI.

4.3 RCx report content (normalized roles)

- Elapsed fault hours by equipment/severity
- SAT vs setpoint, duct static vs setpoint
- Fan/motor runtime analytics
- Missing data warnings
- Duplicate/model warnings from edge audit APIs
- **Never** severity-only cards without equipment context

4.4 Prerequisites

- ACME validation PR merged and **3.0.33** deployed (equipment context on live FDD alerts).
- Follow-ups: FDD run-history equipment grouping, RTU role mapping (see [ACME validation follow-ups]({{ “/ops/acme-validation-follow-ups/” | relative_url }})).

4.5 Safety

Central is read-only toward edges. No BACnet writes, commands, or BAS changes from Central.

5 Run with Docker images

6 Run with Docker images

Deploy Open-FDD on a Linux edge host using **three published GHCR images**. No git clone required.

6.1 Images

GHCR image	Service	Role
`ghcr.io/bbartling/openfdd-bridge`	`bridge`	Operator API, dashboard, historian ingest
`ghcr.io/bbartling/openfdd-commission`	`commission`	BACnet discover/ read/write **and poll loop**
`ghcr.io/bbartling/openfdd-mcp-rag`	`mcp-rag`	Doc-search sidecar

Edge hosts pull **`latest`** from GHCR (three images: bridge, commission, mcp-rag). The retired **`openfdd-bacnet-poll`** image is no longer published.

6.2 1. Install and validate Docker

```
sudo systemctl enable --now docker
sudo usermod -aG docker "$USER"
newgrp docker   # or log out/in

docker --version
docker compose version
```

```
docker ps
docker run --rm hello-world
```

Expect: `active` / `enabled` for docker, `docker` in `groups`, hello-world succeeds.

6.3 2. Bootstrap the site (one script)

Downloads `docker-compose.yml`, creates `workspace/`, generates `auth.env.local`, and sets `BACNET_BIND` from the host LAN NIC (e.g. `ens192` → `10.200.200.185/24:47808`):

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_edge_bootstrap.sh
bash /tmp/openfdd_edge_bootstrap.sh
```

Bootstrap + pull + start in one go:

```
bash /tmp/openfdd_edge_bootstrap.sh --start
```

Auth — bootstrap writes `~/open-fdd/workspace/auth.env.local` (gitignored). This file holds the API signing secret and one username/password per role. Use it for your first dashboard login:

```
cat ~/open-fdd/workspace/auth.env.local
```

Variable	Role
`OFDD_AUTH_SECRET`	Signs session tokens (do not share)
`OFDD_OPERATOR_*`	Read-only operator
`OFDD_INTEGRATOR_*`	**Default dashboard login** — commissioning + BACnet
`OFDD_AGENT_*`	Agent / automation

Example shape (passwords are **random on first bootstrap only** — existing file is kept on re-runs):

```
OFDD_AUTH_SECRET=<random>
OFDD_OPERATOR_USER=operator
OFDD_OPERATOR_PASSWORD=<random>
OFDD_INTEGRATOR_USER=integrator
OFDD_INTEGRATOR_PASSWORD=<random>
OFDD_AGENT_USER=agent
OFDD_AGENT_PASSWORD=<random>
```

Sign in as **integrator** with the password from that file → First login and health check({{ “/quick-start/health-check/” | relative_url }}).

Action	Touches `auth.env.local`?
First `bash … bootstrap.sh --start`	Creates file with random passwords (if missing)
`bash … bootstrap.sh --start` again	**Keeps** your file — does not overwrite
`bash … bootstrap.sh --restart`	**Keeps** your file — only restarts bridge to reload it
`nano auth.env.local` + `--restart`	Uses **your** edited passwords
`--force-auth`	**Regenerates** random passwords (opt-in only)

The **bridge** container reads `auth.env.local` at startup only. After you change passwords (manual edit or `--force-auth`), recreate bridge (env files are not reloaded by `restart` alone):

```
cd ~/open-fdd && docker compose up -d --force-recreate bridge
```

Regenerate passwords and reload (stack already running):

```
bash /tmp/openfdd_edge_bootstrap.sh --force-auth --restart --show-secrets
```

Manual edit then reload:

```
nano ~/open-fdd/workspace/auth.env.local
bash /tmp/openfdd_edge_bootstrap.sh --restart
```

Options:

Flag	Purpose
`--start`	`docker compose pull && up -d` after layout; curls `http://127.0.0.1:8765/health`
`--image-tag TAG`	default `latest`
`--repo-ref BRANCH`	branch for `compose.edge.yml` if not on `master` yet
`--force-auth`	regenerate `auth.env.local` (restarts bridge if stack is up)
`--restart`	recreate bridge to reload `auth.env.local`; curls `/health` after

Flag	Purpose
`--show-secrets`	print passwords at end (lab only)

From a repo checkout: `./scripts/openfdd_edge_bootstrap.sh --start`

The script prints **BACnet NIC**, **BACNET_BIND**, and file paths — **validate** `commission.env` before polling OT devices:

```
nano ~/open-fdd/workspace/bacnet/commissioning/commission.env
```

> If you used `--start`, the stack is already up. Next step → First login and health check({{ "/quick-start/health-check/" | relative_url }}).

6.3.1 Manual start (optional)

Use this only if you ran bootstrap **without** `--start` and want to bring the stack up yourself:

```
cd ~/open-fdd
docker compose pull
docker compose up -d
docker compose ps
curl -sf http://127.0.0.1:8765/health && echo
```

Expected: **bridge**, **commission**, **mcp-rag** — all `Up`. Then → First login and health check({{ "/quick-start/health-check/" | relative_url }}).

6.4 Long-term operation

6.4.1 Survive power cycles

All services use `restart: unless-stopped`. After reboot:

```
cd ~/open-fdd && docker compose ps
curl -sf http://127.0.0.1:8765/health
```

6.4.2 Start / stop / restart

```
cd ~/open-fdd
docker compose up -d          # idempotent
docker compose stop
docker compose restart commission
docker compose logs -f --tail 100 commission
```

> Never run `docker compose down -v` or delete `workspace/` on a live site.

6.4.3 LAN access

Put **Caddy** (or nginx) on port 80 → `127.0.0.1:8765`, or expose 8765 through the firewall.

6.5 Deployment choices

Mode	Compose services	When to use
Standard OT / BACnet	`bridge` + `commission` + `mcp-rag`	Real buildings — commission polls BACnet into `samples.csv`; bridge ingests to feather
Non-BACnet / demo	`bridge` + `mcp-rag` (omit `commission`)	JSON API, Modbus-only, or lab without field BACnet — see Driver framework({{ "/drivers/"
TLS on OT LAN	Same stack + **host Caddy** `OFDD_CADDY_MODE=tls`	Encrypt browser↔edge; trust self-signed CA on operator PCs — not for public internet

Do **not** run the retired `openfdd-bacnet-poll` image or legacy `openfdd-bacnet-poll` systemd unit alongside Docker **commission** — that double-polls BACnet.

6.6 Next steps

→ First login and health check({{ "/quick-start/health-check/" | relative_url }})
→ Updating the stack({{ "/quick-start/updating/" | relative_url }}) — `./scripts/openfdd_site_update.sh`

7 First login and health check

8 First login and health check

Quick checks after `docker compose up -d` on the edge host. Services use `restart: unless-stopped` — after a reboot, run `docker compose ps` once Docker is up (see [Run with Docker]({{ “/quick-start/docker/” | relative_url }}#survive-power-cycles)).

8.1 Public health

```
curl -s http://127.0.0.1:8765/health
```

Through Caddy on port 80:

```
curl -s http://127.0.0.1/health
```

Expect `"ok": true` and `"auth_required": true` when auth is configured.

8.2 Login

1. Open the dashboard (`http://<host-lan-ip>/` or `:8765`).
2. Sign in with credentials from `workspace/auth.env.local`.

```
curl -s -X POST http://127.0.0.1:8765/api/auth/login \
  -H 'Content-Type: application/json' \
  -d '{"username":"integrator","password":"<from-auth-env>"}'
```

8.3 Smoke checklist

Check	Pass
`docker compose ps`	bridge, commission, mcp-rag **Up**
`GET /health`	`200`, `ok: true`
Login	Token returned
BACnet poll	`samples.csv` growing (`tail -1 workspace/bacnet/polls/samples.csv`)
Historian	Files under `workspace/data/feather_store/`

8.4 Stack detail (authenticated)

Integrator token required:

```
        TOKEN=$(curl -sf -X POST http://127.0.0.1:8765/api/auth/login \
          -H 'Content-Type: application/json' \
          -d '{"username":"integrator","password":"..."}' | jq -r .token)
        curl -sf http://127.0.0.1:8765/health/stack -H "Authorization:
Bearer $TOKEN" | jq .
```

Look for `bacnet_poll` green/yellow in the services list.

8.5 Troubleshooting

Symptom	Fix
401 everywhere	Configure `workspace/auth.env.local`
Empty BACnet discover	Fix `BACNET_BIND` in `commission.env` — network setup({{ “/bacnet/network-setup/”
No poll rows	Commission container running? `points.csv` has enabled rows?
LAN browser timeout	Open firewall port 80 or 8765

Advanced probes (model graph, Rule Lab, compose profiles): [Developer health check]({{ “/developer/health-check/” | relative_url }}).

9 Quick Start

10 Quick Start

Run Open-FDD on a Linux edge host using **three GHCR Docker images**. No git clone on the host.

Step	Page
1	Run with Docker images({{ “/quick-start/docker/”

Step	Page
2	First login and health check({{ “/quick-start/health-check/”
3	Updating the stack({{ “/quick-start/updating/”

One-liner bootstrap (edge host):

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/scripts/openfdd_edge_bootstrap.sh
bash /tmp/openfdd_edge_bootstrap.sh --start
```

Prerequisites: Linux, Docker enabled on boot, BACnet LAN if polling OT equipment.

> **BACnet writes are disabled by default.** See Write safety({{ “/bacnet/write-safety/” | relative_url }}).

Stack: `bridge` + `commission` (BACnet + poll) + `mcp-rag`. Details: Containers({{ “/architecture/containers/” | relative_url }}).

11 Updating the stack

12 Updating the stack

Upgrade GHCR images on the edge host: **backup `workspace/`, pull new tags, recreate containers**. No git pull on the host.

12.1 Before you start

1. SSH to the edge host (`cd ~/open-fdd`).
2. Pick a tag from GitHub Packages — default is **`latest`**.
3. Short maintenance window (containers restart briefly; `restart: unless-stopped` brings them back after reboot).

12.2 Fetch helper scripts (no git clone)

If you bootstrapped with `openfdd_edge_bootstrap.sh`, scripts are already under `~/open-fdd/scripts/`. Otherwise:

```
mkdir -p ~/open-fdd/scripts
curl -fsSL -o ~/open-fdd/scripts/openfdd_site_backup.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/
```

```
scripts/openfdd_site_backup.sh
        curl -fsSL -o ~/open-fdd/scripts/openfdd_site_update.sh \
          https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_site_update.sh
        chmod +x ~/open-fdd/scripts/openfdd_site_*.sh
```

12.3 One-command upgrade (recommended)

```
        cd ~/open-fdd
        ./scripts/openfdd_site_backup.sh
        ./scripts/openfdd_site_update.sh
```

Pulls **`:latest`** by default, verifies all three images exist on GHCR, recreates containers, and hits `/health`.

12.4 Step by step

12.4.1 1. Backup site state

```
        cd ~/open-fdd
        ./scripts/openfdd_site_backup.sh
```

Archives `workspace/` (feather, BACnet CSVs, model, auth) under `~/openfdd-backups/<timestamp>/`.

Custom backup location:

```
        BACKUP_ROOT=~/openfdd-backups/manual ./scripts/
openfdd_site_backup.sh
```

12.4.2 2. Verify images (optional)

```
        export NEW_TAG="${NEW_TAG:-latest}"

        docker manifest inspect ghcr.io/bbartling/openfdd-bridge:$
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-commission:$
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-mcp-rag:$
{NEW_TAG} >/dev/null &&
        echo "All images exist for ${NEW_TAG}"
```

12.4.3 3. Pull and recreate

```
cd ~/open-fdd
./scripts/openfdd_site_update.sh
```

Manual equivalent (same as the script):

```
cd ~/open-fdd
docker compose pull
docker compose up -d --force-recreate
docker compose ps
curl -sf http://127.0.0.1:8765/health && echo
```

12.4.4 4. Verify

```
docker compose ps
curl -sf http://127.0.0.1:8765/health | jq .
tail -1 workspace/bacnet/polls/samples.csv
ls -lah workspace/data/feather_store/
```

→ [Health check]({{ “/quick-start/health-check/” | relative_url }})

12.5 Dashboard UI updates

If the release changed the React UI, copy a new `workspace/api/static/app/` build onto the host before or after the image pull. Image pull alone may not change the browser bundle hash.

12.6 Rollback

Restore `docker-compose.yml` from backup and pull again, or re-run bootstrap from a known-good `master` commit. GHCR keeps only the **three newest** versions per image; early sites should rely on **`latest`** plus `workspace/` backups.

`workspace/` is untouched by image-only upgrades.

12.7 Safe cleanup

```
docker image prune -f
```

> Never run `docker compose down -v`, `docker volume prune`, or delete `workspace/` on a live site.

12.8 Next steps

→ Run with Docker images({{ “/quick-start/docker/” | relative_url }}) — bootstrap a new host
→ [Health check]({{ “/quick-start/health-check/” | relative_url }})

13 Local development

14 Local development

14.1 Clone and auth

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local
# Edit integrator/operator passwords and OFDD_AUTH_SECRET (32+ chars)
```

14.2 Start stack

```
./scripts/build_and_test.sh          # dashboard build + pytest
./scripts/docker_build.sh
./scripts/openfdd_stack.sh up
./scripts/stack_health_check.sh
```

URL	Service
`http://127.0.0.1:8765`	Bridge (direct)
`http://127.0.0.1:8765/docs`	OpenAPI

14.3 BACnet bench overlay (optional)

```
docker compose -f docker/compose.dev.yml -f docker/compose.bench.yml up -d
# commission.env — set BACNET_BIND to your OT interface
```

14.4 Production-like local stack (Caddy, LAN scans)

Same shape as an Acme edge deploy: **prod React UI**, **Caddy on :80**, bridge on loopback **8765**, auth enabled. Use for OWASP ZAP or nmap from another machine on the LAN.

```
./scripts/pentest_production_stack.sh start    # build prod UI + Caddy + auth.pentest.local
./scripts/pentest_production_stack.sh status   # bridge, commission, caddy, mcp-rag
./scripts/pentest_production_stack.sh verify   # LAN IP, /health, ZAP target URL
./scripts/pentest_production_stack.sh stop     # when done
```

Credentials: `workspace/auth.pentest.local` (generated on first start). ZAP target is usually `http://<lan-ip>/` (benserver example: `http://192.168.204.18/`). If LAN clients cannot reach :80, run `sudo ./scripts/open_lan_port.sh 80`.

See also Health checks (developer)({{ "/developer/health-check/" | relative_url }}) for probe details and [ZAP baseline expectations]({{ "/security/zap-baseline/" | relative_url }}) for accepted Medium/Low findings.

14.5 Docs preview

```
cd docs && bundle install && bundle exec jekyll serve
# http://127.0.0.1:4000/open-fdd/
```

15 Build Docker images

16 Build Docker images

16.1 Local build

```
./scripts/docker_build.sh
# Optional tar for air-gap:
./scripts/docker_build.sh --save
```

Images: `openfdd-bridge`, `openfdd-commission`, `openfdd-mcp-rag`.

16.2 Publish to GHCR (maintainers)

GitHub Actions workflow **Publish Docker images to GHCR** — manual dispatch or git tag `vX.Y.Z`. Publishes **`pyproject.toml` version** (e.g. `3.1.3`), minor alias (`3.1`), and **`latest`**, then prunes old GHCR versions.

Maintainer checklist:

1. Green CI on `master`
2. Run **Publish Docker images to GHCR** workflow (or push tag `vX.Y.Z`)
3. Verify `ghcr.io/bbartling/openfdd-{bridge,commission,mcp-rag}:3.1.3` and `:latest`
4. Edge hosts: `OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_ghcr.sh --limit <host>`

Details live in `.github/workflows/` and `scripts/docker_build.sh` — not duplicated here.

17 Developer Guide

18 Developer Guide

Clone, build, test, and contribute to Open-FDD.

Topic	Page
Local setup	Local development({{ “/developer/local-setup/”
Docker builds	Build Docker images({{ “/developer/docker-build/”
Health checks	Health checks (developer) ({{ “/developer/health-check/”
Tests & CI	Tests and CI({{ “/developer/tests-ci/”
Security testing	[Release & LAN scan cycle] ({{ “/developer/security-testing/”
PRs	Contributing({{ “/developer/contributing/”

AI-assisted contributors: see repository `AGENTS.md` for workspace layout, skill routing, and agent policy (one paragraph in public docs; details stay internal).

19 Tests and CI

20 Tests and CI

20.1 Local tests

```
        ./scripts/build_and_test.sh           # UI + workspace bridge pytest
        python3 -m pytest open_fdd/tests -q   # PyPI package tests
        cd workspace/dashboard && npm test    # dashboard unit tests (if configured)
```

20.2 CI (GitHub Actions)

Workflow	Trigger	Checks
`ci.yml`	PR / push	pytest, bridge security audit, dashboard build, Jekyll docs
`docs-pages.yml`	push `master`	GitHub Pages site
`publish-open-fdd.yml`	tag `v*` / `open-fdd-v*`	PyPI wheel (Trusted Publishing)
`docker-publish.yml`	tag `v*` / manual	GHCR bridge, commission, mcp-rag

20.3 Docs build (same as CI)

```
        cd docs && bundle exec jekyll build -d /tmp/open-fdd-site
```

Fix any template or link errors before opening a PR that touches `docs/`.

20.4 Security-related CI

`ci.yml` includes bridge security tests (`tests/workspace_bridge/test_security.py`). That complements — but does not replace — LAN [ZAP + Nmap scans]({{ “/security/zap-baseline/” | relative_url }}) on each patch revision. See Security testing cycle({{ “/developer/security-testing/” | relative_url }}).

21 DataFusion SQL rule recipes

22 DataFusion SQL recipes

Use when you need simple threshold, `CASE WHEN`, or SQL-readable rules. Requires `pip install open-fdd[datafusion]`. SQL runs **server-side only**.

22.1 Temperature high

```
SELECT
  *,
  "duct-t" > 80.0 AS fault
FROM telemetry
```

Quote hyphenated historian columns.

22.2 Humidity high

```
SELECT
  *,
  "oa-h" > 70.0 AS fault
FROM telemetry
```

22.3 Occupied + temperature

```
SELECT
  *,
  CASE
    WHEN occupied = true AND "stat_zn-t" > 76.0 THEN true
    ELSE false
```

```
  END AS fault
FROM telemetry
```

22.4 With fault confirmation

Add to rule `config`:

```
{"min_true_rows": 5, "poll_interval_s": 60}
```

PyArrow remains better for rolling windows, custom helpers, and ML prep. See PyArrow recipes.

23 Contributing

24 Contributing

24.1 Expectations

- Open an issue or discuss large changes before big refactors.
- Keep PRs focused; include test or docs updates when behavior changes.
- Do not commit secrets (`auth.env.local`, Ansible `secrets/*.env.local`, commissioning CSVs with real site data).
- Follow existing code style in `workspace/api/` and `workspace/dashboard/`.

24.2 Pull request checklist

- [ ] Tests pass locally (`./scripts/build_and_test.sh` or scoped pytest)
- [ ] Docs updated if user-facing behavior changed
- [ ] `docs/` Jekyll build succeeds
- [ ] No credentials in diff
- [ ] Auth/Caddy/header changes: re-run [pentest verify + LAN ZAP]({{ “/developer/security-testing/” | relative_url }}) and

update [ZAP baseline]({{ “/security/zap-baseline/” | relative_url }}) if expectations changed

24.3 Repository layout (short)

Path	Contents
`workspace/api/`	Operator Bridge (FastAPI)
`workspace/dashboard/`	React SPA
`open_fdd/`	PyPI package (`arrow_runtime`, `playground`)
`docker/`	Compose files and image contexts
`infra/ansible/`	Edge deploy playbooks
`docs/`	This site (Jekyll + Just the Docs)

24.4 Docs links (GitHub Pages)

The site is a **project page** at `https://bbartling.github.io/open-fdd/` (`baseurl: /open-fdd` in `docs/_config.yml`).

- **Do not** use {% raw %}`/path/`{% endraw %} in Markdown body text — it omits `baseurl` and 404s on GitHub Pages.
- **Do** use `[label]({{ "/section/page/" | relative_url }})` for internal links.
- After `bundle exec jekyll build`, run `python3 scripts/check_docs_internal_links.py --site _site` from the repo root.

24.5 Security issues

Do not file public issues for vulnerabilities. Contact the maintainer or use GitHub private security advisories.

24.6 Agent-assisted development

Cursor/Claude contributors: read `AGENTS.md` and `skills/` for automation boundaries. Public docs intentionally avoid agent playbook detail.

25 Health checks (developer)

26 Health checks (developer)

Extended validation for engineers with a **full git checkout** — CI, compose overlays, and pre-release gates.

IT operators on edge hosts should use [Quick Start — health check]({{ “/quick-start/health-check/” | relative_url }}) and Deployment validation({{ “/ops/deployment-validation/” | relative_url }}) instead.

26.1 Clone and local stack

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local   # loopback dev only
./scripts/docker_build.sh                                 # or: ./scripts/openfdd_stack.sh prod
./scripts/openfdd_stack.sh up
```

GHCR production pull (no local build):

```
OPENFDD_IMAGE_TAG=2026.06.07-edge ./scripts/openfdd_stack.sh prod
```

26.2 Stack health script

```
./scripts/stack_health_check.sh
OPENFDD_BASE_URL=http://192.168.204.18:8765 ./scripts/stack_health_check.sh
./scripts/stack_health_check.sh --require-ollama
```

Probes: containers up, `/health`, `/api/model/health`, sites, tree, SPARQL graph (with integrator auth when configured).

26.3 Compose overlays

Overlay	Use
`docker/compose.dev.yml`	Local dev bind-mount
`docker/compose.bench.yml`	Host-network BACnet lab
`docker/compose.prod.yml`	GHCR images instead of local build

Overlay	Use
`docker/compose.edge.yml`	IT edge template (copy to `~/open-fdd/docker-compose.yml`)

```
        docker compose -f docker/compose.dev.yml -f docker/
compose.bench.yml ps
        docker compose -f docker/compose.dev.yml config --images
```

26.4 BACnet poll pipeline

```
        tail -1 workspace/bacnet/polls/samples.csv
        cat workspace/data/bacnet_ingest_state.json
        docker compose logs --since 10m commission | grep -i poll
        docker compose logs --since 10m bridge | grep -i ingest
```

26.5 pytest (bridge security / BACnet)

```
        PYTHONPATH=workspace/api pytest tests/workspace_bridge/
test_security.py -q
```

26.6 Pentest production stack (LAN ZAP)

```
        ./scripts/pentest_production_stack.sh start
        ./scripts/pentest_production_stack.sh verify
```

Then from a **LAN workstation**, run the packaged scan:

- Windows: `scripts/security/Run-OpenFddSecurityScan.ps1`
- macOS/Linux: `scripts/security/run_openfdd_security_scan.sh`

See scripts/security/README.md, [Security — ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}), and the full security testing cycle({{ "/developer/security-testing/" | relative_url }}). Host-side notes: `workspace/deploy/PENTEST.md`.

26.7 Post-deploy insurance (inventory hosts)

For automated checks against named inventory hosts, see `infra/ansible/scripts/post_deploy_check.sh` (maintainer tooling — not required for IT docker-pull sites).

27 Security testing cycle

28 Security testing cycle

How Open-FDD validates each **patch revision** before it ships to the live Acme HVAC bench — local automation, AI-assisted PR review, LAN passive scans, and edge post-deploy checks.

Related security docs: [ZAP baseline]({{ “/security/zap-baseline/” | relative_url }}), TLS and certificates({{ “/security/tls-and-certs/” | relative_url }}), LAN hardening({{ “/security/lan-hardening/” | relative_url }}), Linux host hardening({{ “/security/linux-host-hardening/” | relative_url }}), [Tenable remediation]({{ “/security/tenable-remediation/” | relative_url }}), Authenticated scanning (roadmap)({{ “/security/authenticated-scanning/” | relative_url }}).

28.1 Who owns what

Concern	Owner	Where
HTTP security headers (CSP, X-Frame, COOP, …)	**Bridge** middleware	`workspace/api/openfdd_bridge/security_headers.py`
TLS termination + HSTS	**Caddy**	`workspace/deploy/caddy/`, Ansible `Caddyfile.j2`
OT LAN TLS certificates	**Site integrator / maintainer**	`./scripts/setup_caddy_certs.sh` → `workspace/deploy/caddy/certs/`; Ansible `/etc/openfdd/caddy/` on edge
Pentest credentials	Generated per stack	`workspace/auth.pentest.local` (never commit)
LAN ZAP + Nmap scans	**Developer workstation** on test LAN	`scripts/security/`

See [TLS and certificate ownership]({{ “/security/tls-and-certs/” | relative_url }}) for cert lifecycle.

28.2 Per-revision workflow

Typical cycle on a **control machine** (local dev) through a **lab edge host** (live bench):

```
flowchart LR
  A[Local dev + pytest] --> B[Pentest stack verify]
  B --> C[LAN ZAP + Nmap]
  C --> D[PR + GitHub CI]
  D --> E[CodeRabbit AI review]
  E --> F[Merge + tag open-fdd-vX.Y.Z]
  F --> G[GHCR + PyPI publish]
  G --> H[Acme upgrade + post_deploy_check --full]
  H --> I[Bump next patch locally]
```

28.2.1 1. Local functional tests

```
        ./scripts/build_and_test.sh
        PYTHONPATH=workspace/api pytest tests/workspace_bridge/
test_security.py -q
        infra/ansible/scripts/bench_5007_arrow_battery.sh   # when FDD/
Arrow paths change
```

28.2.2 2. Pentest production stack (on Open-FDD host)

```
        ./scripts/pentest_production_stack.sh start
        ./scripts/pentest_production_stack.sh verify
```

Confirms Caddy `:80` fronts the bridge, auth is enabled, BACnet writes disabled, and security headers are singular (no Caddy duplicates). Details: [Health checks — pentest]({{ “/developer/health-check/” | relative_url }}#pentest-production-stack-lan-zap).

28.2.3 3. Host + exposure checks (on edge VM)

After IT Nessus scans or before Acme deploy sign-off:

```
        ./scripts/security/check_host_security.sh
        ./scripts/security/check_openfdd_exposure.sh --edge-ip <lan-ip>
```

Host remediation (dry-run first): `./scripts/security/remediate_ubuntu_host.sh`

28.2.4 4. LAN security scan (from laptop)

Use the packaged scripts — do not hand-roll one-off docker/nmap commands unless debugging:

Platform	Command
Windows	`.\scripts\security\Run-OpenFddSecurityScan.ps1`
macOS / Linux	`./scripts/security/run_openfdd_security_scan.sh`

Setup (Docker Desktop, Nmap): scripts/security/README.md.

Target URL is usually **`http://<lan-ip>/`** (Caddy), not `:8765`.

Review `openfdd-security-report/90-quick-findings-summary.txt` and compare against [ZAP baseline expectations]({{ "/security/zap-baseline/" | relative_url }}).

28.2.5 5. Pull request + automated review

- Open PR against `master`
- **GitHub Actions** (`ci.yml`): pytest, bridge security audit, dashboard build, Jekyll docs
- **CodeRabbit** (or similar AI PR review): style, security footguns, missing tests/docs

Do not merge with failing CI or unresolved duplicate-header regressions.

28.2.6 6. Release

```
git tag open-fdd-v3.0.4   # example
git push origin open-fdd-v3.0.4
```

Tag triggers GHCR `:latest` and PyPI publish workflows.

28.2.7 7. Lab edge validation

After pulling new images on a live bench VM:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full
```

Re-run LAN ZAP + Nmap against the edge LAN IP if Caddy or auth behavior changed. Site-specific example: [GL36 lab note]({{ "/examples/acme-gl36-lab/" | relative_url }}).

28.2.8 8. Next cycle

Bump patch version locally (`pyproject.toml`, `open_fdd/__init__.py`) on `master` for the next development round — do not tag until the next release is ready.

28.3 What each scan layer covers

Layer	Tool	Depth
Unit / integration	pytest, `test_security.py`	Header middleware, auth gates
Stack smoke	`pentest_production_stack.sh verify`	Live headers, health, auth file
Passive web	ZAP baseline	Public routes, CSP, cookies, CORS
Port exposure	Nmap scoped	One host, known service ports
Authenticated API/dashboard	ZAP full + login	**Roadmap** — [Authenticated scanning]({{ “/security/authenticated-scanning/”

Baseline ZAP does **not** exercise integrator login or protected `/api/*` routes deeply.

28.4 Open-source contributors

The `scripts/security/` folder is part of the public repo so any fork can run the same LAN smoke against their bench. PRs that change Caddy, auth, or `security_headers.py` should update [ZAP baseline]({{ “/security/zap-baseline/” | relative_url }}) and re-run the scan scripts.

29 Python recipes (full Arrow library)

30 Python recipes — full Arrow library

Copy-paste **`apply_faults_arrow`** modules for Rule Lab. Replaces the legacy pandas/YAML Expression Rule Cookbook.

- **No pandas** on the edge — PyArrow + `pyarrow.compute` only
- Set **`fault_code`** in Rule Lab metadata (letter codes or Grade-A `AHU-ECON-001`)
- Map historian column names in the data model / rule bindings (`ma-t`, `oa-t`, `stat_zn-t`, …)

Quick patterns: [Arrow recipes]({{ “/rule-cookbook/arrow-recipes/” | relative_url }}) · Index: Expression cookbook (Arrow-native)({{ “/rule-cookbook/expression-cookbook/” | relative_url }})

30.1 Shared helpers (paste once per rule file)

```python
import pyarrow as pa
import pyarrow.compute as pc

from open_fdd.arrow_runtime.cookbook import mixing_envelope_mask
from open_fdd.arrow_runtime.windows import (
    arrow_rolling_min,
    arrow_rolling_max,
    arrow_abs_diff,
)


def _norm_cmd(col: pa.ChunkedArray) -> pa.ChunkedArray:
    c = pc.cast(col, pa.float64())
    return pc.if_else(pc.greater(c, 1.0), pc.divide(c, 100.0), c)


def _col(table, name: str) -> pa.ChunkedArray:
    return pc.cast(table[name], pa.float64())
```

```
        def _hypot_tol(a: float, b: float) -> float:
            return (a * a + b * b) ** 0.5
```

30.2 GL36 AHU Rules A–M

30.2.1 Rule A — duct static low at full fan speed (`AHU-A`)

```
        FAULT_CODE = "AHU-A"
        SP, SP_SP, FAN = "duct-static-pressure", "duct-static-pressure-sp", "supply-fan-speed-command"
        SP_MARGIN, DRV_HI, DRV_NEAR = 0.12, 0.93, 0.06


        def apply_faults_arrow(table, cfg, context=None):
            sp, sp_sp = _col(table, SP), _col(table, SP_SP)
            fan = _norm_cmd(table[FAN])
            return pc.and_(
                pc.less(sp, pc.subtract(sp_sp, SP_MARGIN)),
                pc.greater_equal(fan, DRV_HI - DRV_NEAR),
            )
```

30.2.2 Rules B & C — blended air outside OAT/RAT band (`AHU-D`)

Uses `mixing_envelope_mask` (covers below-band and above-band).

```
        FAULT_CODE = "AHU-D"


        def apply_faults_arrow(table, cfg, context=None):
            return mixing_envelope_mask(
                table,
                {**cfg, "mixing_tol": 1.15, "normalize_cmd_percent": 1},
                mat_col="ma-t",
                oat_col="oa-t",
                rat_col="ra-t",
                fan_col="supply-fan-speed-command",
            )
```

30.2.3 Rule D — discharge cold when heating commanded (`AHU-B`)

```
        FAULT_CODE = "AHU-B"
        BLEND_TOL, SAT_TOL, FAN_DT = 1.15, 1.15, 0.55
```

```
        def apply_faults_arrow(table, cfg, context=None):
            mat, sat = _col(table, "ma-t"), _col(table, "sa-t")
            valve = _norm_cmd(table["heating-valve-command"])
            fan = _norm_cmd(table["supply-fan-speed-command"])
            rhs = pc.add(pc.subtract(mat, BLEND_TOL), FAN_DT)
            cold = pc.less_equal(pc.add(sat, SAT_TOL), rhs)
            return pc.and_(cold, pc.greater(valve, 0.01), pc.greater(fan,
0.01))
```

30.2.4 Rule E — SAT too low with full heating (`AHU-C`)

```
        FAULT_CODE = "AHU-C"
        SUPPLY_ERR = 1.0


        def apply_faults_arrow(table, cfg, context=None):
            sat = _col(table, "sa-t")
            sp = _col(table, "sa-t-sp") if "sa-t-sp" in
table.column_names else _col(table, "supply-air-temperature-
setpoint")
            valve = _norm_cmd(table["heating-valve-command"])
            fan = _norm_cmd(table["supply-fan-speed-command"])
            return pc.and_(
                pc.less(sat, pc.subtract(sp, SUPPLY_ERR)),
                pc.greater(valve, 0.9),
                pc.greater(fan, 0.01),
            )
```

30.2.5 Rule F — SAT/MAT mismatch in economizer mode (`AHU-E`)

```
        FAULT_CODE = "AHU-E"
        FAN_DT, BLEND_TOL, SAT_TOL, ECON_MIN = 0.55, 1.15, 1.15, 0.12
        TOL = _hypot_tol(SAT_TOL, BLEND_TOL)


        def apply_faults_arrow(table, cfg, context=None):
            mat, sat = _col(table, "ma-t"), _col(table, "sa-t")
            damper = _norm_cmd(table["oa-damper-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            mismatch = pc.greater(pc.abs(pc.subtract(pc.subtract(sat,
FAN_DT), mat)), TOL)
            econ = pc.and_(pc.greater(damper, ECON_MIN), pc.less(cool,
0.1))
            return pc.and_(mismatch, econ)
```

30.2.6 Rule G — ambient too warm for free cooling (AHU-E)

```
        FAULT_CODE = "AHU-E"
        OAT_TOL, FAN_DT, SAT_TOL, ECON_MIN = 1.15, 0.55, 1.15, 0.12


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            sp = _col(table, "sa-t-sp")
            damper = _norm_cmd(table["oa-damper-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            warm = pc.greater(oat, pc.add(pc.subtract(sp, FAN_DT),
SAT_TOL - OAT_TOL))
            econ = pc.and_(pc.greater(damper, ECON_MIN), pc.less(cool,
0.1))
            return pc.and_(warm, econ)
```

30.2.7 Rule H — OAT/MAT mismatch (econ + mech cooling) (AHU-E)

```
        FAULT_CODE = "AHU-E"
        OAT_TOL, BLEND_TOL = 1.15, 1.15
        TOL = _hypot_tol(OAT_TOL, BLEND_TOL)


        def apply_faults_arrow(table, cfg, context=None):
            oat, mat = _col(table, "oa-t"), _col(table, "ma-t")
            cool = _norm_cmd(table["cooling-valve-command"])
            damper = _norm_cmd(table["oa-damper-command"])
            return pc.and_(
                pc.greater(pc.abs(pc.subtract(mat, oat)), TOL),
                pc.greater(cool, 0.01),
                pc.greater(damper, 0.9),
            )
```

30.2.8 Rule I — OAT/MAT mismatch (economizer only) (AHU-E)

```
        FAULT_CODE = "AHU-E"
        TOL = _hypot_tol(1.15, 1.15)


        def apply_faults_arrow(table, cfg, context=None):
            oat, mat = _col(table, "oa-t"), _col(table, "ma-t")
            damper = _norm_cmd(table["oa-damper-command"])
            return
pc.and_(pc.greater(pc.abs(pc.subtract(mat, oat)), TOL),
pc.greater(damper, 0.9))
```

30.2.9 Rule J — discharge above blended in cooling (`AHU-B`)

```
        FAULT_CODE = "AHU-B"
        FAN_DT, BLEND_TOL, SAT_TOL, ECON_MIN = 0.55, 1.15, 1.15, 0.12
        TOL = _hypot_tol(SAT_TOL, BLEND_TOL)


        def apply_faults_arrow(table, cfg, context=None):
            sat, mat = _col(table, "sa-t"), _col(table, "ma-t")
            cool = _norm_cmd(table["cooling-valve-command"])
            damper = _norm_cmd(table["oa-damper-command"])
            hot = pc.greater(sat, pc.add(mat, pc.add(TOL, FAN_DT)))
            mode_a = pc.and_(pc.greater(damper, 0.9), pc.greater(cool,
0.0))
            mode_b = pc.and_(pc.less_equal(damper, ECON_MIN),
pc.greater(cool, 0.9))
            return pc.and_(hot, pc.or_(mode_a, mode_b))
```

30.2.10 Rule K — discharge above setpoint in full cooling (`AHU-C`)

```
        FAULT_CODE = "AHU-C"
        SAT_TOL, ECON_MIN = 1.15, 0.12


        def apply_faults_arrow(table, cfg, context=None):
            sat = _col(table, "sa-t")
            sp = _col(table, "sa-t-sp")
            cool = _norm_cmd(table["cooling-valve-command"])
            damper = _norm_cmd(table["oa-damper-command"])
            high = pc.greater(sat, pc.add(sp, SAT_TOL))
            full = pc.or_(
                pc.and_(pc.greater(damper, 0.9), pc.greater(cool, 0.9)),
                pc.and_(pc.less_equal(damper, ECON_MIN), pc.greater(cool,
0.9)),
            )
            return pc.and_(high, full)
```

30.2.11 Rule L — cooling coil ΔT when inactive (`CH-C`)

```
        FAULT_CODE = "CH-C"
        ENTER_TOL, LEAVE_TOL, ECON_MIN = 1.15, 1.15, 0.12
        TOL = _hypot_tol(ENTER_TOL, LEAVE_TOL)


        def apply_faults_arrow(table, cfg, context=None):
            ent = _col(table, "cooling-coil-entering-air-temperature")
            leave = _col(table, "cooling-coil-leaving-air-temperature")
```

```
            heat = _norm_cmd(table["heating-valve-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            damper = _norm_cmd(table["oa-damper-command"])
            drop = pc.greater(pc.subtract(ent, leave), TOL)
            mode_a = pc.and_(pc.greater(heat, 0), pc.equal(cool, 0),
pc.less_equal(damper, ECON_MIN))
            mode_b = pc.and_(pc.equal(heat, 0), pc.equal(cool, 0),
pc.greater(damper, ECON_MIN))
            return pc.and_(drop, pc.or_(mode_a, mode_b))
```

30.2.12 Rule M — heating coil ΔT when inactive (`AHU-B`)

```
        FAULT_CODE = "AHU-B"
        ENTER_TOL, LEAVE_TOL, FAN_DT, ECON_MIN = 1.15, 1.15, 0.55, 0.12
        TOL = _hypot_tol(ENTER_TOL, LEAVE_TOL) + FAN_DT


        def apply_faults_arrow(table, cfg, context=None):
            ent = _col(table, "heating-coil-entering-air-temperature")
            leave = _col(table, "heating-coil-leaving-air-temperature")
            heat = _norm_cmd(table["heating-valve-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            damper = _norm_cmd(table["oa-damper-command"])
            rise = pc.greater(pc.subtract(leave, ent), TOL)
            m1 = pc.and_(pc.equal(heat, 0), pc.equal(cool, 0),
pc.greater(damper, ECON_MIN))
            m2 = pc.and_(pc.equal(heat, 0), pc.greater(cool, 0),
pc.greater(damper, 0.9))
            m3 = pc.and_(pc.equal(heat, 0), pc.greater(cool, 0),
pc.less_equal(damper, ECON_MIN))
            return pc.and_(rise, pc.or_(m1, m2, m3))
```

30.3 Starter pack (VAV / AHU baseline)

30.3.1 `01_vav_zone_temp_bounds_occupied` (`VAV-C`)

```
        from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask,
_unoccupied_mask

        FAULT_CODE = "VAV-C"


        def apply_faults_arrow(table, cfg, context=None):
            occupied = pc.invert(_unoccupied_mask(table, cfg))
            oob = sensor_bounds_mask(table, "zone_temp", cfg)
            return pc.and_(oob, occupied)
```

30.3.2 `02_vav_zone_temp_flatline_occupied` (VAV-C)

```
        from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask,
_unoccupied_mask

        FAULT_CODE = "VAV-C"


        def apply_faults_arrow(table, cfg, context=None):
            occupied = pc.invert(_unoccupied_mask(table, cfg))
            flat = sensor_flatline_mask(table, "zone_temp", cfg)
            return pc.and_(flat, occupied)
```

30.3.3 `03_vav_damper_command_extreme_flatline` (VAV-D)

```
        from open_fdd.arrow_runtime.cookbook import flatline_1h_mask

        FAULT_CODE = "VAV-D"
        DAMPER = "damper-position-command"


        def apply_faults_arrow(table, cfg, context=None):
            stuck = flatline_1h_mask(table, {**cfg, "column": DAMPER})
            wide = pc.greater(_norm_cmd(table[DAMPER]), 0.975)
            return pc.and_(stuck, wide)
```

30.3.4 `04_ahu_runtime_outside_schedule` (BLD-C)

Use run-hours analytics script + schedule compare; boolean mask:

```
        from open_fdd.arrow_runtime.cookbook import _fan_on_mask,
_unoccupied_mask

        FAULT_CODE = "BLD-C"


        def apply_faults_arrow(table, cfg, context=None):
            return pc.and_(_fan_on_mask(table, cfg),
_unoccupied_mask(table, cfg))
```

30.3.5 `05` / `06` / `07` — see Rules A, `sensor_bounds_mask` on `sa-t` / `sensor_flatline_mask` on `sa-t`

30.4 Economizer starters

30.4.1 ahu_econ_100oa_temp_tracking_fault (AHU-E)

```
        FAULT_CODE = "AHU-E"
        DAMPER_FULL, FAN_ON, TRACK_TOL, FAN_HEAT = 0.95, 0.5, 2.0, 0.5


        def apply_faults_arrow(table, cfg, context=None):
            oat, mat, sat = _col(table, "oa-t"), _col(table, "ma-t"),
_col(table, "sa-t")
            damper = _norm_cmd(table["oa-damper-command"])
            fan = _col(table, "supply-fan-status")
            econ = pc.and_(pc.greater(fan, FAN_ON),
pc.greater_equal(damper, DAMPER_FULL))
            mat_bad = pc.greater(pc.abs(pc.subtract(mat, oat)), TRACK_TOL)
            sat_bad = pc.greater(pc.abs(pc.subtract(pc.subtract(sat,
FAN_HEAT), mat)), TRACK_TOL)
            return pc.and_(econ, pc.or_(mat_bad, sat_bad))
```

30.4.2 ahu_mech_cooling_when_free_cooling_available (AHU-E)

```
        FAULT_CODE = "AHU-E"
        MARGIN, DAMPER_LOW, COOL_ON = 2.0, 0.50, 0.10


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            sp = _col(table, "sa-t-sp")
            damper = _norm_cmd(table["oa-damper-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            free = pc.less_equal(oat, pc.subtract(sp, MARGIN))
            return pc.and_(free, pc.less(damper, DAMPER_LOW),
pc.greater_equal(cool, COOL_ON))
```

30.4.3 ahu_oa_damper_excess_open_extreme_ambient (AHU-E)

```
        FAULT_CODE = "AHU-E"
        DAMPER_MIN, HI_OAT, LO_OAT = 0.50, 70.0, 40.0


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            damper = _norm_cmd(table["oa-damper-command"])
            extreme = pc.or_(pc.greater_equal(oat, HI_OAT),
pc.less_equal(oat, LO_OAT))
            return pc.and_(pc.greater_equal(damper, DAMPER_MIN), extreme)
```

30.5 VAV zones

30.5.1 `zone_reheat_warm_ambient` (VAV-A)

```
        FAULT_CODE = "VAV-A"
        T_CUTOFF, REHEAT_MIN = 78.0, 0.52


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            reheat = _norm_cmd(table["reheat-valve-command"])
            return pc.and_(pc.greater(oat, T_CUTOFF), pc.greater(reheat,
REHEAT_MIN))
```

30.5.2 `zone_damper_valve_full_open` (VAV-D)

```
        FAULT_CODE = "VAV-D"
        FULL_OPEN, ROLL = 97.5, 105


        def apply_faults_arrow(table, cfg, context=None):
            d = _norm_cmd(table["damper-position-command"])
            hi = pc.greater(d, FULL_OPEN)
            rmin = arrow_rolling_min(d, ROLL)
            return pc.and_(hi, pc.greater(rmin, FULL_OPEN))
```

30.5.3 Zone / IAQ bounds

Use `sensor_bounds_mask(table, "zone_temp", cfg)` or `sensor_bounds_mask` with CO_2 profile — see [sensor validation]({{ “/rule-cookbook/expression-cookbook/” | relative_url }}#sensor-validation-bounds-flatline-rate-of-change).

30.6 Central plant

30.6.1 `dp_below_sp_pump_max` (CH-E)

```
        FAULT_CODE = "CH-E"
        DP_MARGIN, PMP_HI, PMP_NEAR = 2.2, 0.93, 0.06


        def apply_faults_arrow(table, cfg, context=None):
            dp = _col(table, "differential-pressure")
            sp = _col(table, "differential-pressure-sp")
            pump = _norm_cmd(table["pump-speed-command"])
            return pc.and_(
                pc.less(dp, pc.subtract(sp, DP_MARGIN)),
```

```
                pc.greater_equal(pump, PMP_HI - PMP_NEAR),
            )
```

30.6.2 `flow_high_pump_max` (CH-B)

```
        FAULT_CODE = "CH-B"
        FLOW_HI, PMP_HI, PMP_NEAR = 1100.0, 0.93, 0.06


        def apply_faults_arrow(table, cfg, context=None):
            flow = _col(table, "water-flow")
            pump = _norm_cmd(table["pump-speed-command"])
            return pc.and_(pc.greater(flow, FLOW_HI),
pc.greater_equal(pump, PMP_HI - PMP_NEAR))
```

30.6.3 `plant_supply_temp_deadband` (CH-D)

```
        FAULT_CODE = "CH-D"
        SP_BAND = 2.2


        def apply_faults_arrow(table, cfg, context=None):
            temp = _col(table, "chilled-water-supply-temperature")
            sp = _col(table, "chilled-water-supply-temperature-sp")
            pump = _norm_cmd(table["pump-speed-command"])
            low = pc.less(temp, pc.subtract(sp, SP_BAND))
            high = pc.greater(temp, pc.add(sp, SP_BAND))
            return pc.and_(pc.greater(pump, 0.01), pc.or_(low, high))
```

30.6.4 `chiller_excessive_runtime` (CH-F)

```
        from open_fdd.arrow_runtime.cookbook import arrow_rolling_mean

        FAULT_CODE = "CH-F"
        WINDOW, MAX_ON = 276, 264  # 5-min data ≈ 23 h window, 22 h max on


        def apply_faults_arrow(table, cfg, context=None):
            on = pc.greater(_col(table, "chiller-status"),
0.5).cast(pa.float64())
            roll = arrow_rolling_mean(on, WINDOW)
            count = pc.multiply(roll, float(WINDOW))
            return pc.greater(count, float(MAX_ON))
```

30.7 Heat pumps

30.7.1 `hp_discharge_cold_when_heating` (HP-D)

```
        FAULT_CODE = "HP-D"
        MIN_SAT, ZONE_COLD, FAN_ON = 85.0, 69.0, 0.01


        def apply_faults_arrow(table, cfg, context=None):
            sat = _col(table, "sa-t")
            zone = _col(table, "stat_zn-t")
            fan = _col(table, "supply-fan-status")
            return pc.and_(
                pc.greater(fan, FAN_ON),
                pc.less(zone, ZONE_COLD),
                pc.less(sat, MIN_SAT),
            )
```

30.8 Opportunistic / ventilation

30.8.1 `econ_active_warm_ambient` (AHU-E)

```
        FAULT_CODE = "AHU-E"
        T_CUTOFF, DPR_MIN = 63.0, 0.42


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            damper = _norm_cmd(table["oa-damper-command"])
            return pc.and_(pc.greater(oat, T_CUTOFF), pc.greater(damper,
DPR_MIN))
```

30.8.2 `mech_cool_when_econ_available` (AHU-E)

```
        FAULT_CODE = "AHU-E"
        T_CUTOFF, DPR_MAX = 63.0, 0.32


        def apply_faults_arrow(table, cfg, context=None):
            oat = _col(table, "oa-t")
            damper = _norm_cmd(table["oa-damper-command"])
            cool = _norm_cmd(table["cooling-valve-command"])
            return pc.and_(pc.less(oat, T_CUTOFF), pc.less(damper,
DPR_MAX), pc.greater(cool, 0.01))
```

30.8.3 `low_oa_fraction_estimated` (VAV-B)

```
        FAULT_CODE = "VAV-B"
        OA_MIN, GAP = 21.0, 2.2


        def apply_faults_arrow(table, cfg, context=None):
            mat, rat, oat = _col(table, "ma-t"), _col(table, "ra-t"),
_col(table, "oa-t")
            fan = _norm_cmd(table["supply-fan-speed-command"])
            denom = pc.subtract(oat, rat)
            safe = pc.greater(pc.abs(denom), GAP)
            num = pc.subtract(mat, rat)
            oa_pct = pc.multiply(pc.divide(num, denom), 100.0)
            low = pc.less(oa_pct, OA_MIN)
            return pc.and_(pc.greater(fan, 0.01), safe, low)
```

30.8.4 `preheat_excess_temp` (AHU-B)

```
        FAULT_CODE = "AHU-B"
        EXCESS = 2.2


        def apply_faults_arrow(table, cfg, context=None):
            pre = _col(table, "preheat-coil-leaving-air-temperature")
            sp = _col(table, "sa-t-sp")
            oat = _col(table, "oa-t")
            valve = _norm_cmd(table["preheat-valve-command"])
            hot_oa = pc.and_(
                pc.greater(oat, sp),
                pc.greater(pc.subtract(pre, oat), EXCESS),
            )
            cold_oa = pc.and_(
                pc.less(oat, sp),
                pc.greater(pc.subtract(pre, sp), EXCESS),
            )
            return pc.and_(pc.greater(valve, 0.01), pc.or_(hot_oa,
cold_oa))
```

30.9 Weather station

30.9.1 `weather_temp_spike` (BLD-B)

```
        from open_fdd.arrow_runtime.cookbook import rate_of_change_mask

        FAULT_CODE = "BLD-B"
```

```
        def apply_faults_arrow(table, cfg, context=None):
            return rate_of_change_mask(
                table,
                {**cfg, "max_per_sample": 16.0, "column": "oa-t"},
                col="oa-t",
            )
```

30.9.2 `weather_gust_lt_wind` (BLD-B)

```
        FAULT_CODE = "BLD-B"


        def apply_faults_arrow(table, cfg, context=None):
            gust = _col(table, "wind-gust-speed")
            wind = _col(table, "wind-speed")
            return pc.and_(
                pc.is_valid(gust),
                pc.is_valid(wind),
                pc.less(gust, wind),
            )
```

30.9.3 RH bounds

```
        from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

        FAULT_CODE = "DC-C"


        def apply_faults_arrow(table, cfg, context=None):
            merged = {**cfg, "value_kind": "rh"}
            return sensor_bounds_mask(table, "relative_humidity", merged)
```

30.10 Schedule + weather gating (occupied hours)

```
        from open_fdd.arrow_runtime.cookbook import _unoccupied_mask,
_fan_on_mask

        FAULT_CODE = "BLD-C"


        def apply_faults_arrow(table, cfg, context=None):
            """Fan running when schedule says unoccupied."""
            return pc.and_(_fan_on_mask(table, cfg),
_unoccupied_mask(table, cfg))
```

Pass `occupied_start_hour`, `occupied_end_hour`, `tz_offset_hours` in Rule Lab `cfg`.

30.11 FC4 — PID hunting (legacy GL36)

Ports pandas `FaultConditionFour` / `check_hunting` to Arrow. Two modes:

Mode	`cfg["hunting_mode"]`	Use on
Per-loop command	`command` (default)	VAV damper, reheat, pump VFD, RTU cooling %
AHU operating-state bitmap	`ahu_os`	RTU/AHU with economizer + fan + heat + cool columns in wide frame

Fault codes: VAV-F (VAV outputs), **CH-G** (plant pump), **RTU-E** (RTU cooling capacity), **AHU-G** (multi-signal OS / legacy `FC4` alias).

```
        """pid_hunting_fc4.py — deploy via setup_gl36_fdd.py or Rule
Lab."""

        import pyarrow as pa

        from open_fdd.arrow_runtime.cookbook import
pid_hunting_ahu_os_mask, pid_hunting_command_mask

        FAULT_CODE = "VAV-F"  # or AHU-G, CH-G, RTU-E per binding


        def apply_faults_arrow(table, cfg, context=None):
            mode = str(cfg.get("hunting_mode") or
"command").strip().lower()
            if mode in ("ahu", "ahu_os", "os"):
                return pid_hunting_ahu_os_mask(table, cfg)
            col = str(cfg.get("value_column") or cfg.get("column") or
"").strip()
            if col and col not in table.column_names:
                return pa.array([False] * table.num_rows, type=pa.bool_)
            return pid_hunting_command_mask(table, cfg, col=col or None)
```

Tuning (Acme 60 s poll, 1 h window):

```
        cfg = {
            "poll_interval_s": 60,
            "hunting_window_hours": 1.0,
            "delta_os_max": 10,          # max command steps or OS bitmap
changes per window
            "min_command_delta": 0.03,   # 3 % minimum step to count as a
reversal
            "min_active_command": 0.05,
# loop must be modulating above 5 %
        }
```

Legacy interlink: `LEGACY_CODE_MAP["FC4"]` → **AHU-G**; per-loop VAV/ plant rules use family-specific codes above.

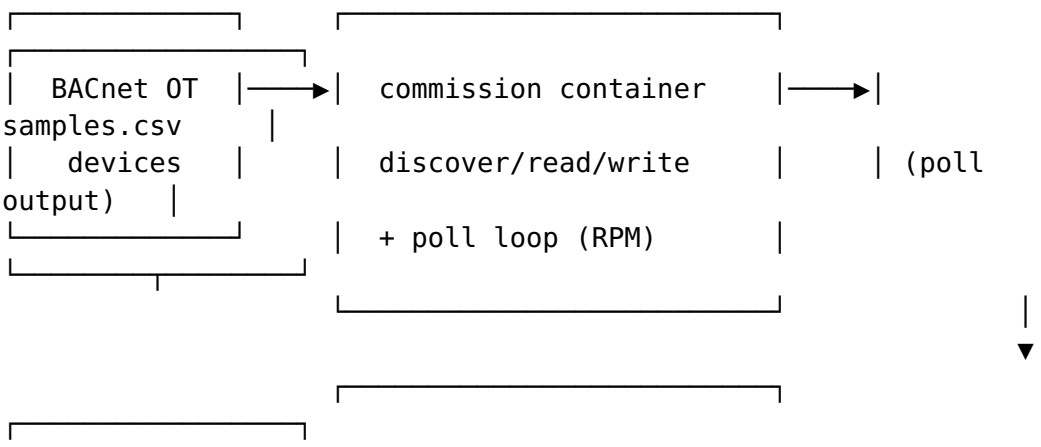
30.12 Next steps

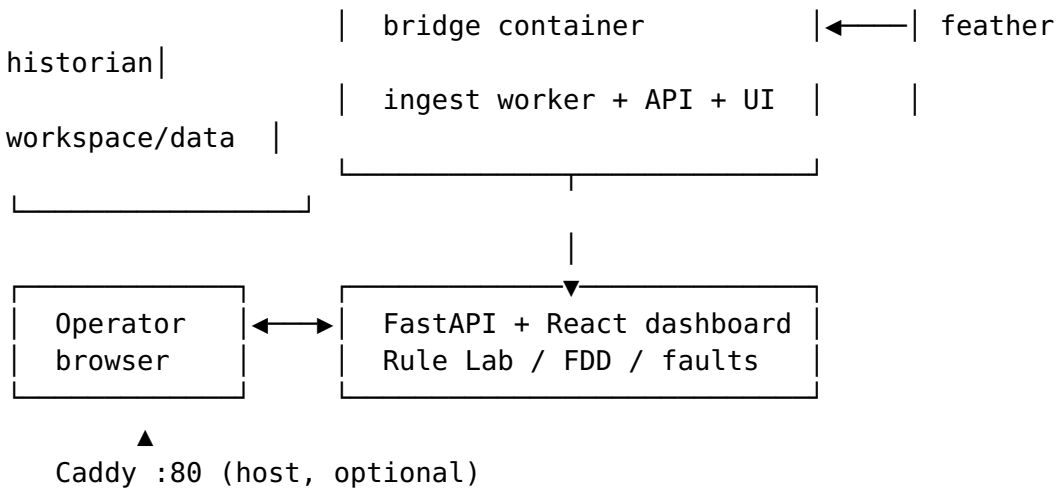
1. Copy a module into `workspace/data/rules_py/<site>_<rule>.py`
2. Bind feather columns on the Rule Lab row
3. Set `fault_code` metadata
4. Quick-test on 3–24 h feather window
5. Ship via `setup_gl36_fdd.py` or Rule Lab save

See also: [Expression cookbook]({{ “/rule-cookbook/expression-cookbook/” | relative_url }}) · Fault codes({{ “/fault-codes/” | relative_url }}) · Rule Lab({{ “/operator-bridge/rule-lab/” | relative_url }})

31 System overview

32 System overview





32.1 Docker edge stack (production)

Three GHCR images — see Containers({{ “/architecture/containers/” | relative_url }}):

Container	BACnet OT	Operator LAN
bridge	No (HTTP only)	Yes, via Caddy `:80` or loopback `:8765`
commission	Yes (host network, `:47808`)	No (internal `:8767`)
mcp-rag	No	No (internal `:8090`)

BACnet **polling** is not a separate container — it runs inside **commission**. The bridge only **ingests** poll CSV into feather.

32.2 Components

Layer	Responsibility
Operator Bridge	Auth, REST/WebSocket API, compiled dashboard, Rule Lab, historian ingest
BACnet commission	Who-Is, read/write (gated), **scheduled poll loop** → `samples.csv`
Historian	Feather files per point; local-first retention
FDD runner	`workspace/data/rules_py/*.py` on timer or `POST /api/rules/batch`
Model	Brick TTL + `model.json` bindings (`fdd_input`, equipment tree)
MCP RAG	Optional doc retrieval for agent tools
Caddy	Host reverse proxy `:80` → bridge (typical edge)

Layer	Responsibility
Ollama	Optional host LLM for building insight (not required for FDD)

32.3 Deployment

IT operators: [Quick Start — Docker]({{ “/quick-start/docker/” | relative_url }}). Modes and Caddy: Deployment modes({{ “/architecture/deployment-modes/” | relative_url }}).

32.4 Optional PyPI-only path

`pip install open-fdd` ships Arrow runtime + playground lint helpers **without** the Operator Bridge UI. Use for offline rule tests or CI. See [Appendix — Python package]({{ “/appendix/python-package/” | relative_url }}).

33 Deployment modes

34 Deployment modes

How Open-FDD is typically run: local development, lab LAN trials, and production edge on a trusted OT/IT network. Open-FDD is designed for **building LAN or OT edge** deployment — not direct exposure to the public internet.

34.1 Mode summary

Mode	Stack	BACnet	Front door	Auth
Local dev	Compose on control machine	Optional bench simulator	`127.0.0.1:8765`	`OFDD_AUTH_DISABLED=1` optional on loopback
Lab LAN	Three GHCR containers	Real or simulator VLAN	Caddy `:80` or direct `:8765`	Required unless explicit insecure lab flags
Edge production	Three GHCR containers	OT NIC via **commission**	**Caddy** `:80` or `:443` → loopback bridge	**Required** — `workspace/auth.env.local`

Mode	Stack	BACnet	Front door	Auth
	+ host Caddy			

34.2 v3 container stack (all edge modes)

Image	Role
`openfdd-bridge`	Operator Bridge API, dashboard, **feather historian ingest**
`openfdd-commission`	BACnet discover/read/write + **poll loop**
`openfdd-mcp-rag`	Doc-search sidecar for agent tools

> **`openfdd-bacnet-poll` is retired.** BACnet polling runs inside **commission** only. Do not run the legacy poll container or `openfdd-bacnet-poll` systemd unit alongside Docker commission — that double-polls the OT network.

See Containers({{ “/architecture/containers/” | relative_url }}) for ports, networks, and persistence.

34.3 Caddy HTTP (typical edge)

- Caddy listens on building LAN `:80`
- Reverse-proxies to bridge at `127.0.0.1:8765`
- Bridge stays loopback-only; operators bookmark `http://<edge-ip>/`

Bootstrap and compose: [Quick Start — Docker]({{ “/quick-start/docker/” | relative_url }}).

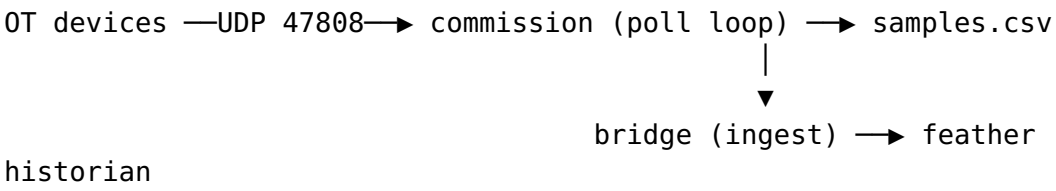
34.4 Caddy TLS (OT LAN)

For HTTPS on the edge LAN (self-signed or site CA):

1. Generate certs: `./scripts/setup_caddy_certs.sh` (control machine) or site PKI
2. Start with `OFDD_CADDY_MODE=tls`
3. Trust the CA on operator PCs

Details: TLS and certificates({{ “/security/tls-and-certs/” | relative_url }}). Security headers are set by the **Operator Bridge**; Caddy adds **HSTS** only in TLS mode.

34.5 BACnet data path



The bridge thread named `bacnet_poll_worker` is **ingest only** — it does not bind BACnet.

34.6 What not to do on production edges

- Port-forward `:8765` to the public internet without TLS and strong auth
- Run `docker compose down -v` or delete `workspace/` on a live site
- Deploy the retired `openfdd-bacnet-poll` image alongside commission

34.7 Related

Topic	Page
First deploy	[Quick Start — Docker]({{ “/quick-start/docker/”
Upgrade	Updating the stack({{ “/quick-start/updating/”
LAN security	LAN hardening({{ “/security/lan-hardening/”
Lab example site	[Examples — GL36 lab note]({{ “/examples/acme-gl36-lab/”

35 Containers

36 Containers

Open-FDD v3 edge sites run **three** GHCR containers plus optional **host** services (Caddy, Ollama). There is no separate BACnet poll container — **field BACnet polling lives in commission**.

Quick answer: Who binds BACnet UDP 47808? **`openfdd-commission`**. Bridge only ingests `samples.csv` into the feather historian.

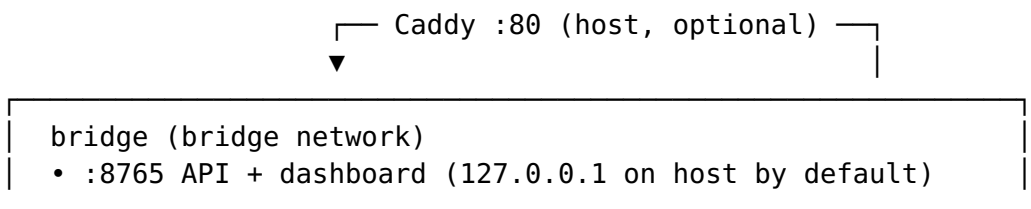
36.1 GHCR images

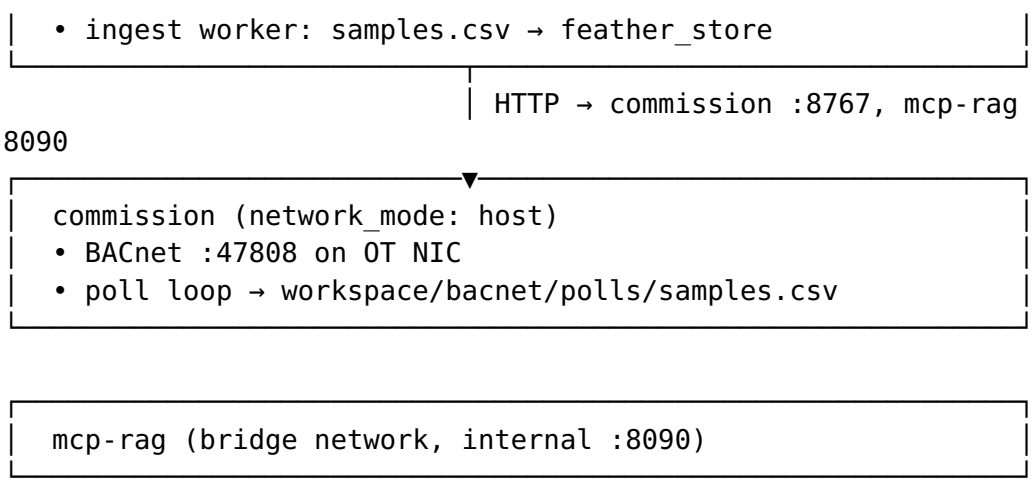
Image	Compose service	Role
`ghcr.io/bbartling/openfdd-bridge`	`bridge`	Operator API, React dashboard, feather **ingest**
`ghcr.io/bbartling/openfdd-commission`	`commission`	BACnet discover / read / write + **poll loop**
`ghcr.io/bbartling/openfdd-mcp-rag`	`mcp-rag`	Doc-search sidecar for agent tools

Tag: **`latest`** on GitHub Packages. Retired: `openfdd-bacnet-poll` (poll runs inside **commission**). GHCR retains only the **three newest** versions per image.

Template: `docker/compose.edge.yml` → copy to `~/open-fdd/docker-compose.yml`.

36.2 Stack diagram





36.3 Responsibilities

Concern	Container
BACnet Who-Is / read / write	**commission**
BACnet scheduled RPM reads	**commission** (`bacnet_poll_loop`)
CSV → feather historian	**bridge** (feather ingest worker — `bacnet_poll_worker` module name is historical)
Operator login, Rule Lab, faults UI	**bridge**
Agent doc search	**mcp-rag**

The bridge thread named `bacnet_poll_worker` is **ingest only** — it does not bind BACnet.

36.4 Network and ports (typical edge)

Endpoint	Reachable from	Notes
Caddy `:80`	Building LAN	Reverse proxy to bridge
Bridge `:8765`	Loopback (default)	Bind `127.0.0.1` in compose
Commission : `8767`	Bridge / loopback	HTTP API for BACnet ops
Commission : `47808`	OT BACnet LAN	UDP; requires

Endpoint	Reachable from	Notes
		`network_mode: host`
MCP `:8090`	Bridge container network	Not exposed to LAN by default

36.5 Long-lived deployment

All compose services set `restart: unless-stopped`. After host reboot:

```
sudo systemctl enable docker
cd ~/open-fdd && docker compose ps
```

If containers are down: `docker compose up -d` (idempotent).

See Run with Docker images({{ “/quick-start/docker/” | relative_url }}).

36.6 Persistent data

State lives on the **host** under `workspace/` (bind mount). Containers are replaceable; **backup `workspace/`** before image upgrades.

Path	Contents
`workspace/data/feather_store/`	Historian
`workspace/data/*.json`	Model, rules, FDD results
`workspace/bacnet/commissioning/`	`commission.env`, `points.csv`
`workspace/bacnet/polls/samples.csv`	Poll output
`workspace/auth.env.local`	Login secrets
`workspace/api/static/app/`	Dashboard bundle (if updated separately)

36.7 Optional host services

Service	Role
Caddy	TLS or plain HTTP `:80` → `127.0.0.1:8765`

Service	Role
Ollama	Optional LLM on `:11434` (not in the three-image stack)
FDD loop timer	Optional `openfdd-fdd-loop.timer` on host (`docker exec` batch)

37 Data flow

38 Data flow

38.1 Telemetry ingest

Open-FDD supports three commissioning drivers. Each appends long-format samples to workspace CSVs and writes **feather shards** tagged by `source` (`bacnet`, `modbus`, `json_api`).

38.1.1 BACnet

1. **Commission poll loop** (inside the `commission` container) reads enabled BACnet points from `points.csv` on a schedule (host network, BACnet `:47808`).
2. RPM results append to `workspace/bacnet/polls/samples.csv`.
3. **Bridge ingest worker** loads new rows into `feather_store/bacnet/`.

```
BACnet devices → commission (poll loop) → samples.csv → bridge (ingest) → feather_store/bacnet
```

38.1.2 Modbus TCP

1. Operator configures registers on the **Modbus** tab (or `read_and_store` API).
2. Poll worker reads holding/input registers via Modbus TCP.
3. Samples append to `workspace/modbus/polls/samples.csv` → `feather_store/modbus/`.

38.1.3 JSON API (HTTP/HTTPS)

1. Operator configures REST endpoints on the **JSON API** tab — GET/POST, Bearer or Basic auth, `${ENV:VAR}` placeholders from

`json_api.env.local`, optional TLS verify off for self-signed OT gateways.
2. Poll worker issues HTTP requests (one GET can fan out to multiple JSON paths, e.g. OpenWeather `web-oat-t` / `web-rh` / `web-weather-desc`).
3. Samples append to `workspace/json_api/polls/samples.csv` → `feather_store/json_api/`.
4. FDD batch **merges** `json_api` columns with BACnet/Modbus on the same site (nearest timestamp, 30 min tolerance) for cross-source rules.

```
OT REST / weather API → bridge JSON API worker → samples.csv → feather_store/json_api
                                                                ↘ merged into FDD frame with bacnet/modbus
```

Showcase: [OpenWeatherMap bundle]({{ "/drivers/json-api/" | relative_url }}#openweathermap-showcase-recommended-demo). Details: Driver framework({{ "/drivers/" | relative_url }}), [BACnet polling]({{ "/bacnet/polling/" | relative_url }}), JSON API driver({{ "/drivers/json-api/" | relative_url }}), Containers({{ "/architecture/containers/" | relative_url }}).

38.2 Rule evaluation

1. Rules are Python files in `workspace/data/rules_py/` with metadata in `rules_store.json`.
2. **Bindings** map rules to points via the Brick model (`fdd_input` tags).
3. Batch runner (`POST /api/rules/batch` or host timer) produces `fdd_results.json`.
4. Dashboard **faults** view and check-engine status read aggregated results.

38.3 Model sync

- Commissioning imports update `model.json` / TTL graph.
- `POST /api/model/sync-ttl` refreshes SPARQL-backed tree used by the UI.

38.4 Retention

Feather files grow on disk; retention is configurable (`workspace/data.env.local`). Image upgrades must **preserve** `workspace/data/` — backup before `docker compose up -d --force-recreate`. See Updating the stack({{ "/quick-start/updating/" | relative_url }}).

39 Architecture

40 Architecture

Open-FDD v3 edge: **three Docker containers** (`bridge`, `commission`, `mcp-rag`), BACnet polling in **commission**, feather ingest in **bridge**, optional host Caddy and Ollama.

Page	Topic
System overview({{ “/architecture/overview/”	relative_url }})
Deployment modes({{ “/architecture/deployment-modes/”	relative_url }})
Data flow({{ “/architecture/data-flow/”	relative_url }})
Containers({{ “/architecture/containers/”	relative_url }})
Driver framework({{ “/drivers/”	relative_url }})

Operators: [Quick Start — Docker]({{ “/quick-start/docker/” | relative_url }}) — no git clone on the edge host.

BACnet polling: Polling({{ “/bacnet/polling/” | relative_url }}) — commission container only.

41 Fault confirmation / minimum duration

42 Fault confirmation

A raw rule condition can flicker true for one sample. **Fault confirmation** delays the final fault mask until the condition stays true for N consecutive rows or N minutes.

42.1 Config fields

Field	Type	Meaning
`min_true_rows`	int	Consecutive true samples required
`min_elapsed_minutes`	float	Elapsed time from streak start (uses `timestamp`)
`poll_interval_s`	float	Fallback when timestamps missing
`timestamp_column`	str	Default `timestamp`

42.2 Example

At **60-second polling**, `min_true_rows: 5` means the condition must persist about **5 minutes** before `fault` becomes true.

PyArrow:

```
def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["duct-t"], 80.0)
```

Rule config:

```
{"min_true_rows": 5, "poll_interval_s": 60}
```

DataFusion SQL (same config — confirmation runs after SQL):

```
SELECT
  *,
  "duct-t" > 80.0 AS fault
FROM telemetry
```

The runtime applies confirmation in `open_fdd.arrow_runtime.confirmation` for both backends.

43 Rule Cookbook

44 Rule Cookbook

Practical patterns for **Arrow-native Rule Lab** on feather historian PyArrow tables — **no pandas on the IoT edge**.

Page	Use when
[Rule Lab quick start]({{ “/operator-bridge/rule-lab/”	relative_url }})
[PyArrow rule recipes]({{ “/rule-cookbook/python-recipes-arrow/”	relative_url }})
DataFusion SQL recipes({{ “/rule-cookbook/datafusion-sql-recipes/”	relative_url }})
Fault confirmation({{ “/rule-cookbook/fault-confirmation/”	relative_url }})
Windowing & debugging({{ “/rule-cookbook/windowing-debugging/”	relative_url }})
[Expression cookbook]({{ “/rule-cookbook/expression-cookbook/”	relative_url }})
GL36 algorithm stubs({{ “/rule-cookbook/gl36-algorithm-stubs/”	relative_url }})

PyArrow is the primary path for complex HVAC FDD and ML prep. **DataFusion SQL** is optional, Rust-ready, and best for simple expression-style rules.

Programmatic sensor defaults: `open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES`.

```
        import pyarrow.compute as pc
        from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

        def apply_faults_arrow(table, cfg, context=None):
            return sensor_bounds_mask(table, "zone_temp", cfg,
min_true_rows=5)
```

45 Windowing & debugging

46 Windowing & debugging

46.1 Lookback vs rolling window

	Historian lookback	Rolling window
Set by	Batch `lookback_hours`, FDD loop, Quick test UI	`WINDOW_SAMPLES` constant in `rule.py`
Typical	1 h scheduled; 24 h Update all	12 samples ≈ 1 h @ 5 min poll
Validate with	`_kit_lookback_stats(table)` — see [Lookback window helper]({{ "/rule-cookbook/lookback-window/" relative_url }})	

46.2 Rolling windows (Arrow)

Use `open_fdd.arrow_runtime.windows` for sample-based rolling min/max and consecutive-true streaks:

```
from open_fdd.arrow_runtime.windows import arrow_rolling_min, arrow_rolling_max, arrow_consecutive_true
```

Cookbook masks (`flatline_1h_mask`, `spread_1h_mask`, `oob_mask`) default to ~12 samples (~1 h at 5 min poll).

Bench rules use constants: `WINDOW_SAMPLES`, `FLATLINE_TOLERANCE`, `OAT_LOW` / `OAT_HIGH`, etc. Legacy `cfg` keys still work for uploaded rules that read `cfg.get(...)`.

46.3 Rule Lab console

Quick-test and batch responses include `backend: arrow`, `ms`, and flagged row counts. Arrow summary events appear in the event console on fault hits.

47 Expression cookbook (Arrow-native)

48 Expression cookbook (Arrow-native)

Reference for **Open-FDD 3.x Rule Lab**: every rule is a Python module with **`apply_faults_arrow(table, cfg, context)`** using **`pyarrow.compute`** — **no pandas**, **no YAML expression files**, **no NumPy DataFrames on the IoT edge**.

This is the **only** expression cookbook for Open-FDD 3.x. The pandas/YAML `RuleRunner` path is retired — use Arrow `apply_faults_arrow` and the operator bridge Rule Lab. Legacy GL36-style recipes map to **fixed fault codes({{ "/fault-codes/" | relative_url }})** and Arrow patterns below.

Topic	Page
Full copy-paste library (GL36 A–M, VAV, plant)	**Python recipes (full Arrow library) ({{ "/rule-cookbook/python-recipes-arrow/"
Quick templates	[Arrow recipes]({{ "/rule-cookbook/arrow-recipes/"
Shared imports	[Python recipes]({{ "/rule-cookbook/python-recipes/"
Console window stats	[Lookback window]({{ "/rule-cookbook/lookback-window/"
Fault code catalog	Fault codes({{ "/fault-codes/"
Programmatic defaults	`open_fdd.arrow_runtime.sensor_catalog`

48.1 Legacy → modern translation

Legacy (pandas / YAML)	Arrow-native (3.x)
`type: expression` + `expression:` string	`apply_faults_arrow()` in `rules_py/*.py`
`type: hunting` (GL36 **FC4**)	`pid_hunting_fc4.py` → **AHU-G** / **VAV-F** / **CH-G** / **RTU-E**
`RuleRunner.run(df, column_map=…)`	Rule Lab batch on **feather PyArrow table**
`np.maximum(a, b)`	`pc.max(a, b)`

Legacy (pandas / YAML)	Arrow-native (3.x)
`series.rolling(n).min()`	`arrow_rolling_min(vals, n)` from `open_fdd.arrow_runtime.windows`
`series.diff().abs()`	`arrow_abs_diff(vals, 1)`
`normalize_cmd(x)`	`norm_cmd_array()` / `pc.if_else(pc.greater(x, 1), pc.divide(x, 100), x)`
`params.max_temp` in YAML	Module constant or `cfg["bounds_high"]`
`flag: rule_a_flag`	Rule metadata **`fault_code`** → `AHU-A` … `VAV-C`
Numeric codes `VAV-03`	Letter codes **`VAV-C`** (see `LEGACY_CODE_MAP` in bridge)
Pandas `FC4` / `fc4_flag`	`LEGACY_CODE_MAP`: `FC4` → **AHU-G**

Edge constraint: Docker / BACnet poll path never imports pandas. Central portfolio Dash may use pandas for CSV analytics only.

48.2 How Arrow rules are structured

1. **Historian columns** — Feather column names from BACnet poll (`oa-t`, `sa-t`, `stat_zn-t`, …) or Brick labels via model export.
2. **Module constants** — Thresholds at top of file (bench style) or read from `cfg` for site tuning.
3. **`apply_faults_arrow`** — Returns a **boolean PyArrow array** (True = fault sample).
4. **`fault_code`** — Set in Rule Lab metadata; must exist in `GET /api/faults/catalog`.

```python
import pyarrow.compute as pc
from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

FAULT_CODE = "VAV-C"  # metadata in Rule Lab UI


def apply_faults_arrow(table, cfg, context=None):
    return sensor_bounds_mask(table, "zone_temp", cfg)
```

48.2.1 Command scaling (0–1 vs 0–100 %)

BACnet often exposes **0–100** for damper/VFD commands. Scale explicitly:

```
        def _norm_cmd(col):
            return pc.if_else(pc.greater(col, 1.0), pc.divide(col,
100.0), col)
```

48.2.2 Occupied hours (schedule gating)

Use `open_fdd.arrow_runtime.cookbook._unoccupied_mask` or compare local hour from `timestamp` column. Example: **fan on when unoccupied → BLD-C** / `schedule_compare`.

48.3 Sensor validation (bounds, flatline, rate of change)

Use **`open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES`** for defaults, or copy constants into `rules_py`. Tune per site in Rule Lab `cfg` or building-agent tuning brief.

48.3.1 Bounds (out of range) — `oob_rolling`

Sensor kind	Min	Max	Flatline tol	Max Δ / hour	Max Δ / 15 min	Typical fault code
Zone temp (°F)	55	90	0.10	4.0	2.0	**VAV-C**
Supply air temp	50	110	0.15	8.0	3.0	**AHU-C**, **RTU-C**
Return air temp	55	95	0.10	3.0	1.5	**AHU-D** (also mixing)
Mixed air temp	40	110	0.15	6.0	2.5	**AHU-D**
Outdoor air temp	−40	130	0.10	12.0	6.0	**BLD-B**
Duct static (inH_2O)	−0.5	3.0	0.02	0.5	0.25	**AHU-A**
Relative humidity (%)	0	100	1.0	15.0	8.0	**DC-C**
Chilled water (°F)	40	90	0.10	4.0	2.0	**CH-D**
Hot water (°F)	70	200	0.15	6.0	3.0	**CH-D**

Sensor kind	Min	Max	Flatline tol	Max Δ / hour	Max Δ / 15 min	Typical fault code
Condenser water (°F)	50	110	0.15	5.0	2.5	**CH-A**
CO_2 (ppm, occupied)	400	1000	5.0	200	80	**VAV-B** (ventilation)
Discharge air temp	45	120	0.15	10.0	4.0	**VAV-E**, **HP-D**

Return air uses a **narrow band** vs zone/OAT because RAT should track building return conditions, not outdoor extremes. When OAT/RAT/MAT are all present, prefer **mixing envelope** (below) over RAT bounds alone.

CO_2 upper bound is an occupied ventilation target; unoccupied rules may use 400–2000 ppm per policy.

48.3.2 Flatline (stuck sensor) — `flatline_1h`

Default **12 samples ≈ 1 hour** at 5-minute poll. True when rolling max − min ≤ tolerance.

```
        from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask

        def apply_faults_arrow(table, cfg, context=None):
            return sensor_flatline_mask(table, "outdoor_air_temp", cfg)
# BLD-B
```

Bench references: `workspace/data/rules_py/bench_oa-t_flatline_1h.py`, `bench_stat_zn-t_flatline_1h.py`.

48.3.3 Rate of change (spike) — `rate_of_change`

Flags physically impossible step changes (comms glitch, substituted value).

```
        from open_fdd.arrow_runtime.cookbook import rate_of_change_mask
        from open_fdd.arrow_runtime.sensor_catalog import cfg_from_profile

        def apply_faults_arrow(table, cfg, context=None):
            merged = cfg_from_profile("outdoor_air_temp", cfg)
            merged["samples_per_hour"] = 12  # 5-min poll
            return rate_of_change_mask(table, merged, col="oa-t")
```

Legacy **weather_temp_spike** (`spike_limit: 16` °F per step) → **BLD-B** with `max_per_15min: 6` or stricter per-step limit.

48.3.4 Mixing envelope (MAT vs OAT/RAT) — `mixing_envelope`

Legacy GL36 **Rule B / C** and **AHU-D**. MAT should sit between OAT and RAT (± tolerance) while fan runs.

```python
from open_fdd.arrow_runtime.cookbook import mixing_envelope_mask

def apply_faults_arrow(table, cfg, context=None):
    return mixing_envelope_mask(
        table,
        {**cfg, "mixing_tol": 1.15},
        mat_col="ma-t",
        oat_col="oa-t",
        rat_col="ra-t",
        fan_col="supply-fan-speed-command",
    )
```

48.4 GL36-inspired AHU rules (legacy expression → fault code)

ASHRAE Guideline 36-style rules from the old YAML cookbook, translated to Arrow. Assign `fault_code` per row in Rule Lab.

Legacy rule	Summary	Code	Full Arrow module
Rule A — duct static low @ full fan	SP below SP setpoint at high VFD	**AHU-A**	[Rule A]({{ “/rule-cookbook/python-recipes-arrow/”
Rule B — blend below band	MAT below OAT/RAT envelope	**AHU-D**	[Rules B & C]({{ “/rule-cookbook/python-recipes-arrow/”
Rule C — blend above band	MAT above OAT/RAT envelope	**AHU-D**	[Rules B & C]({{ “/rule-cookbook/python-recipes-arrow/”
Rule D — discharge cold when heating	SAT low vs MAT, heat valve open	**AHU-B**	[Rule D]({{ “/rule-cookbook/python-recipes-arrow/”
Rule E — SAT low, full heating	SAT below SP, valve > 90%	**AHU-C**	[Rule E]({{ “/rule-cookbook/python-recipes-arrow/”
Rule F — SAT/MAT mismatch econ	Econ mode, SAT ≠ MAT	**AHU-E**	[Rule F]({{ “/rule-cookbook/python-recipes-arrow/”

Legacy rule	Summary	Code	Full Arrow module
Rule G — ambient warm free cool	OAT > SAT SP, econ open, cool off	**AHU-E**	[Rule G]({{ “/rule-cookbook/python-recipes-arrow/”
Rule H — OAT/MAT mismatch econ+mech	Mech + econ, MAT ≠ OAT	**AHU-E**	[Rule H]({{ “/rule-cookbook/python-recipes-arrow/”
Rule I — OAT/MAT mismatch econ-only	Econ only, MAT ≠ OAT	**AHU-E**	[Rule I]({{ “/rule-cookbook/python-recipes-arrow/”
Rule J — discharge above blend cooling	SAT > MAT in cooling	**AHU-B**	[Rule J]({{ “/rule-cookbook/python-recipes-arrow/”
Rule K — discharge above SP full cool	SAT > SP, full cooling	**AHU-C**	[Rule K]({{ “/rule-cookbook/python-recipes-arrow/”
Rule L — cooling coil ΔT when off	CHW drop when valves closed	**CH-C**	[Rule L]({{ “/rule-cookbook/python-recipes-arrow/”
Rule M — heating coil ΔT when off	HW rise when valves closed	**AHU-B**	[Rule M]({{ “/rule-cookbook/python-recipes-arrow/”
FC4 — PID hunting	Excessive command reversals or AHU OS changes	**AHU-G** / **VAV-F** / **CH-G** / **RTU-E**	[FC4 PID hunting]({{ “/rule-cookbook/python-recipes-arrow/”

All rules A–M have **full `apply_faults_arrow` modules** in Python recipes (full Arrow library)({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}).

48.5 Starter pack (VAV / AHU baseline)

Maps to legacy YAML starter filenames; use as first deploy bundle.

Legacy YAML starter	Arrow approach	Fau
`01_vav_zone_temp_bounds_occupied`	`sensor_bounds_mask(..., "zone_temp")` + occupied mask	**VAV**

Legacy YAML starter	Arrow approach	Fau
`02_vav_zone_temp_flatline_occupied`	`sensor_flatline_mask(..., "zone_temp")` + occupied	**VAV**
`03_vav_damper_command_extreme_flatline`	`flatline_1h_mask` on damper cmd column	**VAV**
`04_ahu_runtime_outside_schedule`	`after_hours_fan_satisfied_mask` / run-hours script	**BL**
`05_ahu_duct_static_pressure_not_maintained`	[Rule A]({{ “/rule-cookbook/python-recipes-arrow/”	rela #ru stat full- ahu
`06_ahu_internal_temp_sensor_bounds`	`sensor_bounds_mask` on SAT/MAT	**AH**
`07_ahu_internal_temp_sensor_flatline`	`sensor_flatline_mask` on SAT	**AH**

48.5.1 Economizer starters

Full modules: [economizer section]({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#economizer-starters) (`ahu_econ_100oa_temp_tracking_fault`, `ahu_mech_cooling_when_free_cooling_available`, `ahu_oa_damper_excess_open_extreme_ambient`).

48.6 VAV, central plant, heat pump

Full modules: [VAV]({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#vav-zones) · Central plant({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#central-plant) · Heat pumps({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#heat-pumps).

Rolling persistence: `arrow_rolling_min` + `pc.and_` (replaces pandas `.rolling(n).min()`).

48.7 Opportunistic / ventilation

Full modules: [Opportunistic section]({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#opportunistic–ventilation) · [Weather]({{ “/rule-cookbook/python-recipes-arrow/” | relative_url }}#weather-station).

48.8 Data quality vs equipment faults

Symptom	Pattern	Code	Do not confuse with
Flatline 1 h	`flatline_1h`	**VAV-C**, **AHU-C**, **BLD-B**	Equipment off / damper closed
OOB sample	`oob_rolling`	sensor-specific	True process excursion
Spike between polls	`rate_of_change`	**BLD-B**	Fast legitimate weather front
No new samples	`stale_points`	**BLD-D**	Equipment fault
MAT ∉ [OAT, RAT]	`mixing_envelope`	**AHU-D**	Stratification during startup

Always gate sensor faults with **fan on**, **valve open**, or **occupied** where appropriate — see [Sensor & data quality]({{ “/fault-codes/sensor-quality/” | relative_url }}).

48.9 Metric (°C) equivalents

Set `cfg["temp_unit"] = "metric"` in Rule Lab or scale constants:

Imperial	Metric (approx)
55 °F	12.8 °C
90 °F	32.2 °C
50 °F	10.0 °C
110 °F	43.3 °C
2.0 °F tol	1.1 °C
0.15 inH_2O	~37 Pa

`open_fdd.playground.temp_units` documents UI field keys for °F/°C toggle.

48.10 Binding rules to the fault catalog

1. Pick a **letter code** from Fault codes({{ “/fault-codes/” | relative_url }}) — never invent suffixes.
2. In Rule Lab, set **fault_code** on the rule row.

3. Batch FDD aggregates into `GET /api/faults/status` and portfolio rollup.
4. Legacy numeric codes (`AHU-03`) migrate via `LEGACY_CODE_MAP` in the bridge.

```
curl -s http://127.0.0.1:8765/api/faults/catalog | jq
'.families[].codes[] | select(.code=="VAV-C")'
```

48.11 Pandas YAML → Arrow checklist (commissioning)

- [ ] Replace each `type: expression` YAML with a `rules_py` module
- [ ] Map `inputs` Brick labels → feather column names in model/poll CSV
- [ ] Set `fault_code` per rule (letter suffix)
- [ ] Run Rule Lab kit on 7-day feather window; confirm flag rate sane
- [ ] Apply sensor_catalog defaults; tune bounds for site climate
- [ ] Enable building-agent check-in; verify faults in portfolio rollup

Next: Python recipes (full Arrow library)({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}) · [Arrow recipes]({{ "/rule-cookbook/arrow-recipes/" | relative_url }}) · Fault codes({{ "/fault-codes/" | relative_url }}) · Rule Lab({{ "/operator-bridge/rule-lab/" | relative_url }})

49 GL36 algorithm stubs

50 GL36 algorithm stubs (doc-only)

Draft **supervisory** PyArrow patterns for the future Algorithms tab. These mirror Niagara GL36 blocks documented in:

README_TRIM_RESPOND.md

FDD rules return **fault masks**. Algorithms return **setpoints or request integers** — the stubs below use `apply_algorithm_arrow` as the planned entrypoint. For early testing in Rule Lab, you can adapt the logic into advisory FDD rules (boolean “sequence unhealthy” flags).

50.1 Shared lookback helper

```
        import pyarrow.compute as pc

        LOOKBACK_HOURS = 6


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")
```

50.2 VAV zone cooling requests (0–3)

Simplified from GL36 VAV box request generator — zone temp vs cooling setpoint bands.

```
        """GL36 VAV cooling requests — doc stub."""

        import pyarrow.compute as pc

        ZONE_TEMP = "zn-t"
        ZONE_SP = "zn-t-sp"
        ZONE_DEMAND = "zn-cool-loop"
        HIGH_DIFF_F = 5.0
        MED_DIFF_F = 3.0
        LOOP_ON = 95.0
        LOOP_OFF = 85.0


        def apply_algorithm_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            tz = pc.cast(table[ZONE_TEMP], "float64")
            sp = pc.cast(table[ZONE_SP], "float64")
            diff = pc.subtract(tz, sp)
            demand = pc.cast(table[ZONE_DEMAND], "float64")
            # Last row request level for kit demo (full port needs timer
state)
```

```
            last_diff = diff[-1].as_py() if table.num_rows else 0.0
            last_demand = demand[-1].as_py() if table.num_rows else 0.0
            if last_diff >= HIGH_DIFF_F:
                req = 3
            elif last_diff >= MED_DIFF_F:
                req = 2
            elif last_demand > LOOP_ON:
                req = 1
            else:
                req = 0
            print(f"cooling_requests={req} dT={last_diff:.1f}F
loop={last_demand:.0f}%")
            return {"cooling_requests": req}
```

50.3 AHU duct static trim & respond (advisory fault)

Detect **duct static stuck high** while VAV pressure requests are low — trim & respond may not be trimming.

```
        """AHU duct static trim advisory — doc stub."""

        import pyarrow.compute as pc

        DUCT_SP = "duct-static"
        DUCT_SP_MAX = 1.50
        DUCT_SP_TRIM_TARGET = 0.80
        VAV_PRESS_REQ_SUM = "vav-press-req-sum"
        LOOKBACK_HOURS = 12


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            sp = pc.cast(table[DUCT_SP], "float64")
            reqs = pc.cast(table[VAV_PRESS_REQ_SUM], "float64")
            stuck_high = pc.and_(pc.greater(sp, DUCT_SP_TRIM_TARGET),
pc.less(reqs, 1.0))
            return pc.and_(stuck_high, pc.greater(sp, DUCT_SP_MAX * 0.9))
```

50.4 Chiller plant enable (valve request counting)

From GL36 §5.18.15.2 — count AHUs with CHW valve above threshold.

```
        """Chiller plant enable advisory — doc stub."""

        import pyarrow.compute as pc
```

```
        AHU_VALVE_COLS = ["ahu1-clg-vlv", "ahu2-clg-vlv", "ahu3-clg-vlv"]
        REQ_THRESH = 95.0
        DIS_THRESH = 10.0
        MIN_AHUS_REQ = 2


        def apply_algorithm_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            requesting = 0
            for col in AHU_VALVE_COLS:
                if col not in table.column_names:
                    continue
                v = pc.cast(table[col], "float64")[-1].as_py()
                if v >= REQ_THRESH:
                    requesting += 1
            enable = requesting >= MIN_AHUS_REQ
            print(f"chiller_enable={enable} requesting_ahus={requesting}/
{MIN_AHUS_REQ}")
            return {"chiller_enable": enable, "requesting_ahus":
requesting}
```

50.5 Hot water plant HWST trim & respond (advisory)

Flags **HWST trimmed too low** while heating requests remain high.

```
        """HWST trim advisory — doc stub."""

        import pyarrow.compute as pc

        HWST = "hw-supply-t"
        HWST_SP = "hw-supply-t-sp"
        HEAT_REQ_SUM = "hw-reset-req-sum"
        LOW_HWST_F = 120.0
        LOOKBACK_HOURS = 6


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            temp = pc.cast(table[HWST], "float64")
            reqs = pc.cast(table[HEAT_REQ_SUM], "float64")
            return pc.and_(pc.less(temp, LOW_HWST_F), pc.greater(reqs,
2.0))
```

50.6 Chilled water plant (DP + CHWST reset advisory)

Flags **CHWST at design cold** with low cooling requests — possible stuck 100% plant loop.

```
        """CHW plant reset advisory — doc stub."""

        import pyarrow.compute as pc

        CHWST = "chw-supply-t"
        CHWST_DESIGN_COLD_F = 44.0
        COOL_REQ_SUM = "chw-reset-req-sum"
        LOOKBACK_HOURS = 6


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            temp = pc.cast(table[CHWST], "float64")
            reqs = pc.cast(table[COOL_REQ_SUM], "float64")
            return pc.and_(pc.less(temp, CHWST_DESIGN_COLD_F + 1.0),
pc.less(reqs, 1.0))
```

50.7 Testing later

1. Bind historian columns in **Model & assignments** (`fdd_input` / brick types).
2. Copy a stub into Rule Lab as an FDD advisory rule, or wait for the **Algorithms** tab kit export.
3. Run `python run_test.py` from a downloaded kit and confirm `_kit_lookback_stats` prints expected `start` / `stop` / `span`.

Reference implementation (Niagara Java): README_TRIM_RESPOND.md.

51 Central plants (CHW / CTW / BLR)

52 Central plant Arrow recipes

Chiller plant, condenser water, cooling tower, boiler, and hot-water distribution faults use the same Arrow-native primitives as air-side rules.

52.1 Starter fault codes

Code	Primitive	Recipe
CHW-DT-001	low_delta_t_mask	Low chilled-water ΔT syndrome
CHW-DP-001	reset_stuck_mask	CHW differential pressure reset stuck high

52.2 Required point roles (CHW)

- chw_supply_temp, chw_return_temp — ΔT rules
- chw_dp, chw_dp_setpoint — reset stuck rules
- pump_command, pump_feedback — command/feedback mismatch

52.3 Synthetic test pattern

```
import pyarrow as pa
from open_fdd.arrow_runtime.primitives import low_delta_t_mask

table = pa.table({
    "chw_supply_temp": [44.0] * 10,
    "chw_return_temp": [44.5] * 10,
})
mask = low_delta_t_mask(
    table,
    {"min_delta_t": 4.0},
    supply_col="chw_supply_temp",
    return_col="chw_return_temp",
```

```
        )
        assert mask.to_pylist()[-1] is True
```

See [Fault codes — chiller plant]({{ “/fault-codes/” | relative_url }}) (expanded in 3.0.19).

53 Dashboard overview

54 Dashboard overview

54.1 Primary areas

Area	Purpose
Home / Building insight	Check-engine status; fault cards show equipment, rule, point, historian column, BACnet
Trends	Point history from feather store
Faults	Rule hits grouped by equipment family
Rule Lab	Edit, lint, and test Python FDD rules (PyArrow kit zip)
Algorithms	Supervisory GL36 trim & respond (coming soon)
Model & assignments	Equipment tree, FDD rule pins, commissioning JSON
BACnet	Discovery, reads, commission workflows
Settings / health	Stack status, auth mode

54.2 Live updates

Dashboard tiles use REST polling and `WS /ws/dashboard` for selected live views (ticket auth via `POST /api/auth/ws-ticket`).

54.3 Roles

Read-only operators see trends and faults; integrators access Rule Lab and model import. See Auth and roles({{ “/operator-bridge/auth-roles/” | relative_url }}).

55 Auth and roles

56 Auth and roles

56.1 Production (LAN / edge)

- Set `OFDD_AUTH_SECRET` and role passwords in `workspace/auth.env.local`.
- Bridge on non-loopback addresses **requires** auth unless explicit insecure dev flags are set (lab only).

Role	Typical access
`viewer`	Read trends, faults, model tree
`operator`	Run batch FDD, acknowledge views
`integrator`	Rule Lab, bindings, model import
`commission`	BACnet writes (plus write guard)
`agent`	Agent tools / app-edit gates
`admin`	Full configuration

Login: `POST /api/auth/login` → Bearer JWT on API calls.

56.2 Localhost dev

`OFDD_BRIDGE_HOST=127.0.0.1` with `OFDD_AUTH_DISABLED=1` is acceptable on a single machine. **Never** use auth-disabled mode on a building LAN.

Full hardening: Security({{ “/security/” | relative_url }}).

57 Rule Lab

58 Rule Lab

Rule Lab authors **Arrow-native** Python FDD rules against the feather historian.

58.1 Equipment-scoped test and export

Use **Equipment-scoped test & export** on Rule Lab:

1. Select site, equipment, and sensor (point).
2. **Test selected rule** or **Test all for equipment** against recent feather data.
3. **Download equipment kit** — zip with `manifest.json`, `equipment.json`, `points.json`, per-rule samples, expanded helper source, and commissioning export subset.

API: `GET /api/rules/export-equipment-kit?equipment_id=…` · expanded helpers: `GET /api/playground/rules/{id}/source-expanded`.

58.2 Export all rules

Integrators can download **Export all rules** — one zip with per-rule kits, `manifest.json`, catalog snapshots, and model TTL. API: `GET /api/rules/export-all-kit`. Bench rules use **module constants** at the top of `rule.py` — no browser config panel, no `config.json` in the dev kit zip.

```
        """Bench OA-T out of bounds (Arrow)."""

        import pyarrow.compute as pc

        VALUE_COLUMN = "oa-t"
        OAT_LOW = 68.0
        OAT_HIGH = 88.0
        LOOKBACK_HOURS = 1


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def _kit_value_stats(table):
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            print(
                f"column={VALUE_COLUMN} min={pc.min(vals).as_py():.2f} "
                f"max={pc.max(vals).as_py():.2f}
mean={pc.mean(vals).as_py():.2f}"
            )
```

```
        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            _kit_value_stats(table)
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            return pc.or_(pc.less(vals, OAT_LOW), pc.greater(vals,
OAT_HIGH))
```

58.3 Workflow

1. **Download kit** — PyArrow zip (see below)
2. **Edit constants** locally (`VALUE_COLUMN`, limits, `LOOKBACK_HOURS`, `WINDOW_SAMPLES`)
3. **Run** `pip install -r requirements.txt` then `python run_test.py`
4. **Upload** `rule.py` on Rule Lab (integrator)

Upload validation (Phase B): AST parse, forbidden imports (**no pandas/numpy**), `apply_faults_arrow(table, cfg, context=None)` signature for fault rules. Script-mode rules must use **`table`** (PyArrow), not legacy **`df`** DataFrames — lint errors explain this for AI agents.

The bridge runs rules on **PyArrow tables** via `open_fdd.arrow_runtime`, and persists `.py` sources under `workspace/data/rules_py/`.

- **Quick test** — `POST /api/playground/test-rule` (default 3 h lookback)
- **Batch** — `POST /api/rules/batch` / `openfdd-fdd-loop` timer (**1 h** default lookback)
- **Update all records** — 24 h lookback, 6 h chunks (`use_chunks: true`)
- **Templates** — `GET /api/playground/arrow-templates`

Pin points to rules via **Model & assignments** (`/model`) commissioning JSON or BACnet tree right-click. Export shows Rule Lab **names** on each point (`fdd_rules_linked`); import uses rule **ids** (`fdd_rule_ids`).

Bench seed: `python3 scripts/setup_bench_afdd.py` — imports `bench_import_model.json` and bench cookbook rules with matching id/name pairs.

58.4 Dev kit zip (download)

Download kit on Rule Lab exports `openfdd-rule-kit-<rule_id>.zip` with:

File	Purpose
`rule.py`	Arrow rule + **constants** at top; optional `_kit_lookback_stats` / `_kit_value_stats` helpers
`data.py`	`SITE_ID`, `LOOKBACK_HOURS`, `load_table()` reading `sample.feather`
`sample.feather`	Full historian window for the kit lookback (all rows, not a CSV preview)
`run_test.py`	Runs `rule.apply_faults_arrow(table, {}, context=...)` and prints `flagged=N`
`requirements.txt`	`open-fdd>=3.0.1` and `pyarrow`
`README.md`	Install and run instructions
`column_map.json`	BRICK logical keys → feather column names
`kit_meta.json`	Export metadata (site, columns, row count, lookback hours)

Not included: `config.json` (retired — tune thresholds as Python constants).

Local test:

```
unzip openfdd-rule-kit-*.zip
cd openfdd-rule-kit-*
pip install -r requirements.txt
python run_test.py
```

Expected console output:

```
site=demo lookback_h=3 rows=…
lookback=3h rows=… start=… stop=… span=2.98h
column=oa-t min=… max=… mean=…
flagged=…
```

58.5 Lookback windows

Rules that need multi-hour history should call `_kit_lookback_stats(table)` (see [Lookback window helper]({{ “/rule-cookbook/lookback-window/” | relative_url }})). Pass `hours=1`, `3`, `6`, `12`, or `24` to validate timestamp span in the console.

Rolling-window rules (flatline, spread) use `WINDOW_SAMPLES` (~12 samples ≈ 1 h at 5 min poll) — see Windowing & debugging({{ “/rule-cookbook/windowing-debugging/” | relative_url }}).

58.6 Algorithms (supervisory — coming soon)

GL36 trim & respond and plant reset sequences will live on the **Algorithms({{ "/operator-bridge/algorithms/" | relative_url }})** tab with the same zip kit shape but `apply_algorithm_arrow` outputs. Reference: README_TRIM_RESPOND.

58.7 Related

- Model workflow({{ "/operator-bridge/model-workflow/" | relative_url }})
- [Arrow recipes]({{ "/rule-cookbook/arrow-recipes/" | relative_url }})
- GL36 algorithm stubs({{ "/rule-cookbook/gl36-algorithm-stubs/" | relative_url }}) (doc-only)

59 JSON API driver

60 JSON API driver

Poll **HTTP or HTTPS** REST endpoints on the OT LAN (or public APIs such as OpenWeatherMap) and store extracted JSON values in the feather historian (`source=json_api`).

Use this driver for gateways, micro-PLCs, IoT hubs, vendor APIs, and **web weather** when you want to cross-check shaky local outdoor-air sensors against a reference.

60.1 Supported features

Feature	Support
Methods	`GET`, `POST`
Transport	`http://` and `https://`
TLS certificate verify	On by default; disable for self-signed OT gateways
Authentication	None, **Bearer token**, **HTTP Basic**, or **query-string API keys via env**
Env placeholders	

Feature	Support
	`${ENV:VAR}` or `${VAR}` in URL, headers, body, bearer token
Custom headers	Optional `headers` map (merged with auth headers)
Value extraction	Dot-path into JSON body (`title`, `main.temp`, `weather.0.description`)
Multi-extract poll	One HTTP request per URL group; fan-out to multiple `json_path` rows
Polling	**1 / 5 / 15 / 30 min** or **1 hour** (shared with BACnet/Modbus)
Historian	`workspace/json_api/polls/samples.csv` → `feather_store/json_api/`
FDD merge	Scheduled FDD merges `bacnet` + `modbus` + `json_api` columns by nearest timestamp

60.2 Secrets: `json_api.env.local`

Copy `workspace/json_api.env.example` → `workspace/json_api.env.local` (gitignored). The bridge loads this file at startup and when `run_local.sh` starts the stack.

Use placeholders in commissioning URLs so API keys never land in git:

```
https://api.openweathermap.org/data/2.5/weather?q=${ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=${ENV:OPENWEATHER_UNITS}
```

Check which vars are configured: `GET /api/json-api/env/status`.

60.3 OpenWeatherMap showcase (recommended demo)

This is the flagship example of what the generic JSON API driver can do — **one URL, three historian columns**, env-based API key, browser tree like BACnet/Modbus.

60.3.1 1. Configure env

```
cp workspace/json_api.env.example workspace/json_api.env.local
# Edit: OPENWEATHER_API_KEY, OPENWEATHER_CITY, OPENWEATHER_UNITS
(imperial = °F)
```

Restart the bridge (or `./scripts/run_local.sh restart`) so env vars load.

60.3.2 2. Register from the UI

1. Sign in as **integrator** → **JSON API** tab.
2. Open the **OpenWeatherMap showcase** panel.
3. Use the **OpenWeatherMap** preset → **Register** (default poll 30 min; choose interval in the form).

Three endpoints appear under `api.openweathermap.org` in the commissioning tree:

Label	JSON path	Historian column	Units
web-oat-t	`main.temp`	`web-oat-t`	degF / degC
web-rh	`main.humidity`	`web-rh`	%
web-weather-desc	`weather.0.description`	`web-weather-desc`	text

Poll worker deduplicates: **one GET** serves all three extractions per cycle.

60.3.3 3. Headless / script

```
# Standalone fetch (like the original playground tester)
python3 scripts/openweather_json_api_example.py

# Register via REST (integrator login)
python3 scripts/openweather_json_api_example.py --register --base
http://127.0.0.1:8000
```

60.3.4 4. FDD: local OA-T vs web weather

With BACnet `oa-t` and JSON API `web-oat-t` in the merged site frame, scheduled FDD compares local outdoor air to the weather service.

Item	Detail
Rule module	`oat_vs_web_spread_1h.py`
Fault code	**BLD-B** (outdoor air sensor fault)

Item	Detail
Default threshold	`|oa-t − web-oat-t| > 8 °F` (`max_spread_f` in rule cfg)
Acme rule id	`acme-oat-vs-web-spread` (via `setup_gl36_fdd.py`)
Local OAT source	RTU `1100-unknown-2` (Outdoor Air Temperature Local) — same value as boiler/AHU networked OAT
Web column	`web-oat-t` from OpenWeather bundle

Model alias: local BACnet historian column must be `oa-t`:

```
python3 scripts/acme_patch_oat_column.py --point-id 1100-unknown-2 --host "$ACME_HOST" --token "$TOKEN"
```

Or set `external_id` / `fdd_input` to `oa-t` on the chosen `Outside_Air_Temperature_Sensor` in the model.

Units: keep `OPENWEATHER_UNITS=imperial` when BACnet OAT is °F.

Enable the rule in Rule Lab (or `setup_gl36_fdd.py` push) and run an FDD batch. Missing either column → rule returns all-false (no crash).

60.4 Commissioning (UI)

The **JSON API** tab follows Home Assistant `sensor.rest`:

1. Sign in → **RESTful sensor (JSON API)** tab.
2. Pick a **preset** (JSONPlaceholder, DummyJSON, httpbin, Open-Meteo, OpenAQ, OpenWeather) or enter a **resource URL**.
3. Add one or more **sensors** — each row is a `value_json_path` + historian **label** (multi-value from one GET, like HA `json_attributes`).
4. Click **Test resource** — probes without saving; shows extracted values and raw JSON.
5. Click **Register & poll** — writes `endpoints.csv`, enables polling at the selected interval.
6. Poll choices match BACnet/Modbus: **1 / 5 / 15 / 30 min** or **1 hour**.

60.4.1 API for agents

Endpoint	Purpose
`GET /api/json-api/presets`	Full preset catalog + poll intervals (human + AI readable)

Endpoint	Purpose
`POST /api/json-api/test`	Probe resource + multi-sensor extraction
`POST /api/json-api/register-bundle`	Register one resource with many sensors
`POST /api/json-api/presets/{id}/register`	One-click preset registration

60.4.2 Bench preset

Use **JSONPlaceholder — todo title** → **Test resource** to verify outbound HTTPS without field hardware.

60.5 Authentication examples

60.5.1 Query-string API key via env (OpenWeather)

```
        {
          "url": "https://api.openweathermap.org/data/2.5/weather?q=${ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=${ENV:OPENWEATHER_UNITS}",
          "method": "GET",
          "json_path": "main.temp",
          "label": "web-oat-t",
          "auth_type": "none"
        }
```

60.5.2 Bearer token (API key)

```
        {
          "url": "https://192.168.10.50/api/v1/status",
          "method": "GET",
          "json_path": "sensors.zone_temp",
          "label": "zone-temp",
          "auth_type": "bearer",
          "bearer_token": "${ENV:GATEWAY_API_KEY}",
          "verify_tls": false
        }
```

60.5.3 HTTP Basic

```
        {
          "url": "http://192.168.10.50/data.json",
          "method": "GET",
          "json_path": "value",
          "label": "sensor-value",
```

```
  "auth_type": "basic",
  "basic_user": "operator",
  "basic_password": "changeme"
}
```

60.5.4 POST with JSON body

```
{
  "url": "https://gateway.local/api/read",
  "method": "POST",
  "json_path": "result.pv",
  "label": "pv",
  "body": { "point": "AHU-1_SAT", "command": "read" },
  "auth_type": "bearer",
  "bearer_token": "…"
}
```

Credentials in `endpoints.csv` are redacted in API responses (`***`). Prefer `${ENV:…}` for production keys.

60.6 REST API

Method	Path	Role	Description
GET	`/api/json-api/env/status`	operator+	Which env vars are configured (no secret values)
POST	`/api/json-api/presets/openweather`	integrator+	Register 3-point OpenWeather bundle + optional poll
GET	`/api/json-api/driver/tree`	operator+	Commissioning tree grouped by host
GET	`/api/json-api/poll/status`	operator+	Poll worker status
POST	`/api/json-api/poll/once`	integrator+	Run one poll cycle
POST	`/api/json-api/request`	operator+	One-shot HTTP request (no store)
POST	`/api/json-api/read_and_store`	integrator+	Request + CSV + feather ingest
POST	`/api/json-api/refresh`	operator+	Re-fetch one endpoint
PATCH	`/api/json-api/endpoint/poll`	integrator+	Enable/disable poll interval
DELETE		integrator+	Remove endpoint

Method	Path	Role	Description
	`/api/json-api/endpoint/{point_id}`		

Full route list: API routes({{ “/appendix/bridge_api/” | relative_url }}).

60.7 Typical OT URLs

Device type	Example URL	Notes
Web weather	`https://api.openweathermap.org/data/2.5/weather?...`	Env-based `appid`; 3 historian columns
Gateway status	`http://192.168.1.50/api/status`	Often plain HTTP on LAN
HTTPS BMS API	`https://10.0.0.12/rest/points/zone1`	May need `verify_tls: false`
Local mock	`https://jsonplaceholder.typicode.com/todos/1`	Bench / CI only

60.8 Limitations

- Response body must be **JSON** (not XML or plain text).
- Numeric paths (e.g. `main.temp`) are stored as strings in poll CSV; FDD rules cast with `pc.cast(..., "float64")`.
- Outbound requests originate from the **bridge** container/host — ensure routing and firewall rules allow the edge box to reach the target (OT subnet or public internet for weather).

60.9 Disable polling (save API quota)

Stop scheduled calls without deleting endpoints: JSON API tree → select OpenWeather points → **Stop polling**, or integrator API:

```
curl -X PATCH http://127.0.0.1:8765/api/json-api/endpoint/poll \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"point_id":"ja-api-openweathermap-org-get-web-oat-t","enabled":false,"poll_interval_s":0}'
```

API keys live in `json_api.env.local` (gitignored) — never committed; audit logs do not record expanded URLs with `appid=`. See Logging and audit({{ “/ops/logging/” | relative_url }}).

60.10 Smoke test

```
python3 scripts/smoke_multi_driver_stack.py
```

Validates JSON API ingest alongside BACnet and Modbus, feather isolation, BRICK scope, and FDD batch.

61 Algorithms

62 Algorithms (coming soon)

The **Algorithms** dashboard tab (`/algorithms`) is a placeholder for **supervisory Python sequences** — the mirror image of Rule Lab FDD:

	FDD (Rule Lab)	**Algorithms (planned)**
Purpose	Detect faults	Compute setpoints, request levels, plant enables
Entrypoint	`apply_faults_arrow(table, cfg, context)`	`apply_algorithm_arrow(table, cfg, context)` (planned)
Output	Boolean fault mask	Numeric outputs / structured dict
Historian	Same `feather_store` PyArrow tables	Same

Until the tab ships, draft patterns live in this doc and in GL36 algorithm stubs({{ “/rule-cookbook/gl36-algorithm-stubs/” | relative_url }}).

62.1 GL36 Trim & Respond reference

Open-FDD algorithms will follow the same ASHRAE Guideline 36 trim & respond methodology already implemented for Niagara in:

README_TRIM_RESPOND.md

That guide covers:

- VAV zone request generators (cooling + static pressure)

- AHU duct static pressure reset
- AHU supply air temperature reset
- Central plant chiller / boiler enable and HWST / CHWST trim & respond

Porting those blocks to PyArrow is in progress; the dashboard tab shows **COMING SOON** until upload, batch, and BACnet write paths are wired.

62.2 Planned workflow (same kit shape as Rule Lab)

1. Download **algorithm kit** zip from the Algorithms tab (future)
2. Edit **constants** at the top of `algorithm.py` (no `config.json`)
3. Run `python run_test.py` locally against `sample.feather`
4. Upload `algorithm.py` when satisfied

Kit files will match the Rule Lab zip layout (see [Rule Lab — dev kit zip]({{ “/operator-bridge/rule-lab/” | relative_url }}#dev-kit-zip-download)).

62.3 Lookback windows

Rules and algorithms that need multi-hour history should use the shared **lookback stats helper** documented in [Lookback window helper]({{ “/rule-cookbook/lookback-window/” | relative_url }}). Pass `hours=1`, `3`, `6`, `12`, or `24` to print row count, timestamp start/stop, and span for console validation.

Scheduled batch defaults: **1 h** lookback on the FDD loop; **Update all records** in Rule Lab uses **24 h** with 6 h chunks.

62.4 AHU supervisory checks (doc-only stubs)

These are **not** uploaded rules yet — copy-paste references for future algorithm or FDD work.

62.4.1 Duct static pressure too high

Flags sustained high duct static (possible stuck setpoint, blocked filter, or trim/respond not trimming).

```
"""AHU duct static high — doc stub (Arrow constants)."""

import pyarrow.compute as pc
```

```
        VALUE_COLUMN = "duct-static"
        HIGH_INWC = 1.20
        LOOKBACK_HOURS = 3
        MIN_SAMPLES_ABOVE = 6


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            high = pc.greater(vals, HIGH_INWC)
            # Require several consecutive highs (~30 min at 5 min poll)
            from open_fdd.arrow_runtime.windows import
arrow_consecutive_true

            streak = arrow_consecutive_true(high, MIN_SAMPLES_ABOVE)
            return streak
```

62.4.2 Supply air temperature too cold (cooling coil over-delivering)

```
        """AHU SAT too cold vs setpoint — doc stub."""

        import pyarrow.compute as pc

        SAT_COLUMN = "sa-t"
        SP_COLUMN = "sa-t-sp"
        COLD_DELTA_F = 5.0
        LOOKBACK_HOURS = 1


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")
```

```
        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            sat = pc.cast(table[SAT_COLUMN], "float64")
            sp = pc.cast(table[SP_COLUMN], "float64")
            return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

62.5 Plant supervisory checks (doc-only)

See GL36 algorithm stubs({{ “/rule-cookbook/gl36-algorithm-stubs/” | relative_url }}) for chiller plant enable, HWST trim & respond, and CHW DP/CHWST reset patterns aligned with the Niagara README.

62.6 Related

- Rule Lab({{ “/operator-bridge/rule-lab/” | relative_url }}) — live FDD authoring
- Model workflow({{ “/operator-bridge/model-workflow/” | relative_url }}) — point bindings
- Windowing & debugging({{ “/rule-cookbook/windowing-debugging/” | relative_url }}) — rolling windows

63 Model workflow

64 Model workflow

64.1 Brick model

- **Equipment tree** — SPARQL over synced TTL (`workspace/data/data_model.ttl`)
- **Point registry** — BACnet inventory → `series_id` for historian bindings
- **Health score** — orphan points, missing `fdd_input`, stale telemetry

64.2 Typical integrator flow

1. Import or sync BACnet point registry.

2. Map equipment classes (AHU, VAV, …) in the model UI — set `brick_type` on points and `feeds` on equipment (AHU→VAV).
3. Tag points with `fdd_input` roles (zone temp, SAT, damper cmd, …).
4. Bind Rule Lab rules to points, equipment, or BRICK classes (`bindings` in `rules_store.json`).
5. Run `POST /api/model/sync-ttl` after bulk imports.
6. Validate on **Trend plot** with FDD overlays (`?fdd=1`) — faults render on a right-hand 0/1 axis per device scope.

AI Agent and commissioning JSON import can bulk-apply steps 2–4; integrator should review model health before production FDD. See FDD and assignments({{ “/operator-bridge/fdd-assignments/” | relative_url }}) and Ollama and analytics({{ “/operator-bridge/ollama-analytics/” | relative_url }}).

Export: `GET /api/model/export` (TTL/JSON). Commissioning bundle: `GET /api/model/commissioning-export` — includes per-point `fdd_rules_linked` (Rule Lab names) plus `fdd_rules[]` catalog. Use **Import / export** tab preview table or **Point → FDD rule pins** before editing raw JSON. API detail in Appendix({{ “/appendix/bridge_api/” | relative_url }}).

65 Driver framework

66 Driver framework

Open-FDD edge commissioning uses a **shared driver pattern** for OT data sources. Each driver polls field devices on a schedule, stores long-format samples in workspace CSVs, and ingests into the **feather historian** under a distinct `source` tag for FDD and plots.

66.1 Drivers

Driver	Protocol	Feather `source`	Commissioning tab
BACnet({{ “/bacnet/”	relative_url }})	MS/TP / IP (:47808)	`bacnet`
Modbus	Modbus TCP	`modbus`	Modbus
[JSON API]({{ “/drivers/json-api/”	relative_url }})	HTTP/HTTPS REST (+ [OpenWeather showcase]	relative_url }} #openweathermap-showcase-

Driver	Protocol	Feather `source`	Commissioning tab
		({{ “/drivers/	recommended-
		json-api/”	demo))

All three drivers share the same operator workflow:

1. **Discover or configure** — add devices/endpoints to the driver registry.
2. **Request once** — on-demand read to verify connectivity and JSON path / register map.
3. **Request & store** — append to poll CSV and write a feather shard.
4. **Enable polling** — 1 / 5 / 10 / 15 minute intervals via tree right-click or bulk toolbar.
5. **Poll all now** — force one worker cycle without waiting for the interval.

66.2 Data layout

```
workspace/
  bacnet/commissioning/   points.csv, discovered inventory
  bacnet/polls/           samples.csv
  modbus/commissioning/   registers.csv
  modbus/polls/           samples.csv
  json_api/commissioning/ endpoints.csv
  json_api/polls/         samples.csv
  data/feather_store/
    bacnet/{site_id}/     shard-*.feather
    modbus/{site_id}/
    json_api/{site_id}/
```

66.3 Historian and FDD

- **Plot tab** — numeric columns from `bacnet` and `modbus` sources (`GET /api/timeseries/plot?source=…`).
- **String JSON fields** — stored in feather under `json_api`; use the JSON API tree or feather directly (plot is numeric-only).
- **FDD batch** — merges `bacnet` + `modbus` + `json_api` historian columns by nearest timestamp; rules use BRICK-bound `external_id` names (e.g. compare BACnet `oa-t` vs JSON API `web-oat-t` — see [JSON API]({{ “/drivers/json-api/” | relative_url }})).
- **BRICK model** — bind points in **Model & assignments** commissioning JSON; sync TTL for SPARQL scope.

66.4 Rule Lab (Arrow upload/download)

1. **Download kit** — zip with `rule.py` (constants at top), `data.py`, `sample.feather` (~3h), `run_test.py`, `requirements.txt`.
2. Local test: `pip install -r requirements.txt` then edit constants in `rule.py` and `python run_test.py`.
3. **Upload rule.py** — Arrow-only; must define `apply_faults_arrow(table, cfg, context=None)`.
4. Browser shows **read-only** source; use Quick test + Update all records.

API: `GET /api/rules/export-kit`, `POST /api/rules/upload`.

66.5 Smoke validation

From repo root on a running stack (`http://127.0.0.1:8765`):

```
python3 scripts/smoke_multi_driver_stack.py
python3 scripts/validate_modbus_temp_e2e.py --no-server
```

66.6 Related docs

- Data flow({{ “/architecture/data-flow/” | relative_url }}) — poll → CSV → feather → FDD
- API routes({{ “/appendix/bridge_api/” | relative_url }}) — REST reference
- [BACnet polling]({{ “/bacnet/polling/” | relative_url }}) — commission container loop

67 Operator Bridge

68 Operator Bridge

The Operator Bridge is the FastAPI service and React dashboard operators use daily.

Page	Topic
Dashboard overview({{ “/operator-bridge/dashboard/”	relative_url }})
Auth and roles({{ “/operator-bridge/auth-roles/”	relative_url }})
	relative_url }})

Page	Topic
Rule Lab({{ “/operator-bridge/rule-lab/”	
Algorithms({{ “/operator-bridge/algorithms/”	relative_url }})
Model workflow({{ “/operator-bridge/model-workflow/”	relative_url }})
FDD and assignments({{ “/operator-bridge/fdd-assignments/”	relative_url }})
Ollama and analytics({{ “/operator-bridge/ollama-analytics/”	relative_url }})

REST details: [Appendix — API routes]({{ “/appendix/bridge_api/” | relative_url }}). Env files: Workspace env files({{ “/appendix/workspace-env-files/” | relative_url }}).

69 Ollama and analytics

70 Ollama and analytics

70.1 What runs without login

The **Building status** home page (`/`) and `GET /openfdd-agent/building-insight` are **public** (same policy as fault status). Anyone on the LAN can read the briefing sentence and underlying numeric levers; only the **AI Agent** chat and stack revision endpoints require sign-in.

70.2 Ollama setup

1. Copy `workspace/ollama.env.example` → `workspace/ollama.env.local`.
2. Pick a model for available RAM (comments in the example file):

RAM tier	Suggested model
4 GB Pi	`qwen3:0.6b`
8 GB	`qwen3:1.7b`

RAM tier	Suggested model
16 GB	`qwen3:4b`
32 GB	`qwen3:8b`
64 GB	`qwen3:14b`

1. Set `OFDD_OLLAMA_BASE_URL` (default `http://127.0.0.1:11434`).
2. Optional: `OFDD_OLLAMA_GPU_MODE` (`cpu` / `auto` / `gpu`), `OFDD_OLLAMA_TIMEOUT_S` for slow CPUs.
3. Bootstrap helper: `./scripts/bootstrap_ollama.sh`.

Bridge reads env via `run_local.sh` / compose. Model resolution: `workspace/api/openfdd_bridge/ollama_profiles.py`.

70.3 Automatic analytics (deterministic, always on)

These pandas calculations run on a refresh interval and feed the home dashboard **whether or not Ollama is up**:

Lever	Module	Refresh env	What it compu
Zone temperature snapshot	`zone_temp_analytics.py`	`OFDD_ZONE_TEMP_INTERVAL_S` (default 3600s)	Site-wide and p AHU zone temp averages, sprea out-of-band cou using BRICK `fe` (AHU→VAV) wh modeled; falls b to all `Zone_Air_Tempera` points
Device poll health	`device_poll_health.py`	stack health cycle	BACnet device l seen / stale poll detection
Operational brief	`building_insight.py` → `get_operational_brief`	same as insight	Compact numer status for integrators
Root-cause hints	`root_cause_hints.py`	on insight refresh	Multi-zone fault chains using `brick:feeds` fron

Shared lookback window: `OFDD_ANALYTICS_LOOKBACK_DAYS` (default 14) in `operational_analytics.py`.

API routes (auth optional unless noted):

- `GET /openfdd-agent/building-insight` — sentence + zone snapshot + fault sentences

- `GET /openfdd-agent/zone-temp-analytics`
- `GET /openfdd-agent/device-poll-health`
- `GET /openfdd-agent/ollama/health` — integrator diagnostics

70.4 When Ollama is used

`building_insight.get_building_insight()`:

1. Refreshes zone + device snapshots (pandas).
2. Calls `ollama_client.health()`.
3. If Ollama is reachable and `should_use_ollama_for_insight()` passes, sends a **compact context** (status, zone levers, feeds chains) to the configured model for a one-sentence briefing (`source: "ollama"`).
4. On failure, uses `_fallback_sentence()` (`source: "deterministic"`).

Context assembly: `building_insight._compact_context()`. Prompt template references BRICK feeds and `root_cause_hints`.

70.5 AI Agent (authenticated)

`/agent` page uses `POST /openfdd-agent/chat` with tool access (`agent_tools.py`): BRICK graph, scope bundles, rule binding patches, model health, etc. This is separate from the public home briefing.

70.6 Code map

```
workspace/api/openfdd_bridge/
  building_insight.py      # cached home briefing + Ollama
sentence
  zone_temp_analytics.py   # zone temp levers (BRICK feeds)
  device_poll_health.py    # BACnet poll staleness
  operational_analytics.py # lookback window + methodology dict
  root_cause_hints.py      # multi-fault BRICK chain hints
  ollama_client.py         # HTTP client to Ollama
  ollama_profiles.py       # RAM tiers / model names
```

Dashboard consumers: `BuildingStatusPage.tsx`, `StackStatusStrip.tsx` (WebSocket `/ws/dashboard`).

71 FDD and assignments

72 FDD and assignments

72.1 What the product actually does

Open-FDD separates **data modeling** (BRICK equipment, points, `feeds` relationships) from **rule assignment** (which FDD rules apply to which points, equipment, or BRICK classes). Both are editable in **Model & assignments** and via commissioning JSON import.

72.1.1 BRICK modeling (integrator + AI-assisted)

Concern	Where it lives	Notes
Equipment tree	`ModelService` + TTL sync	AHU, VAV, sensors; `equipment_type` and `brick_type` on points
`brick:feeds`	`model_feeds.py`, `ttl_service.py`	AHU→VAV→zone relationships; used by zone analytics and root-cause hints
Point registry	BACnet inventory → model points	`fdd_input` roles (`oa-t`, `stat_zn-t`, …), `series_id` / historian column
Commissioning import	`POST /api/model/commissioning-import`	Bulk JSON: sites, equipment, points, optional `fdd_rules`

AI Agent (`agent_tools.py`) can read `slim_brick_graph`, `scope_bundle`, patch rule bindings, and upsert points — but it does **not** auto-generate a full site model without operator review. Expected workflow:

1. BACnet discover/poll populates point registry.
2. Integrator or agent tags `brick_type`, `fdd_input`, and `feeds` on equipment.
3. `POST /api/model/sync-ttl` writes `workspace/data/data_model.ttl`.

4. Model health (`model_health.py`) flags orphan points and missing `fdd_input`.

72.1.2 Rule assignment (three binding kinds)

Saved rules in `rules_store.json` carry `bindings`:

```
{
  "point_ids": ["5007-analog-input-1173"],
  "equipment_ids": ["bench-1"],
  "brick_types": ["Outside_Air_Temperature_Sensor"]
}
```

Binding kind	Effect
`point_ids`	Rule runs for that historian point / series
`equipment_ids`	Rule runs for all points on equipment (mass bind)
`brick_types`	Rule runs for any point with that BRICK class on site

Assign via:

- **Model & assignments** UI — equipment/point chips, rule pin menu on trend plot
- **Rule Lab** — bindings panel
- **Commissioning import** — `fdd_rules[].bindings` and per-point `fdd_rule_ids`
- **AI Agent** — `patch_rule_binding` tool

Runtime evaluation: `fdd_runner.py` (batch FDD) and `plot_readings.evaluate_fault_plots()` (trend overlays). Plot scope filters rules to those bound to **selected telemetry** (point, equipment, or BRICK class); unbound rules still evaluate site-wide.

Brick bindings (3.0.20+): when a rule binds `brick_types` (or `equipment_ids` / `point_ids`), the batch runner resolves **every** matching historian column via `historian_columns_for_rule`, runs the Arrow rule per column with `value_column` set, and OR-combines fault masks. One rule config therefore applies consistently to all zone temps, discharge temps, etc.

72.1.3 CLASS vs device

- **BRICK class** (`brick_type` on points, or `brick_types` on rule bindings) — template-style assignment ("all `Zone_Air_Temperature_Sensor`").
- **Device / equipment** (`equipment_id`, BACnet device instance) — instance assignment ("this VAV-12").
- **Point** (`point_ids`) — single-sensor assignment.

`applies_to` on legacy rules is deprecated; use `bindings` in 3.0+.

72.2 Commissioning JSON shape

Export: `GET /api/model/commissioning-export`. Import accepts `import_ready_json` wrappers (see `commissioningImport.ts`).

72.2.1 Rule names vs ids (Rule Lab alignment)

Rule Lab shows **human names** (`Bench OA-T flatline 1h`). Saved rules use stable **ids** (`bench-oa-t-flatline-1h` from setup scripts, or a UUID for uploaded custom rules). Commissioning export includes both so you never have to cross-reference sections by hand:

Field	Direction	Purpose
`points[].fdd_rule_ids`	import + export	Assignment surface — array of rule ids
`points[].fdd_rules_linked`	**export only**	`[{id, name}]` — Rule Lab names beside each point
`fdd_rules[].id`	import + export	Stable rule key (matches Rule Lab / `rules_store.json`)
`fdd_rules[].name`	import + export	Same label as Rule Lab dropdown
`fdd_rules[].source_file`	export	Basename of deployed `.py` when present (e.g. `bench_oa-t_flatline_1h.py`)

On import, `fdd_rules_linked` is stripped (like `fdd_rule_ids` — not stored in `model.json`). Use the **Point → FDD rule pins** table on Model & assignments to read names without parsing JSON.

Minimum fields integrators/AI should produce:

- `sites`, `equipment[]` with `id`, `equipment_type`, optional `feeds[]`
- `points[]` with `site_id`, `equipment_id`, `brick_type`, `fdd_input`, BACnet IDs, optional `fdd_rule_ids`
- `fdd_rules[]` with `id`, `name`, `bindings`, `enabled` (optional `source_file` on export)

After import, verify in **Trend plot** with `?fdd=1` — fault lanes appear on the right-hand 0/1 axis for bound rules.

72.3 Related

- Model workflow({{ “/operator-bridge/model-workflow/” | relative_url }})
- Rule Lab({{ “/operator-bridge/rule-lab/” | relative_url }})
- [Trend plot / fault overlays]({{ “/operator-bridge/dashboard/” | relative_url }}#trend-plot)

73 Network setup

74 Network setup

74.1 Bind address

Commission and poll containers need the **OT interface** IP and UDP **47808**:

```
# workspace/bacnet/commissioning/commission.env
BACNET_BIND=192.168.1.50/24:47808
BACNET_INSTANCE=599999
```

- Use the NIC that can reach controllers (not only the management VLAN unless routed).
- Poll driver uses `network_mode: host` on Linux edges.

74.2 Discovery range

```
DISCOVER_LOW=1
DISCOVER_HIGH=4194302
DISCOVER_TIMEOUT=30
```

Narrow the range in large sites to avoid broadcast storms.

74.3 Verify

From commission container logs or API: Who-Is → I-Am responses. If zero devices, check firewall, bind IP, and VLAN routing before tuning rules.

75 Discover and read

76 Discover and read

The **commission** container exposes HTTP APIs for discovery and read-property workflows used by the dashboard BACnet tools.

Capability	Notes
Who-Is / I-Am	Device discovery on OT subnet
Read property	Present value, object name, units
RPM	Bulk reads for inventory export
Write property	**Gated** — see Write safety({{ “/bacnet/write-safety/”

Exported inventory feeds `points_discovered.csv` and the Brick point registry.

76.1 Operator workflow

1. Confirm network setup({{ “/bacnet/network-setup/” | relative_url }}).
2. Run discover from dashboard or API.
3. Review discovered objects; enable points for poll profiles.
4. Sync model bindings for FDD inputs.

77 Polling

78 Polling

BACnet field reads run in the `openfdd-commission` container. The Operator Bridge does not bind UDP 47808.

> **Retired:** `openfdd-bacnet-poll` image and `openfdd-bacnet-poll` systemd unit. Poll loop lives in **commission** only — do not run both.

BACnet polling has two stages:

1. **Field reads** — commission poll loop → `workspace/bacnet/polls/samples.csv`
2. **Historian ingest** — bridge background worker → `workspace/data/feather_store/`

78.1 Commission poll loop (default)

The **commission** container runs the poll loop at startup. It reads enabled rows from `points.csv` and appends BACnet RPM results to `samples.csv`.

Setting	Location
BACnet bind	`workspace/bacnet/commissioning/commission.env`
Enabled points	`workspace/bacnet/commissioning/points.csv`
Poll output	`workspace/bacnet/polls/samples.csv`
Historian	`workspace/data/feather_store/`

Check activity:

```
tail -1 workspace/bacnet/polls/samples.csv
docker compose logs --since 5m commission | grep -i poll
```

78.2 Bridge ingest worker

Inside **bridge**, a background thread watches `samples.csv` and ingests new rows into feather. Disable only for debugging: `OFDD_DISABLE_POLL_WORKER=1`.

78.3 Health

Dashboard stack health (`GET /health/stack`) reports `bacnet_poll` status from the commission agent — not a separate container.

Stale points trigger data-quality fault patterns — see [Sensor quality faults]({{ “/fault-codes/sensor-quality/” | relative_url }}).

78.4 Poll strategy (edge)

Open-FDD intentionally keeps BACnet field traffic **minimal**:

- **One enabled row per object** in `points.csv` — poll only what the model/FDD needs.
- **Uniform cycle** — every enabled point is RPM-polled each cycle; the loop sleeps `max(15, min(poll_interval_s))` (typically **60 s** on Acme).
- Per-point `poll_interval_s` in the UI is the **commissioned target**; throughput analytics compare configured vs observed samples.
- **Unit conversion** (e.g. Trane °C → °F) uses `device_poll_profiles.csv` at ingest — not a poll-rate file.

Acme ships **FDD-minimal polling** (~76 points on a typical GL36 lab when 8 rules are enabled), not the full discovery tree (~340+ GL36 objects or 566 discovered objects). Disable unneeded objects in BACnet commissioning rather than raising global poll rates.

> **AI agents / MCP:** Never enable BACnet polling for every discovered object. Poll **only** historian columns bound to **enabled** FDD rules (`rules_store.json` brick_types, point_ids, and rule config column hints). Use `bacnet_toolshed.fdd_minimal_poll` or `acme_commission_fdd_minimal.sh` — not bulk `--all` enable scripts.

78.5 Async driver and single-stack I/O

Layer	Behavior
Dedicated `bacnet-io` thread	One asyncio loop + serial lock for all field I/O
RPM batching	Present-value reads chunked (25 objects) to cut APDUs
Poll loop	Sequential per-device RPM when `interruptible=True` so operator UI can preempt

Layer	Behavior
Operator UI	**INTERACTIVE** priority — cancels in-flight background work
Override scans	**BACKGROUND**, 15 min timeout — see Operator override scans({{ “/bacnet/override-scans/”
Bridge ingest	Async notify after each poll cycle → feather historian

Do **not** run a second BACnet bind on 47808 (no parallel `openfdd-bacnet-poll` container). See Containers({{ “/architecture/containers/” | relative_url }}).

78.6 Operator override scans

Hourly P8 supervisory scans rotate one device at a time. Documented in Operator override scans({{ “/bacnet/override-scans/” | relative_url }}).

79 Operator override scans

80 Operator override scans (P8)

The commission agent rotates **one BACnet device per hour** (default) through a supervisory priority-array scan. Results are persisted for operator review and portfolio rollup — not for automatic writes.

80.1 Schedule

Setting	Default	Meaning
`OFDD_OVERRIDE_SCAN_INTERVAL_S`	`3600`	Seconds between single-device scans
`OFDD_OPERATOR_OVERRIDE_PRIORITY`	`8`	BACnet priority counted as

Setting	Default	Meaning
		“operator override”
Device list	`points_discovered.csv`	Unique `device_instance` rows
Rotation	`registry.json` cursor	Full plant rotation ≈ `device_count × interval`

Example: **33 devices × 1 h ≈ 33 h** for a full pass. Dashboard shows **next device** and **last scan** time.

80.2 What each scan does

1. `point_discovery` on the target device (commandable objects).
2. `read_property` on each object’s **priority-array** (not RPM — RPM loses priority slots).
3. Merge into `workspace/bacnet/overrides/registry.json` and append `overrides_export.csv`.
4. Advance cursor to the next device.

Fault code **operator_override** (P8) surfaces active overrides on the BACnet tree and building status.

80.3 BACnet I/O priority

Override scans run at **BACKGROUND** priority on the single shared BACnet stack (UDP 47808). Operator reads/writes use **INTERACTIVE** and preempt background work. Scheduled RPM polling also uses BACKGROUND and shares the same asyncio lock — polls and override scans take turns; neither starves the other indefinitely.

Override scans allow up to **15 minutes** per device (`_run_bacnet_override_scan` timeout). Bridge `POST /api/bacnet/overrides/scan-once` proxies with a matching 900 s timeout.

80.4 Manual trigger

```
curl -X POST -H "Authorization: Bearer $TOKEN" \
  http://EDGE:8765/api/bacnet/overrides/scan-once
```

Dashboard: BACnet page → **Scan next device now**.

80.5 Health checks

```
curl -H "Authorization: Bearer $TOKEN" http://EDGE:8765/api/bacnet/overrides/status
```

Expect `device_count` > 0, `last_scan_at` within the last few hours on a healthy edge, and `export_row_count` growing over time.

80.6 Data paths

Path	Content
`workspace/bacnet/overrides/registry.json`	Per-device last scan + override points
`workspace/bacnet/overrides/overrides_export.csv`	Flat export for spreadsheets

Preserved across Docker upgrades when `workspace/` is on the edge volume — see [Backup & restore]({{ “/ops/backup-restore/” | relative_url }}).

81 Write safety

82 Write safety

> **BACnet writes can change live plant behavior.** Open-FDD disables writes by default. Enable only with operator sign-off, a tested allowlist, and audit logging.

82.1 Defaults

Control	Default
Supervisory writes	**Off**
Write allowlist	Empty — must enumerate device/object
Audit	Commands logged when writes enabled

82.2 Enable writes (commission role)

1. Set env flag documented in Security({{ “/security/” | relative_url }}) (`OFDD_BACNET_WRITE_ENABLED` or site equivalent).
2. Populate write allowlist CSV / config with explicit object IDs.
3. Use **commission** role JWT; every write should include reason/note where API supports it.
4. Test on bench hardware before production OT.

82.3 Operator rules

- Never write setpoints or overrides on life-safety or fire/smoke interlocks through experimental rules.
- Prefer read-only polling for FDD until rules are validated in Rule Lab.
- Revert overrides through BACnet priority relinquish where supported.

83 BACnet

84 BACnet

Open-FDD integrates with field controllers via BACnet/IP (and bench MSTP overlays in dev compose).

Page	Topic
Network setup({{ “/bacnet/network-setup/” | relative_url }})	
Discover and read({{ “/bacnet/discover-read/” | relative_url }})	
Polling({{ “/bacnet/polling/” | relative_url }})	
Operator override scans({{ “/bacnet/override-scans/” | relative_url }})	
Write safety({{ “/bacnet/write-safety/” | relative_url }})	

85 Convention

86 Fault code convention

86.1 Format (Grade-A, 3.0.18+)

Short stable code: `FAMILY-SUBSYSTEM-NNN` — e.g. `AHU-ECON-001`, `CHW-DT-001`, `DATA-STAL-001`.

Semantic ID: dotted `canonical_id` — e.g. `ahu.economizer.not_using_free_cooling`.

Legacy letter codes (`AHU-E`, `VAV-C`) remain **aliases** in `open_fdd/faults/catalog/*.yaml` until bridge sync completes.

Field	Example
`code`	`AHU-ECON-001`
`canonical_id`	`ahu.economizer.not_using_free_cooling`
`family`	`AHU`, `VAV`, `CHW`, `DATA`, `CRS`, …
`rule_doc_path`	Link to rule cookbook recipe

Full schema: `open_fdd/faults/schema.py`. Catalog loader: `open_fdd/faults/catalog.py`.

86.2 Categories (Grade-A)

Category	Meaning
`energy`	Waste, reset stuck, plant inefficiency
`comfort`	Setpoint not met, poor zone performance
`reliability`	Short cycling, staging, equipment stress
`maintenance`	Leaking valves, filters, fouling proxies
`indoor_air_quality`	Ventilation shortfall, humidity
`healthcare_risk`	Pressure, isolation, critical-space excursions
`data_quality`	Stale, flatline, OOB, missing historian
`controls_integrity`	Overrides, chatter, schedule violations

86.3 Severity

Level	Operator response
`info`	Log; trend for maintenance window
`low`	Review in routine rounds
`medium`	Investigate within days
`high`	Prompt attention; comfort/energy impact likely
`critical`	Immediate operational risk (especially `healthcare_risk`)

86.4 Linking rules

In Rule Lab, set `fault_code` on the rule metadata. Batch FDD aggregates hits into `GET /api/faults/status` by family.

86.5 Cookbook mapping

Each catalog entry lists `cookbook_patterns` (e.g. `flatline_1h`, `mixing_envelope`, `rate_of_change`) — see Expression cookbook (Arrow-native)({{ “/rule-cookbook/expression-cookbook/” | relative_url }}).

Legacy pandas `type: expression` YAML rules are **not** used on the edge in 3.x; translate to `rules_py` Arrow modules and bind **`fault_code`** in Rule Lab.

87 Live site update

88 Live site update

Upgrade GHCR images on a **live edge host** that already has `~/open-fdd/` — no `git pull` on the host.

> **IT operators:** prefer [Quick Start — Updating the stack]({{ “/quick-start/updating/” | relative_url }}) and `openfdd_site_backup.sh` / `openfdd_site_update.sh` on the edge host. This page adds control-machine UI deploy and Ansible paths.

88.1 Layout on the edge host

```
~/open-fdd/
  docker-compose.yml
  workspace/          # site state — backup first
```

`workspace/` holds BACnet commissioning, poll CSV, feather historian, BRICK model, FDD rules, and `auth.env.local`. Image upgrades must **preserve** it.

See Backup and restore({{ “/ops/backup-restore/” | relative_url }}) before any upgrade.

88.2 Concepts

Term	Meaning
Container	Running instance (`bridge`, `commission`, `mcp-rag`) — recreated on upgrade
Image	GHCR package (tag `latest` or dated tag)
`workspace/`	Open-FDD site state on disk — **do not delete**

88.3 1. Verify new images on GHCR

Three images must exist for every tag:

```
        export NEW_TAG="${NEW_TAG:-latest}"

        docker manifest inspect ghcr.io/bbartling/openfdd-bridge:$
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-commission:$
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-mcp-rag:$
{NEW_TAG} >/dev/null &&
        echo "All three images exist for ${NEW_TAG}"
```

88.4 2. Update dashboard UI (when release changed React)

> **Image pull alone does not update the browser UI.** The Operator Bridge serves `workspace/api/static/app/` from the bind mount **before** image-baked assets.

From control machine (full repo):

```
cd /path/to/open-fdd
./scripts/build_operator_dashboard.sh prod
cd infra/ansible
RUN_POST_CHECK=0 ./deploy.sh ui --limit <inventory_host>
```

Confirm hash on edge:

```
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.js' | head -1
```

One command (UI + images + insurance check):

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
```

88.5 3. Pull and recreate (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

Manual equivalent:

```
cd ~/open-fdd
docker compose pull
docker compose up -d --force-recreate
docker compose ps
curl -sf http://127.0.0.1:8765/health
```

88.6 4. Ansible image-only (control machine)

```
OPENFDD_IMAGE_TAG=latest RUN_POST_CHECK=1 ./scripts/
upgrade_edge_ghcr.sh --limit <inventory_host>
```

88.7 5. Validate

→ Deployment validation({{ “/ops/deployment-validation/” | relative_url }})

88.8 Rollback

Restore `docker-compose.yml` from backup, pull previous tag, `docker compose up -d --force-recreate`. `workspace/` is unchanged. Details: Backup and restore({{ “/ops/backup-restore/” | relative_url }}).

88.9 Related

- Updating the stack({{ “/quick-start/updating/” | relative_url }}) — edge-host helper scripts
- Containers({{ “/architecture/containers/” | relative_url }}) — three-image architecture

89 Central collection

90 OpenFDD RCx Central — collection

OpenFDD Edge sites emit **interval summaries** (not raw historian by default). **OpenFDD RCx Central** ingests rollups over Tailscale/VPN, stores them locally, and feeds the Dash overview and validation jobs.

RCx Central is read-only toward Edge/BACnet. It runs on an analyst workstation or Docker Desktop — not on the OT edge by default.

90.1 Interval summary schema

Defined in `portfolio/store/interval_schema.py` as `FaultIntervalSummary`:

- Identity: `site_id`, `building_id`, `equipment_id`, `equipment_family`
- Window: `timestamp_start`, `timestamp_end`
- Fault: `fault_code`, `canonical_id`, `severity`, `confidence`
- Activity: `active_minutes`, `percent_active`, `occurrence_count`
- Evidence: `evidence_json` (summarized, not full trend)
- Tuning: `tuning_version`, `tuning_params_hash`
- Operator: `operator_status` (`new` | `acknowledged` | `resolved` | `false_positive` | …)

90.2 Edge export formats

- JSONL (append-only events)
- Parquet / Feather (batch intervals)

90.3 Collect today (rollup API)

```
source infra/ansible/secrets/acme.env.local
python3 scripts/portfolio_collect.py
```

90.4 Tuning proposals

Human-reviewable proposals use `TuningProposal` in the same module. The AI agent **never** auto-writes BACnet points on edge — use `GET /api/building-agent/tuning-brief` and `POST /api/building-agent/apply-tuning` with `apply: false` first. Maintainer workflow: `skills/openfdd-edge-deploy-tune/SKILL.md`.

91 AHU faults

92 AHU faults

Air handling unit codes (sample from catalog).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
AHU-A	Supply fan performance degradation	warning	Spread/ runtime vs baseline	Belt slip, dirty filters, VFD limit	Trend speed vs airflow; filter dP
AHU-B	Simultaneous heating and cooling	critical	Heat + cool outputs active together	Sequencing bug, leaking valve	Trend coil commands; valve feedback
AHU-C	SAT sensor fault	warning	Flatline / OOB SAT	Failed sensor, bad calibration	SAT trend; compare to coil discharge
AHU-D	MAT inconsistent with OAT/ RAT	warning	MAT outside OAT–RAT envelope	MAT/OAT/ RAT sensor errors	Verify damper positions; cross-check OAT
AHU-E	Economizer not economizing	warning	Free cool available, OA minimum	Stuck damper, lockout	OA damper vs OAT trend

Code	Title	Severity	Detection summary	Likely causes	Operator checks
AHU-F	Damper command vs feedback mismatch	warning	Cmd ≠ feedback	Stuck actuator, linkage	Compare cmd/ feedback trends

Example rule: `flatline_1h` on SAT → tag **AHU-C**. Recipe: [Python recipes]({{ “/rule-cookbook/python-recipes/” | relative_url }}).

93 Standards crosswalk

94 Standards crosswalk

Grade-A fault definitions carry a `standards_crosswalk` block in `open_fdd/faults/catalog/*.yaml`.

Field	Source	Use in Open-FDD
`ornl_lbnl_taxonomy`	ORNL/ LBNL unified HVAC fault taxonomy	Canonical ontology backbone
`ashrae_g36_reference_type`	ASHRAE Guideline 36	Trim-and-respond, plant requests, supervisory sequences
`ashrae_207_reference_type`	ASHRAE Standard 207	Economizer FDD test categories
`brick_classes`	Brick Schema	Equipment/point semantic roles
`haystack_tags`	Project Haystack (optional)	Interoperability tags

94.1 Example

```yaml
standards_crosswalk:
  ornl_lbnl_taxonomy: ahu/economizer/free_cooling
  ashrae_g36_reference_type: supply_air_temp_reset
  ashrae_207_reference_type: economizer_fault_detection
```

```
    brick_classes: [brick:Air_Handler_Unit, brick:Economizer]
    haystack_tags: [equip, ahu, econ]
```

Export full catalog: `python3 -c "from open_fdd.faults.catalog import catalog_export; import json; print(json.dumps(catalog_export(), indent=2))"`

95 Backup and restore

96 Backup and restore

`workspace/` on the edge host is the **live site state** — feather historian, BACnet commissioning files, BRICK model, FDD rules, auth, and dashboard static bundle. Container images are replaceable; **backup `workspace/` before every upgrade.**

96.1 Quick backup (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Archives default to **`~/openfdd-backups/latest`** on the edge host (overwrites each run — one rolling copy for rigorous testing).

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Custom location (also overwritten each run unless you use a new path):

```
BACKUP_ROOT=~/openfdd-backups/manual ./scripts/openfdd_site_backup.sh
```

Fast pre-upgrade backup (skips large `workspace/bacnet/polls/` CSV history; feather/model/rules kept):

```
BACKUP_INCLUDE_POLL_SAMPLES=0 ./scripts/openfdd_site_backup.sh
```

On **bensserver**, site packs under `edge_backup/local/<site>/<building>/` are likewise overwritten by `edge_site_backup.sh` — no timestamp pile-up during validation.

From bensserver for any edge host (Ansible — backup runs **on the edge**, no workspace transfer over Tailscale):

```
OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_site.sh --limit acme_vm_bbartling
```

```
OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_site.sh --limit
acme_vm_bbartling --fast-backup
```

Acme wrapper (same flow):

```
OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_acme_site.sh --fast-
backup
```

96.2 Manual backup (SSH)

Container processes may write files as **`root:root`** with mode **`0600`**.
Use `sudo` for a full archive:

```
cd ~/open-fdd

export BACKUP_ROOT="$HOME/openfdd-backups/$(date +%Y%m%d-%H%M%S)"
mkdir -p "$BACKUP_ROOT"

cp docker-compose.yml "$BACKUP_ROOT/docker-compose.yml.before"
docker compose ps > "$BACKUP_ROOT/docker-compose-ps-before.txt"
docker compose config --images > "$BACKUP_ROOT/docker-images-
before.txt"

sudo tar --xattrs --acls -czf "$BACKUP_ROOT/workspace-full.tgz"
workspace
sudo chown "$USER:$USER" "$BACKUP_ROOT/workspace-full.tgz"

du -h "$BACKUP_ROOT/workspace-full.tgz"
```

96.3 What to protect

Path	Contents
`workspace/data/feather_store/`	Feather historian
`workspace/bacnet/commissioning/`	`points.csv`, `commission.env`
`workspace/bacnet/polls/samples.csv`	Poll output
`workspace/data/*.json`	Model, rules store, FDD results
`workspace/auth.env.local`	Login secrets
`workspace/api/static/app/`	Dashboard bundle (if deployed separately)

96.4 Restore after a bad upgrade

1. Stop containers: `docker compose stop` (do **not** use `down -v`)
2. Restore `docker-compose.yml` from backup
3. If data corruption: extract selective files from `workspace-full.tgz` — model, rules, feather shards
4. `docker compose pull && docker compose up -d --force-recreate`
5. Verify: Deployment validation({{ "/ops/deployment-validation/" | relative_url }})

96.5 Safe cleanup

```
docker image prune -f
```

> **Never on live sites:** `docker compose down -v`, `docker volume prune`, `docker system prune -a --volumes`, or deleting `workspace/`.

96.6 Related

- Updating the stack({{ "/quick-start/updating/" | relative_url }})
- Live site update({{ "/ops/live_site_update/" | relative_url }})

97 OpenFDD RCx Central API

98 OpenFDD RCx Central API

Local analyst API for multi-edge RCx workflows over Tailscale/VPN. **Read-only toward OpenFDD Edge** — no BACnet, no commands.

> Package path: `portfolio/central/api.py` — service name **openfdd-rcx-central**.

98.1 Start

```
pip install -r portfolio/requirements.txt
./scripts/run_central_api.sh   # http://127.0.0.1:8060/health
./scripts/run_portfolio_dash.sh # http://127.0.0.1:8050
```

Bind defaults: `0.0.0.0` (`OPENFDD_CENTRAL_API_HOST`, `OPENFDD_RCX_DASH_HOST`). LAN: `sudo ./scripts/open_rcx_central_lan_ports.sh`.

Agent guide: docs/agent-skills/rcx-central-dash-agent.md

98.2 Edge registry (browser auth)

Method	Path	Purpose
GET	`/api/central/edges`	Saved Edge instances (secrets masked)
POST	`/api/central/edges`	Add/update Edge
POST	`/api/central/edges/test`	Test connection (health + model)
PUT	`/api/central/edges/{site_id}`	Update Edge
DELETE	`/api/central/edges/{site_id}`	Remove Edge

98.3 Analytics & RCx

Method	Path	Purpose
GET	`/api/central/overview/{site_id}`	Dash payload: KPIs, fault pie, building summary, P8
GET	`/api/central/mechanical-summary/{site_id}`	Equipment counts, BACnet, narrative
GET	`/api/central/fdd-analytics/{site_id}`	FDD rules + fault descriptions
GET	`/api/central/fdd-preset/{site_id}/{preset_id}`	Proxy Edge FDD/BRICK preset (Data Model tab parity)
POST	`/api/central/rcx/preview`	Data readiness + chart catalog
POST	`/api/central/rcx/charts/preview`	Matplotlib PNG previews (base64)
POST	`/api/central/rcx/report`	Download DOCX report

98.4 Validation (legacy collect workflow)

Method	Path	Purpose
GET	`/api/central/sites`	Edge registry + last check-in
POST	`/api/central/sites/{site_id}/collect-validate`	Collect rollup + validation
POST	`/api/central/validation/run`	One-off or scheduled plan
GET	`/api/central/validation/jobs`	Stored jobs
GET	`/api/central/validation/jobs/{id}`	Job detail

98.5 RCx report body

```
{
  "site_id": "acme",
  "hours": 24,
  "charts": ["fault_hours_by_severity", "ahu_sat_vs_setpoint"],
  "sections": ["executive_summary", "mechanical_summary"],
  "save_to_volume": true
}
```

Reports save under `portfolio/data/reports/` (configurable via `OPENFDD_RCX_CENTRAL_DATA`).

Legacy endpoint: `POST /api/central/rcx/report-legacy`

99 VAV faults

100 VAV faults

Variable air volume terminal codes (sample).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
VAV-A	Zone heating/ cooling overlap	warning	Heat and cool at terminal	Reheat valve stuck, SAT too low	Zone temp vs SAT; valve cmd
VAV-B	Poor zone temperature control	warning	Persistent deviation from setpoint	Oversized box, bad tuning	PI loop trends; occupancy
VAV-C	Zone temperature sensor fault	warning	Flatline / OOB zone temp	Failed zone sensor	Flatline test; compare to neighbor zones
VAV-D	Minimum airflow not met	warning	Flow below minimum cmd	Damper stuck, pressure issue	Airflow vs damper cmd
VAV-E	Discharge temp deviation from SAT	warning	DAT far from SAT setpoint	Coil issue, sensor drift	DAT vs SAT SP

Example rule: `oob_rolling` on zone temp → **VAV-C** or comfort bounds per site policy.

101 Docker images (GHCR)

102 Docker images (GHCR)

Full **Open-FDD edge/operator stack** — not the PyPI library.

102.1 Images

GHCR image	Service	Role
`ghcr.io/bbartling/openfdd-bridge`	`bridge`	API, dashboard, historian
	`commission`	

GHCR image	Service	Role
`ghcr.io/bbartling/openfdd-commission`		BACnet discover/ read/poll
`ghcr.io/bbartling/openfdd-mcp-rag`	`mcp-rag`	MCP + doc-search sidecar

Image name `openfdd-mcp-rag` is retained for compatibility; the container runs the unified MCP server.

102.2 Tags

Tag	Meaning
`3.0.30`	Exact release (preferred for production)
`3.0`	Minor alias on release tags
`latest`	Latest **tagged** release only
`edge`	Not published by default — use pinned versions on OT hosts

```
export OPENFDD_IMAGE_TAG=3.0.30
docker pull ghcr.io/bbartling/openfdd-bridge:${OPENFDD_IMAGE_TAG}
docker pull ghcr.io/bbartling/openfdd-commission:${OPENFDD_IMAGE_TAG}
docker pull ghcr.io/bbartling/openfdd-mcp-rag:${OPENFDD_IMAGE_TAG}
```

102.3 Publish

Triggered by git tag `vX.Y.Z` → workflow **Publish Docker images to GHCR** (tags `X.Y.Z`, `X.Y`, `latest`).

Manual: Actions → **Publish Docker images to GHCR** (`workflow_dispatch`) — reads `version` from `pyproject.toml` and publishes the same tags (`3.1.3`, `3.1`, `latest`). Optional `image_tag` input overrides the version.

102.4 vs PyPI

Need	Use
Embed FDD in your Python/cloud job	`pip install open-fdd`
BACnet + UI + bridge on an edge host	GHCR images

See Release process({{ “/developer/release-process/” | relative_url }}).

103 RTU faults

104 RTU faults

Packaged rooftop unit codes (sample).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
RTU-A	Compressor short cycling	warning	Rapid on/off cycles	Low charge, oversized unit	Run-time histogram
RTU-B	Economizer / OA damper fault	warning	OA behavior vs conditions	Actuator, sensor	OA damper vs enthalpy/ OAT
RTU-C	Supply air temperature fault	warning	SAT flatline or OOB	Sensor failure	SAT trend vs outdoor conditions
RTU-D	Simultaneous heat and cool	critical	Heat + cool stages together	Control board, relay stuck	Stage status trends

Bind rules to packaged unit `fdd_input` points (SAT, OAT, compressor cmd, …).

105 Deployment validation

106 Deployment validation

For the **live Acme site** after GHCR updates, use the dedicated harness documented in Acme live validation({{ “/ops/acme-live-validation/” | relative_url }}) (`acme_post_deploy_validate.sh`).

Non-destructive checks after deploy or image upgrade. Run from the **edge host** (smoke) or from a **control machine** with the full repo (insurance suite).

106.1 Edge host smoke (5 minutes)

```
cd ~/open-fdd

docker compose ps
curl -sf http://127.0.0.1:8765/health | jq . || curl -sf http://127.0.0.1:8765/health

tail -1 workspace/bacnet/polls/samples.csv
du -sh workspace/data/feather_store/

docker compose logs --since 20m | grep -Ei "error|exception|traceback|critical" | tail -20 || true
```

Via Caddy (building LAN):

```
curl -sf http://<edge-ip>/health
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.js' | head -1
```

106.2 Browser checklist

1. Open `http://<edge-ip>/` (Caddy `:80`, not only loopback `:8765`)
2. Integrator login (`workspace/auth.env.local` on the edge host)
3. **Host stats** — container image tag / git SHA
4. **BACnet** — driver tree shows devices; last poll timestamp recent
5. **Model & assignments** — commissioning export returns JSON
6. **Rule Lab** — quick-test a saved rule (`backend: arrow`)

106.3 Control-machine insurance check

From a machine with the full `open-fdd` repo and Ansible inventory:

```
cd infra/ansible/scripts
./post_deploy_check.sh --limit <inventory_host> --full
```

Or by IP:

```
./post_deploy_check.sh --host <edge-ip> --ssh-user <deploy-user> --full
```

Checks include: Caddy → React dashboard, `/health/stack`, MCP RAG, BACnet tree, BRICK/SPARQL, Docker log scan.

> **Do not run on live commissioned sites without approval:** `acme_operational_verify.sh` with BACnet rediscover — use `--skip-discover` for smoke only. See [Examples — GL36 lab]({{ "/examples/acme-gl36-lab/" | relative_url }}).

106.4 LAN security smoke (optional)

From a workstation on the same LAN as the edge:

- Windows: `scripts/security/Run-OpenFddSecurityScan.ps1`
- macOS/Linux: `scripts/security/run_openfdd_security_scan.sh`

Expected findings: [ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}).

106.5 Arrow / FDD rules (3.0+)

On control machine:

```
./scripts/validate_fdd_backends.sh --docker
```

On edge after upgrade:

```
grep -R "apply_faults_arrow" workspace/data/rules_py | wc -l
```

106.6 Related

- [Health check]({{ "/quick-start/health-check/" | relative_url }})
- Security testing cycle({{ "/developer/security-testing/" | relative_url }}) (maintainers)

107 Sensor and data quality

108 Sensor and data quality

Sensor faults use cookbook patterns **`flatline_1h`**, **`oob_rolling`**, **`rate_of_change`**, and **`mixing_envelope`**. Full thresholds: [Expression cookbook — sensor validation]({{ "/rule-cookbook/expression-cookbook/" | relative_url }}#sensor-validation-bounds-flatline-rate-of-change).

108.1 Catalog codes

Code	Title	Severity	Pattern	Detection summary
BLD-B	Outdoor air sensor fault	warning	flatline, oob, rate_of_change	OAT flatline, OOB, or unrealistic spike
BLD-D	Stale telemetry	critical	stale_points	No new samples within threshold
AHU-C	SAT sensor fault	warning	flatline_1h, oob_rolling	Supply air temp stuck or out of band
AHU-D	MAT inconsistent with OAT/RAT	warning	mixing_envelope	Mixed air outside OAT–RAT envelope
VAV-C	Zone temperature sensor fault	warning	flatline_1h, oob_rolling	Zone temp stuck or 55–90 °F band breach
RTU-C	Supply air temperature fault	warning	flatline_1h, oob_rolling	Packaged unit SAT fault
CH-D	CHW supply temperature sensor fault	warning	flatline_1h, oob_rolling	Plant CHW sensor fault
DC-C	Datacom sensor anomaly	warning	flatline_1h	CRAH/room sensor flatline or RH OOB
HP-D	Discharge/suction temperature sensor fault	warning	flatline_1h, oob_rolling	Heat pump refrigerant/air temp sensor

108.2 Default bounds (imperial, occupied)

Sensor	Min	Max	Flatline tol	Max Δ/hr
Zone temp	55 °F	90 °F	0.10 °F	4 °F

Sensor	Min	Max	Flatline tol	Max Δ/ hr
Supply air temp	50 °F	110 °F	0.15 °F	8 °F
Return air temp	55 °F	95 °F	0.10 °F	3 °F
Duct static	−0.5	3.0 inH₂O	0.02	0.5
RH	0 %	100 %	1.0 %	15 %
CHW	40 °F	90 °F	0.10 °F	4 °F
HW	70 °F	200 °F	0.15 °F	6 °F
Condenser water	50 °F	110 °F	0.15 °F	5 °F
CO_2 (occupied)	400 ppm	1000 ppm	5 ppm	200 ppm

Return-air checks use **narrow bands**; when MAT/OAT/RAT are available, prefer **mixing envelope** over RAT bounds alone.

108.3 Communication quality

- Monitor poll cycle timestamps in dashboard.
- Use `stale_points` for dropout detection (**BLD-D**).
- Do not confuse **comm loss** with **equipment fault** — verify BACnet device online before dispatching mechanical.

108.4 Operator workflow

1. Confirm **BLD-D** (stale) is resolved before trusting other sensor codes.
2. Cross-check **BLD-B** OAT against weather service or JSON API driver.
3. For **VAV-C** / **AHU-C**, compare neighbor zones / coil behavior before replacing sensor.

109 Logging and audit

110 Logging and audit

Open-FDD uses a **two-tier logging model** familiar to IT and security teams: structured **audit/error JSONL** for forensics and compliance, plus **service stdout** for engineering troubleshooting. Defaults include **age pruning**, **size rotation in MB**, and **Docker log caps** — no operator cron required.

110.1 Log types

Tier	Location	Format	Who reads it
Audit	`workspace/logs/audit.jsonl`	JSON Lines	Security, MSI, integrator API
Errors	`workspace/logs/error.jsonl`	JSON Lines	Support, integrator API
Service stdout	Docker: `docker compose logs` · Dev: `workspace/.local-run/*.log`	Plain text (uvicorn, workers)	Engineering
Historian	`workspace/data/feather_store/`	Feather	FDD, plots — separate retention (`data.env.local`)

Structured audit logs are **not** mixed into uvicorn access logs. That separation matches common practice: SIEM-friendly JSONL for auth and OT commands; unstructured stdout for stack traces and poll worker noise.

110.2 Audit record schema

Each line in `audit.jsonl` / `error.jsonl` is one JSON object:

Field	Example	Notes
`@timestamp`	`2026-06-07T19:57:21+00:00`	UTC ISO-8601
`event_id`	UUID	Unique per event
`event_type`	`auth.login.failure`	Stable taxonomy (see below)
`severity`	`warning`	`debug` … `critical`
`outcome`	`failure`	`success`, `failure`, `denied`, …

Field	Example	Notes
`service`	`openfdd-bridge`	
`host`	hostname	
`action`	`login`	Human-readable verb
`actor`	`{username, role}`	After successful auth
`client`	`{ip, user_agent}`	Respects `OFDD_TRUST_X_FORWARDED_FOR` behind Caddy
`http`	method, path, status	When HTTP-triggered
`resource`	BACnet object, model id, …	When applicable
`detail`	Sanitized map	**Passwords, tokens, secrets stripped by key name**
`request_id`	UUID	On audited HTTP paths

110.2.1 Auth events (industry-standard login trail)

`event_type`	When
`auth.login.success`	Valid credentials
`auth.login.failure`	Bad password (generic client message)
`auth.login.lockout`	Rate limit exceeded (default 5 failures / 5 min)

Login rate limits and lockouts are **audited**; responses never reveal whether the username exists.

110.2.2 OT / commissioning events

`event_type`	When
`bacnet.command`	Supervisory write (allowed, denied, dry-run)
`bacnet.discover`	Who-Is / point discovery
`model.write`	BRICK model mutations
`agent.tool`	Agent tool calls (prompts hashed/truncated)
`api.access`	Sensitive POST/PUT/PATCH/DELETE (BACnet, ingest)
`error.unhandled`	Unhandled exception (also in `error.jsonl`)

Not audited: routine GETs (dashboard reads, plot data, health). This reduces noise while keeping **auth and field commands** on the trail.

110.2.3 Secret redaction

Never logged: passwords, bearer tokens, `OFDD_AUTH_SECRET`, private keys. Detail objects drop keys containing `password`, `token`, `secret`, `authorization`. Agent tool arguments use additional hashing for prompts and rule source.

Opt-in full prompt logging (lab only): `OFDD_AUDIT_LOG_PROMPTS=1`.

110.3 Integrator API (read logs in the UI stack)

Method	Path	Role
GET	`/api/audit/events?limit=100`	integrator
GET	`/api/audit/errors?limit=100`	integrator
GET	`/api/audit/summary`	integrator

Use these for post-incident review without shell access. Forward `workspace/logs/*.jsonl` to SIEM (rsyslog, Filebeat, Wazuh) on production edges.

110.4 Retention and rotation (defaults)

Configured in `workspace/data.env.local` (copy from `data.env.example`):

Variable	Default	Behavior
`OFDD_LOG_RETENTION_DAYS`	**90**	Drop audit/error JSONL lines older than N days; delete aged `.local-run` and archived `*.log` / `audit.*.jsonl`
`OFDD_LOG_MAX_MB`	**50**	When `audit.jsonl` or `error.jsonl` exceeds N MB, archive to `audit.{UTC}.jsonl` and start fresh
`OFDD_LOCAL_RUN_LOG_MAX_MB`	**25**	

Variable	Default	Behavior
		Rotate `workspace/.local-run/*.log` (bridge, Caddy, FDD loop, …)
`OFDD_LOG_ROTATE_INTERVAL_HOURS`	**6**	Background retention while bridge is running (not only at restart)

Rotation runs at **bridge startup** and on the **scheduled interval**. No logrotate.d entry is required for audit JSONL.

Optional path overrides:

```
OFDD_AUDIT_LOG_PATH=/var/log/openfdd/audit.jsonl
OFDD_ERROR_LOG_PATH=/var/log/openfdd/error.jsonl
```

110.5 Docker edge: json-file caps (not journald for app audit)

Production stacks use **Docker Compose** (`docker/compose.edge.yml`). Container stdout uses the **`json-file` driver with size limits**:

```
logging:
  driver: json-file
  options:
    max-size: "25m"
    max-file: "5"
```

That caps each service at roughly **125 MB** of rotated container logs on disk — the same pattern many teams use instead of unbounded `docker logs`.

Why audit JSONL stays on disk (not journald)?

- Append-only files under `workspace/logs/` survive container recreate and bind-mount to the host.
- JSONL lines import cleanly into SIEM without journal field parsing.
- OT incident response often needs **months** of auth/BACnet write history on the edge box.

When journald still applies: native systemd units (Caddy, Ollama on bare metal) — use `journalctl -u <unit>` per your Linux runbook. Bridge audit remains in `workspace/logs/` even when uvicorn stdout goes to Docker or `.local-run/`.

110.6 Local dev (`run_local.sh`)

File	Service
`workspace/.local-run/bridge.log`	Bridge / uvicorn
`workspace/.local-run/commission.log`	BACnet commission agent
`workspace/.local-run/caddy.log`	Caddy
`workspace/.local-run/fdd_loop.log`	Scheduled FDD
`workspace/.local-run/mcp_rag.log`	MCP RAG
`workspace/.local-run/ollama.log`	Ollama

These are **engineering logs**. Structured audit still lands in `workspace/logs/audit.jsonl` when auth is enabled.

110.7 Quick checks

```
# Last auth failures
grep 'auth.login' workspace/logs/audit.jsonl | tail -5

# Integrator API (after login)
curl -s -H "Authorization: Bearer $TOKEN" http://127.0.0.1:8765/api/audit/summary | jq .

# Docker edge
docker compose logs bridge --tail 50
```

110.8 Comparison to typical web apps

Practice	Open-FDD
Structured auth audit	Yes — JSONL with actor, IP, outcome
Login rate limit + lockout audit	Yes
Secret redaction in logs	Yes — key-based + agent sanitization
Request correlation ID	Yes — on audited mutation paths
Centralized stdout JSON	No — stdout is plain text; audit is JSONL

Practice	Open-FDD
Full HTTP access log	No — mutations and auth only (OT noise reduction)
Log rotation / size caps	Yes — defaults above + Docker `max-size`
SIEM-ready export	Yes — file tail / Filebeat on `workspace/logs/`

For BACnet write safety and credential generation, see SECURITY.md and Secrets and auth({{ "/security/secrets-auth/" | relative_url }}).

110.9 Related

- JSON API driver({{ "/drivers/json-api/" | relative_url }}) — OpenWeatherMap showcase, env-based API keys (`json_api.env.local`)
- Live site update({{ "/ops/live_site_update/" | relative_url }}) — `docker compose logs` after upgrades
- Data flow({{ "/architecture/data-flow/" | relative_url }}) — historian retention (`OFDD_FEATHER_*`) is separate from audit logs

111 GHCR image retention

112 GHCR image retention

Open-FDD publishes edge images to **GHCR** (`ghcr.io/bbartling/openfdd-*`). During rapid development, old package versions accumulate. This retention system prunes **noise** while protecting rollback tags for **Acme**, our live validation building.

112.1 Policy (conservative)

Always keep:

- `latest` and `edge`
- Latest **5** SemVer release tags per image (configurable)
- Tags listed in `scripts/ghcr-retention-protected-tags.txt`
- Current Acme deployed tag (workflow input or `--current-acme-tag`)
- Previous Acme known-good tag (`--previous-acme-tag`)

Delete candidates (only with `--confirm-delete`):

- Untagged versions older than **7** days
- SHA-only tags older than **30** days
- Feature/dev-style tags older than **30** days
- SemVer releases beyond the latest N window (unless protected)

Images are defined in `docker/images.yaml`:

- `openfdd-bridge`
- `openfdd-commission`
- `openfdd-mcp-rag`
- `openfdd-cloud-exporter`

112.2 Dry-run first

The script defaults to **dry-run**. It prints a keep/delete table and optional JSON/Markdown reports. Nothing is deleted until you pass **`--confirm-delete`**.

There is **no scheduled automatic deletion** yet — only manual runs locally or via GitHub Actions `workflow_dispatch`.

112.3 Protect Acme rollback tags

Before cleanup, edit `scripts/ghcr-retention-protected-tags.txt`:

```
latest
edge
v3.0.31
v3.0.30
```

Or pass tags at runtime:

```
./scripts/ghcr_prune_packages.sh --all-images --dry-run \
  --current-acme-tag v3.0.31 \
  --previous-acme-tag v3.0.30
```

112.4 Run locally

```
gh auth login   # needs read:packages + delete:packages for confirm-delete

./scripts/ghcr_prune_packages.sh --all-images --dry-run
./scripts/ghcr_prune_packages.sh --image openfdd-bridge --dry-run \
  --json-out reports/ghcr-prune-plan.json \
```

```
  --markdown-out reports/ghcr-prune-plan.md

# After reviewing the plan:
./scripts/ghcr_prune_packages.sh --all-images --confirm-delete \
  --current-acme-tag v3.0.31 --previous-acme-tag v3.0.30
```

112.5 GitHub Actions workflow

Workflow: **Prune old GHCR images** (`.github/workflows/ghcr-prune.yml`)

1. Actions → **Prune old GHCR images** → **Run workflow**
2. Leave **Dry run only** = `true` for the first run
3. Download the `ghcr-prune-report` artifact
4. Re-run with **Dry run only** = `false` only after reviewing the plan
5. Set **Current Acme deployed tag** and **Previous Acme rollback tag** when known

112.6 Permissions

Deleting package versions requires:

- Local: `gh auth login` with `delete:packages` (and `read:packages`)
- Actions: `permissions.packages: write` on the workflow (already set)

If deletion fails, the script prints:

> Deleting GHCR package versions requires package delete permissions...

112.7 Verify images still pull

After pruning:

```
docker pull ghcr.io/bbartling/openfdd-bridge:latest
docker pull ghcr.io/bbartling/openfdd-commission:latest
```

On Acme:

```
export OPENFDD_IMAGE_TAG=v3.0.31
./scripts/upgrade_edge_ghcr.sh --limit acme_vm_bbartling
./scripts/acme_post_deploy_validate.sh --limit acme_vm_bbartling --quick
```

112.8 Rollback example

```
export OPENFDD_IMAGE_TAG=v3.0.30
./scripts/upgrade_edge_ghcr.sh --limit acme_vm_bbartling
```

112.9 Related issues

Track open Acme/deploy blockers in GitHub issue #276.

113 Fault Codes

114 Fault Codes

Open-FDD uses a **check-engine** model: one building-level status (green / yellow / red) with a catalog of fixed fault codes per equipment family.

Page	Content
Convention({{ “/fault-codes/convention/”	relative_url }})
AHU faults({{ “/fault-codes/ahu/”	relative_url }})
VAV faults({{ “/fault-codes/vav/”	relative_url }})
RTU faults({{ “/fault-codes/rtu/”	relative_url }})
[Sensor & data quality]({{ “/fault-codes/sensor-quality/”	relative_url }})
Short lookup({{ “/fault-codes/short-lookup/”	relative_url }})

Live catalog API: `GET /api/faults/catalog` on the Operator Bridge.

115 Operations

116 Operations

Runbooks for live edge hosts after the initial Quick Start({{ “/quick-start/” | relative_url }}) deploy.

Page	When to use
Live site update({{ “/ops/live_site_update/”	relative_url }})
Backup and restore({{ “/ops/backup-restore/”	relative_url }})
Logging and audit({{ “/ops/logging/”	relative_url }})
Deployment validation({{ “/ops/deployment-validation/”	relative_url }})
ACME live validation({{ “/operations/acme-live-validation/”	relative_url }})
Bench 5007 long FDD smoke({{ “/operations/bench-5007-long-fdd-smoke/”	relative_url }})
Bench 5007 dual-source smoke({{ “/operations/bench-5007-dual-source-smoke/”	relative_url }})

Site-specific lab notes (example BACnet scope, GL36 rules): Examples & lab notes({{ “/examples/” | relative_url }}).

For first-time deploy: [Quick Start — Docker]({{ “/quick-start/docker/” | relative_url }}).

117 Portfolio

118 Portfolio (central desk)

Multi-site analytics on **benserver** — not part of the edge Docker stack.

Topic	Page
Collect rollups from edge sites	Central collection({{ “/portfolio/central-collection/”

Repo layout: `portfolio/` (collector, Dash, `sites.json`). Maintainer tuning API flow: `skills/openfdd-edge-deploy-tune/SKILL.md`.

119 Acme VAV/AHU rule guidance

120 Acme VAV/AHU FDD rule bundle

Practical Arrow-native rule set for **acme** (`vm-bbartling`): one RTU/AHU (`acme-vm-bbartling-rtu-01`, BACnet **1100**) feeding **JCI VAVs** (instances 1–100, imperial °F at wire) and **Trane VAVs** (11000–13000, metric °C converted to °F before feather via `device_poll_profiles.csv`).

Install / refresh rules:

```
        python3 scripts/setup_gl36_fdd.py --site-id acme --building-id vm-bbartling \
          --ahu-equipment-id acme-vm-bbartling-rtu-01 --fan-point-id 1100-analog-output-1
```

Validate bundle locally:

```
        python3 scripts/acme_validate_fdd_bundle.py
```

120.1 Mixed manufacturers and units

Manufacturer	BACnet instances	Wire units	Open-FDD handling
JCI VAV	1–100	°F	Rules use `temp_unit: imperial`; no poll conversion
Trane VAV	11000–13000	°C	`metric_temp_f` in `edge_config/acme/vm-bbartling/device_poll_profiles.csv` converts to °F before feather
RTU AHU	1100	°F	AHU rules bind to equipment / fan point

All zone-temperature rules share **imperial bounds** after poll conversion. If a new Trane box is added without a profile row, zone OOB/flatline will be wrong until the profile is added and data re-ingested.

Command scaling (damper, reheat, fan): rules normalize **0–1 vs 0–100%** with `pc.if_else(greater(c, 1.0), divide(c, 100), c)`.

Poll interval: Acme BACnet poll loop is **60 s**. Flatline windows use `poll_interval_s: 60` and `flatline_minutes: 60` → **60 samples per hour**, not the legacy 5-min × 12 assumption.

120.2 Rule matrix

Phase	Priority	Rule ID (prefix `acme-`)	Module	Fault	Equip
0	1	`oat-bounds`	`oat_sensor_bounds.py`	BLD-B	Buildi
0	2	`oat-flatline-1h`	`oat_sensor_flatline.py`	BLD-B	Buildi
0	3	`oat-spike`	`oat_sensor_spike.py`	BLD-B	Buildi
0	3b	`oat-vs-web-spread`	`oat_vs_web_spread_1h.py`	BLD-B	Buildi

Phase	Priority	Rule ID (prefix `acme-`)	Module	Fault	Equip
0	4	`zn-t-oob-occupied`	`vav_zone_temp_bounds_occupied.py`	VAV-C	VAV
0	5	`zn-t-flatline-1h`	`vav_zone_temp_flatline_occupied.py`	VAV-C	VAV
0	6	`da-t-flatline-1h`	`flatline_1h.py`	VAV-E	VAV
0	7	`sat-flatline-1h`	`flatline_1h.py`	AHU-C	AHU
0	8	`sap-flatline-1h`	`flatline_1h.py`	AHU-F	AHU
1	9	`vav-damper-stuck`	`vav_damper_stuck_flatline.py`	VAV-D	VAV
1	10	`vav-airflow-low`	`vav_airflow_low.py`	VAV-D	VAV
1	11	`vav-reheat-warm-ambient`	`zone_reheat_warm_ambient.py`	VAV-A	VAV
1	12	`vav-reheat-leak`	`vav_reheat_dat_vs_sat.py`	VAV-A	VAV
2	13	`ahu-afterhours-runtime`	`ahu_afterhours_runtime.py`	BLD-C	AHU
2	14	`mat-oob-economizer`	`oob_rolling.py`	AHU-D	AHU
2	15	`ahu-run-hours`	`ahu_run_hours.py`	—	AHU
—	—	Stale telemetry	Poll health dashboard	BLD-D	All

120.2.1 Phase 2+ (not enabled by default)

Enable only after point coverage on RTU **1100** is verified (OAT, MAT, RAT, SAT, OA damper, cooling cmd):

- **AHU-A** duct static not maintained — needs static SP + fan speed (future rule)
- **AHU-D** mixed air envelope — `mixing_envelope_mask` + fan on
- **AHU-E** economizer trio — 100% OA tracking, mech cooling when free cooling available, OA damper excess open

120.2.2 Phase 3 rollups (future)

- Many simultaneous **VAV-D** airflow-low → suggest AHU static / fan issue
- Many **VAV-B/C** comfort faults → AHU scheduling or SAT problem
- Hot-water plant support when HW supply temp is modeled

120.3 Data model checks

Before enabling more rules:

1. **No duplicate BACnet device instances** on equipment rows (poll health merges by `bacnet_device_id`).
2. **No duplicate `point_id`** rows with different `external_id` / fdd aliases (causes historian_lag double-count).
3. **Trane devices** in `device_poll_profiles.csv` with `metric_temp_f`.
4. Run `scripts/acme_validate_fdd_bundle.py` after model edits.

120.4 Tuning notes

- **`acme-zn-t-oob-occupied`**: if flag rate > ~70%, use tuning brief / bounds auto-tune (needs ≥85% flagged + analytics on Arrow sweep).
- Disable noisy rules via `enabled: false` in `setup_gl36_fdd.py` rather than deleting modules.
- Economizer and reheat-leak rules stay off until OAT/MAT/SAT columns are confirmed on the 7-day feather window.

120.5 JSON API weather + BACnet OAT

- OpenWeather registered on edge JSON API tab (`web-oat-t`, `web-rh`, ~20–30 min poll).
- Local OAT: bind **`1100-unknown-2`** (RTU outdoor air local) with `external_id: oa-t` — `scripts/acme_patch_oat_column.py`.

- Rule `acme-oat-vs-web-spread` flags when local vs web spread exceeds 8 °F.

120.6 BACnet override scans

Commission agent rotates **one device/hour** through P8 priority-array reads. Status: `GET /api/bacnet/overrides/status`. See Operator override scans({{ “/bacnet/override-scans/” | relative_url }}).

120.7 Docker backup / recovery

Edge upgrades via `scripts/upgrade_edge_ghcr.sh` preserve `workspace/data` and feather volumes on the VM. After upgrade:

1. `curl /health` → expect matching `openfdd_version`
2. `./infra/ansible/scripts/acme_operational_verify.sh`
3. Re-push rules if `rules_store.json` was reset: `setup_gl36_fdd.py --host … --token …`

Feather backup: snapshot `workspace/data/feather/` (or site pack export) before major model changes.

121 Bench 5007 dual-source smoke

122 Bench 5007 dual-source smoke

Validates BACnet MS/TP device **5007** and Niagara baskStream station **bench9065** on the same physical bench.

122.1 Run

```
./scripts/run_local.sh start
./scripts/smoke_bench_5007_dual_source.sh
```

Environment (optional):

Variable	Default
OPENFDD_BASE_URL	http://127.0.0.1:8765
OPENFDD_AUTH_ENV	workspace/auth.env.local
OPENFDD_SMOKE_SITE_ID	demo
OPENFDD_SMOKE_POLL_SECONDS	60
OPENFDD_SMOKE_DURATION_MINUTES	10

The script wraps scripts/smoke_benserver_bench.py and adds DataFusion SQL lab preview when open-fdd[datafusion] is installed.

122.2 Paired semantic points

Semantic	BACnet column	Niagara column
oa-t	oa-t	niagara-oa-t
oa-h	oa-h	niagara-oa-h
duct-t	duct-t	niagara-duct-t
stat_zn-t	stat_zn-t	niagara-stat-zn-t

Import workspace/data/bench_dual_source_model.json if the model is empty.

123 Bench 5007 long FDD smoke

124 Bench 5007 long FDD smoke

Validates **data-source-agnostic FDD**, **PyArrow vs DataFusion SQL equivalence**, and the **fault confirmation window** on BACnet 5007 + Niagara bench9065 (site demo).

Read-only: polls devices; only Open-FDD evaluation thresholds change. Never prints credentials.

124.1 Commands

Mode	Command
CI synthetic	`python scripts/smoke_bench_5007_long_fdd.py --synthetic`
Dev dry-run (~15 min)	`python scripts/smoke_bench_5007_long_fdd.py --dry-run`
Default (2 h)	`./scripts/smoke_bench_5007_long_fdd.sh`
Overnight (12 h)	`python scripts/smoke_bench_5007_long_fdd.py --overnight`

124.2 Key environment

Variable	Default
`OPENFDD_BASE_URL`	`http://127.0.0.1:8765`
`OPENFDD_SMOKE_DURATION_MINUTES`	`120` (720 if `OPENFDD_SMOKE_OVERNIGHT=true`)
`OPENFDD_SMOKE_POLL_SECONDS`	`60`
`OPENFDD_SMOKE_BASELINE_MINUTES`	`20`
`OPENFDD_SMOKE_CONFIRMATION_ROWS`	`10`
`OPENFDD_SMOKE_CONFIRMATION_MINUTES`	`10`
`OPENFDD_SMOKE_PRIMARY_SEMANTIC`	`duct-t`
`OPENFDD_SMOKE_FORCED_THRESHOLD_F`	`80.0`

124.3 Fault confirmation window

At 1-minute polling, `min_true_rows = 10` means the raw condition must stay true for about **10 minutes** before a **confirmed fault** is reported. See fault-confirmation.md.

124.4 Artifacts

- `workspace/reports/bench_5007_long_fdd_<timestamp>.md`
- `workspace/reports/bench_5007_long_fdd_<timestamp>.json`
- `workspace/reports/bench_5007_long_fdd_<timestamp>_events.csv`

124.5 Pass / fail

Verdict	Meaning
PASS	All checks passed; confirmation fully vectorized (not expected today)
WARN	Core FDD uses `pyarrow_compute` / `datafusion_sql`; confirmation engine is `python_loop_over_arrow_mask` (known tech debt)
FAIL	`python_list` core crunching, early confirmed fault, backend mismatch, missing data, or secrets in output

Live mode uses `POST /api/bench/long-fdd/evaluate` (validation/smoke only).

124.6 Related

- Dual-source smoke
- Fault confirmation

125 ACME live validation

126 ACME live validation

ACME (`site_id: acme`, Acme Building) is the **live BACnet HVAC validation site** on the OT LAN. Use it for real VAV/AHU behavior, RCx rule tuning, and overnight FDD runs.

Do **not** confuse ACME with **Bench 5007** dual-source validation (site `demo`, BACnet 5007 + Niagara bench9065). Bench 5007 proves backend equivalence; ACME proves production HVAC rules on field hardware.

Reference model (no secrets): `workspace/data/fixtures/acme_data_model.json`. Validate locally:

```
python scripts/validate_acme_model_context.py
python scripts/validate_acme_model_context.py --json
```

126.1 Model export / import

1. **Export** — `GET /api/model/commissioning-export` (dashboard **Import / export** tab or API).
2. **Validate** — `python scripts/validate_acme_model_context.py` on saved JSON; dashboard dry-run before import.
3. **Import** — `POST /api/model/commissioning-import` only after review. Preserve point `id`, `fdd_input`, `metadata.series_id`, and existing `fdd_rule_ids`. Never invent rule IDs.

Someday ACME will be updated from live export again; until the bench harness is stable, treat the committed fixture as the regression baseline.

126.2 Run FDD (read-only)

Goal	Command / API
Post-upgrade smoke	`./scripts/acme_post_deploy_validate.sh --limit acme_vm_bbartling --full`
Selected rules	Rule Lab preview / `POST /api/playground/test-rule`
All enabled rules	`POST /api/rules/batch` → `workspace/data/fdd_results.json`
Overnight cycles	`OPENFDD_LIVE_ACME=1 python3 scripts/acme_overnight_fdd_validate.py --limit acme_vm_bbartling`

Normal validation is **read-only**: no BACnet writes, no Pooge reset, no setpoint commands. BACnet writes require commission role + explicit enable — see [BACnet write safety]({{ “/security/bacnet-writes/” | relative_url }}).

126.3 Reports

- Post-deploy JSON: `reports/acme-live-validate.json` (gitignored)
- Overnight FDD: under `reports/` from overnight script
- FDD batch: `workspace/data/fdd_results.json` (local edge only)

Redact tokens and hostnames before sharing.

126.4 Fault confirmation

Rules may fire **raw** faults before **confirmed** faults. Configure `min_true_rows` and `min_elapsed_minutes` in rule `cfg` (e.g. 10 rows at 1-minute poll ≈ 10 minutes). See Fault confirmation({{ “/rule-cookbook/fault-confirmation/” | relative_url }}).

126.5 Safe rules on ACME

Prefer read-only sensor bounds, flatline, spread (OAT vs weather), damper stuck, and zone temperature rules. Avoid supervisory writes unless explicitly commissioned and guarded.

126.6 Related

- ACME VAV/AHU rule guidance({{ “/operations/acme_vav_ahu_rule_guidance/” | relative_url }})
- Deployment validation({{ “/ops/deployment-validation/” | relative_url }})
- Bench 5007 long FDD smoke({{ “/operations/bench-5007-long-fdd-smoke/” | relative_url }}) — dual-source bench only

127 AI agent context

128 AI agent context

Canonical map for humans and MCP/RAG agents operating Open-FDD on the OT edge. **No secrets** in docs, exports, reports, or chat context.

128.1 1. Open-FDD in one paragraph

Open-FDD is an **edge FDD platform**: BACnet / Niagara / JSON API drivers ingest telemetry into a **BRICK-style JSON model** and **Feather/Arrow historian**; **PyArrow** (primary) and optional **DataFusion SQL** rules produce boolean fault masks with a shared **confirmation window**; the **Operator Bridge** dashboard and **MCP sidecar** expose commissioning, Rule Lab, batch FDD, and doc search. Deploy via **GHCR Docker** (`openfdd-bridge`, `openfdd-commission`, `openfdd-mcp-rag`) on a trusted LAN — not the public internet.

128.2 2. Deployment mental model

Piece	Role
GHCR stack	`openfdd-bridge` (API + UI), `openfdd-commission` (BACnet poll), `openfdd-mcp-rag` (doc RAG)
Host Caddy	`:80` / `:443` front door to bridge
`workspace/`	Model, rules, feather store, auth env — persist across image upgrades
Auth	`workspace/auth.env.local` (gitignored); JWT via `POST /api/auth/login`
Update	Updating the stack({{ “/quick-start/updating/”

Rebuild MCP index after doc changes: `./scripts/build_mcp_rag_index.sh`.

128.3 3. Source / driver modes

Mode	Feather `source`	When
BACnet direct	`bacnet`	Production OT — commission container polls `:47808`
Niagara baskStream	`niagara_baskstream`	Bench / sites with Niagara HTTP API
JSON API	`json_api`	Weather, REST sensors, OpenWeather demo
Platform API profile	scaffold	Future portfolio central — not production duplicate poll

Do **not** run legacy BACnet poll alongside Docker **commission** on the same device (double poll). Validation smokes may intentionally dual-source **Bench 5007** only.

128.4 4. Data model contract

Commissioning bundle: `GET /api/model/commissioning-export` → `POST /api/model/commissioning-import` (dry-run in UI first).

Field	Rule
`sites[]`, `equipment[]`, `points[]`, `fdd_rules[]`	Required structure
Point `id`	**Preserve** on import — never renumber
`fdd_input`	Rule column role (`oa-t`, `zn-t`, `sat`, …)
`metadata.series_id`	Historian series key
`metadata.external_ref`	Stable feather/BACnet reference
`fdd_rule_ids`	Must reference existing `fdd_rules[].id` — **never invent ru** on import
Column map	`open_fdd.arrow_runtime.build_column_map_from_model_points( site_id)`

Live HVAC reference model: **ACME** (`site_id: acme`) — fixture at `workspace/data/fixtures/acme_data_model.json`. Bench dual-source model: site `demo` / device 5007 — not ACME.

128.5 5. Rule execution contract

Backend	Use when
PyArrow (`apply_faults_arrow`)	Full HVAC logic, rolling windows, ML prep, cookbook helpers
DataFusion SQL (`backend: datafusion_sql`)	Simple threshold / CASE / SQL-readable rules; optional `pip install open-fdd[datafusion]`

Both normalize to **Arrow boolean mask** / `ArrowRuleResult`. SQL runs **server-side only** against registered `telemetry`; unsafe SQL is rejected. No browser-side SQL. No pandas row-loop runtime on edge.

128.6 6. Fault confirmation window

- **Raw fault** — rule mask true on one row.

- **Confirmed fault** — `min_true_rows` consecutive trues and/or `min_elapsed_minutes` elapsed.
- Example: **10 rows** at **1-minute** polling ≈ **10 minutes** confirmed.

See Fault confirmation({{ “/rule-cookbook/fault-confirmation/” | relative_url }}).

128.7 7. Validation modes

Mode	Purpose
Bench 5007	Dual-source BACnet vs Niagara backend equivalence — [long FDD smoke]({{ “/operations/bench-5007-long-fdd-smoke/”
ACME live	Real BACnet HVAC, VAV/AHU rules, RCx — ACME live validation({{ “/operations/acme-live-validation/”
Synthetic CI	`python scripts/smoke_bench_5007_long_fdd.py --synthetic --dry-run`
Model fixture	`python scripts/validate_acme_model_context.py`

128.8 8. Security rules

- **No secrets** in docs, logs, exports, MCP bundle, or validation reports.
- Bridge binds **127.0.0.1** by default; LAN exposure requires explicit operator choice.
- **BACnet writes** disabled/guarded — see [BACnet write safety]({{ “/security/bacnet-writes/” | relative_url }}).
- Niagara / JSON API service accounts: dedicated, read-only where possible.
- Full tracebacks only when `OFDD_DEBUG_TRACEBACKS=1`.

128.9 9. Where to look

Topic	Doc
Start	Quick Start({{ “/quick-start/”
Drivers	Driver framework({{ “/drivers/”
Rules	

Topic	Doc
	Rule Cookbook({{ “/rule-cookbook/”
API	API routes({{ “/appendix/bridge_api/”
Operations	Operations({{ “/ops/”
Architecture	[ADR: Rust-ready Arrow contract]({{ “/adr/adr-rust-ready-arrow-fdd-contract/”
MCP	MCP server({{ “/ai/mcp-server/”
Contributor policy	`AGENTS.md` (repo root)

129 LAN hardening

130 LAN hardening

Open-FDD targets **trusted building LAN / OT edge** deployment. The Operator Bridge is not intended for direct public internet exposure.

Full mode matrix: Deployment modes({{ “/architecture/deployment-modes/” | relative_url }}).

130.1 Auth and bind

Mode	Bridge bind	Auth
Local dev	`127.0.0.1`	Optional `OFDD_AUTH_DISABLED=1` on loopback
Lab LAN	`0.0.0.0`	Required unless explicit insecure lab flags
Edge production	Loopback + **Caddy** on LAN	**Required** — `OFDD_AUTH_SECRET` + role passwords

Startup **fails** on non-loopback bind without credentials (unless insecure lab flags are set).

130.2 Network exposure

- Put **Caddy** on `:80` / `:443`; keep bridge `:8765` on loopback when possible.
- Do not port-forward the Operator Bridge to the public internet without TLS and strong auth.
- Keep BACnet on OT VLANs; management UI on IT VLAN with firewall rules.
- BACnet **writes** are gated by default — [BACnet write guard]({{ “/security/bacnet-writes/” | relative_url }}).

130.3 Rule Lab

Rule Lab runs **operator-authored Python** in a sandbox. Restrict integrator accounts; review rules before enable on production sites.

130.4 TLS and LAN security scans

Topic	Page
Certificates (self-signed / site CA)	TLS and certificates({{ “/security/tls-and-certs/”
ZAP + Nmap bench smoke	[ZAP baseline]({{ “/security/zap-baseline/”
Ubuntu host + Tenable/Nessus	Linux host hardening({{ “/security/linux-host-hardening/”
Release test cycle (maintainers)	[Security testing]({{ “/developer/security-testing/”

130.5 Ollama on the edge

Ollama has **no built-in auth**. Bind `OLLAMA_HOST=127.0.0.1:11434` on the host; never expose `:11434` on the OT LAN. Open-FDD bridge reaches Ollama server-side only. Validation: `scripts/security/check_openfdd_exposure.sh`.

131 ZAP baseline (local bench)

132 ZAP baseline (local bench)

Passive OWASP ZAP baseline against the **pentest production stack** on benserver is the standard LAN security smoke before Acme edge deploys.

Full per-revision workflow (pytest → PR → GHCR → Acme): [Developer — security testing cycle]({{ “/developer/security-testing/” | relative_url }}).

Packaged scan scripts (Windows + Mac/Linux): scripts/security/README.md.

132.1 Run

On the Open-FDD host:

```
./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify
```

From a LAN workstation (after verify passes):

```
# Windows — optional integrator auth probe (copy auth.env.local to scan PC; never commit)
.\scripts\security\Run-OpenFddSecurityScan.ps1 -TargetUrl "http://192.168.204.18" `
  -AuthEnvFile "C:\path\to\auth.env.local"
```

The script writes `40-release-gate-summary.txt` with **SECURITY GATE: PASS/FAIL**: - Fails if production JS bundles contain private LAN IPs or bench Niagara defaults - Warns if auth env is missing (`Auth loaded: False`) — pass `-AuthEnvFile` or shell vars `OFDD_INTEGRATOR_USER` / `OFDD_INTEGRATOR_PASSWORD` (values are never logged)

```
# macOS / Linux
./scripts/security/run_openfdd_security_scan.sh --url http://192.168.204.18
```

ZAP target (example): `http://192.168.204.18/` — Caddy on `:80` only; bridge `:8765` stays loopback. TLS bench: see TLS and certificates({{ “/security/tls-and-certs/” | relative_url }}).

132.2 Expected results (HTTP bench mode)

Finding	Severity	Status
CSP `style-src 'unsafe-inline'`	Medium	**Accepted** — Vite/ React inline styles; TODO to nonce/ hash
Missing COEP	Low	**Accepted** — `require-corp` not enabled (breaks some assets)
Port 80 open (Caddy)	Info	**Expected** — intentional LAN entry
Ports 8765, 5173, 8000, 8080 closed	—	**Good**
Port 443 filtered	—	**Expected** in HTTP mode; use `OFDD_CADDY_MODE=tls` for HTTPS bench
Duplicate `Referrer-Policy` / `X-Frame-Options`	Low	**Fixed** — bridge owns headers; Caddy does not duplicate

132.3 Header ownership

Layer	Sets
Bridge (`security_headers.py`)	CSP, `X-Frame-Options: DENY`, `frame-ancestors 'none'`, COOP, CORP, Permissions-Policy, Referrer-Policy, `X-Content-Type-Options`
Caddy HTTP	Reverse proxy only; strips `Server` banners
Caddy TLS	Adds `Strict-Transport-Security` only

132.4 Authenticated scan (later)

Baseline scan hits ~4 public endpoints (`/health`, public agent insight routes). Deep API/dashboard coverage requires ZAP with integrator login — see Authenticated scanning (roadmap)({{ "/security/authenticated-scanning/" | relative_url }}).

132.5 After fixes

Re-run `pentest_production_stack.sh verify`, then the packaged LAN scan from `scripts/security/`. Confirm response headers show **one** `X-Frame-Options: DENY` and no duplicate `Referrer-Policy`.

133 TLS and certificates

134 TLS and certificates

Who generates, stores, and renews TLS material for Open-FDD on OT LANs — and how that ties into ZAP/Nmap bench testing.

Developer release cycle: Security testing cycle({{ "/developer/security-testing/" | relative_url }}).

134.1 Ownership

Environment	Who owns certs	Storage
Local / benserver bench	Developer / maintainer	`workspace/deploy/caddy/certs/` (`cert.pem`, `key.pem`)
Ansible edge (Acme, customer sites)	Site integrator	`/etc/openfdd/caddy/` on the VM (playbook-managed or manual)
Public internet	**Not in scope**	Open-FDD targets OT LAN; use site

Environment	Who owns certs	Storage
		PKI or self-signed

Open-FDD does **not** ship Let’s Encrypt or public CA automation. OT benches use **self-signed** or customer-provided certs.

134.2 Generate bench certs (local)

```
./scripts/setup_caddy_certs.sh
# or: ./scripts/setup_caddy_certs.sh --cn openfdd.local --out workspace/deploy/caddy/certs
```

Then start with TLS mode:

```
OFDD_CADDY_MODE=tls ./scripts/run_local.sh
# or pentest stack with OFDD_CADDY_MODE=tls
```

Caddy serves `:443` and redirects `:80` → HTTPS. Caddy adds **`Strict-Transport-Security`** only; all other security headers remain on the **bridge** ([ZAP baseline]({{ “/security/zap-baseline/” | relative_url }}#header-ownership)).

134.3 ZAP / Nmap with TLS bench

When `OFDD_CADDY_MODE=tls`:

- Nmap: expect **443 open**, 80 may redirect
- ZAP: use `https://<lan-ip>/` (add `-k` trust or import `cert.pem` on the scan workstation)
- PowerShell: `.\Run-OpenFddSecurityScan.ps1 -TargetUrl "https://192.168.204.18"`
- Bash: `./run_openfdd_security_scan.sh --url https://192.168.204.18`

Self-signed certs produce ZAP TLS warnings — expected on OT LAN.

134.4 Renewal

Self-signed certs from `setup_caddy_certs.sh` default to **365 days**. Re-run the script before expiry or bake renewal into site runbooks.

Edge Ansible deploys should document whether the integrator rotates certs manually or via site PKI.

134.5 Enterprise CA on Caddy (edge)

When IT requires **clean Nessus SSL plugins** (no self-signed / untrusted chain), install a site-provided certificate on the edge VM instead of `setup_caddy_certs.sh` output.

Typical layout (Ansible / manual):

```
/etc/openfdd/caddy/cert.pem   # full chain (leaf + intermediates)
/etc/openfdd/caddy/key.pem    # private key (0600, root or caddy
user)
```

Ansible TLS mode (`caddy_mode: tls`) already points Caddy at those paths — see `infra/ansible/templates/Caddyfile.j2`.

Internal enterprise CA workflow:

1. Generate CSR on the edge host or request cert from site PKI for the operator hostname (e.g. `openfdd.acme.local`).
2. Install `cert.pem` + `key.pem` under `/etc/openfdd/caddy/`.
3. Distribute the **CA root** to admin laptops (trust store) so browsers and ZAP scans validate the chain.
4. Monitor expiry:

```
openssl x509 -in /etc/openfdd/caddy/cert.pem -noout -dates
```

Caddyfile snippet (TLS, enterprise cert):

```
:443 {
    tls /etc/openfdd/caddy/cert.pem /etc/openfdd/caddy/key.pem
    header {
        Strict-Transport-Security "max-age=31536000;
includeSubDomains"
    }
    reverse_proxy 127.0.0.1:8765
}
```

Lab/self-signed mode still produces expected **Medium** Tenable findings — document as accepted risk or switch to enterprise CA. See [Tenable remediation]({{ “/security/tenable-remediation/” | relative_url }}).

134.6 Caddy internal CA mode (lab)

`./scripts/setup_caddy_certs.sh` generates a **self-signed** pair for bench TLS. Trust `cert.pem` on scan workstations (`-k` curl, ZAP import) or accept browser warnings. Not suitable when IT mandates trusted chains.

134.7 Secrets

- **Never commit** `key.pem` or production cert bundles
- `workspace/deploy/caddy/certs/` is for local bench only unless your `.gitignore` already excludes generated keys (verify before push)

135 Authenticated scanning (roadmap)

136 Authenticated scanning (roadmap)

[ZAP baseline]({{ “/security/zap-baseline/” | relative_url }}) is **unauthenticated** — it checks public pages, response headers, and a handful of routes reachable without login. That is the right default for every docker image / patch revision smoke on a test LAN.

For **deep** coverage of integrator dashboards, `/api/*` routes, WebSocket sessions, and role-gated writes, plan a separate **authenticated** scan — only on fake/bench stacks with explicit approval.

Release context: Security testing cycle({{ “/developer/security-testing/” | relative_url }}).

136.1 Current state (3.0.x)

Capability	Status
ZAP baseline via `scripts/security/`	**Shipped**
Pentest creds in `workspace/auth.pentest.local`	**Shipped** (host-local, not in git)
ZAP full / active scan automation	**Not shipped** — manual only
ZAP authenticated context (login script + session)	**Roadmap**
CI running ZAP against PR previews	**Not planned** (LAN bench only)

136.2 Why baseline is not enough

After login, the bridge exposes model editing, BACnet/JSON API configuration, Rule Lab, commissioning export, and write-gated commands. Baseline ZAP never exercises:

- Session cookies / JWT persistence
- CSRF or auth bypass on mutating endpoints
- IDOR on site-scoped resources
- Rate limits and audit logging on failed auth

136.3 Planned approach (contributors welcome)

1. **Bench-only target** — same pentest stack; never point active scans at live OT controllers
2. **Login hook** — ZAP `zap-full-scan.py` or Automation Framework with:
   - `POST /api/auth/login` using integrator creds from env (not committed)
   - Regex or JSON extractor for session token / cookie
3. **Scope file** — include `/api/model/*`, `/api/commissioning/*`, dashboard static assets; exclude BACnet UDP and unrelated LAN hosts
4. **Report parity** — extend `scripts/security/` with opt-in `-Authenticated` / `--zap-full` flags and separate report prefix (`22-zap-authenticated-*`)
5. **Gate** — document accepted findings; fail release only on High/Critical regressions

136.4 Manual authenticated ZAP today

1. Start pentest stack; note creds in `workspace/auth.pentest.local`
2. Open ZAP desktop or docker with API port exposed
3. Configure **Context** → **Authentication** → script or form-based login to `/api/auth/login`
4. Run **Spider** then **Active Scan** on `http://<lan-ip>/` only
5. Compare with baseline report in `openfdd-security-report/`

Do **not** run active scans against production Acme without a maintenance window and BACnet write lock.

136.5 Related

- scripts/security/README.md — baseline setup
- Secrets and auth({{ “/security/secrets-auth/” | relative_url }}) — env file layout

- LAN hardening({{ “/security/lan-hardening/” | relative_url }}) — bind and auth modes

137 Secrets and auth

138 Secrets and auth

138.1 Files (gitignored)

File	Contents
`workspace/auth.env.local`	JWT secret, role passwords
`workspace/json_api.env.local`	REST API keys for JSON API `${ENV:VAR}` URLs (e.g. OpenWeather `OPENWEATHER_API_KEY`)
`workspace/data.env.local`	Historian + **audit log retention** (`OFDD_LOG_*`, `OFDD_FEATHER_*`)
`workspace/ollama.env.local`	Ollama URL, model, RAM tier, GPU mode
`workspace/mcp.env.local`	MCP RAG sidecar
`workspace/caddy.env.local`	Reverse proxy TLS/HTTP mode
`workspace/pentest.env.local`	LAN security scan stack
`infra/ansible/secrets/<host>.env.local`	SSH deploy secrets (maintainers)

Copy from `*.example` templates only. Full inventory: Workspace env files({{ “/appendix/workspace-env-files/” | relative_url }}).

138.2 Required variables (production)

Variable	Purpose
`OFDD_AUTH_SECRET`	HMAC signing (32+ chars)
`OFDD_INTEGRATOR_USER` / `PASSWORD`	Rule Lab, bindings
`OFDD_OPERATOR_USER` / `PASSWORD`	Operations

138.3 GHCR pulls

Use read-only registry credentials on edge if images are private; rotate tokens periodically.

138.4 Backup

Back up `workspace/data/` and auth env in secure operator vault — not in git.

Audit JSONL under `workspace/logs/` may contain client IPs and usernames — treat like security logs. See Logging and audit({{ “/ops/logging/” | relative_url }}).

139 BACnet write allowlists

140 BACnet write allowlists

Supervisory writes require:

1. **Feature flag** enabled in environment (off by default).
2. **Allowlist** file at `workspace/bacnet/write_allowlist.json` (copy from `write_allowlist.example.json`). If the flag is on but the allowlist is missing, writes are **denied**.
3. **Integrator** role on API calls.
4. **Audit** log review after commissioning sessions.

Allowlist entries support `device_instance`, `object_identifier`, `property_identifier`, optional `priority_min`/`priority_max`, `value_min`/`value_max`, and `allowed_values` for binary/multistate targets.

Lab-only override: `OFDD_BACNET_WRITE_ALLOW_ANY=1` (audited; not for production edges).

See operator guide: [BACnet write safety]({{ “/bacnet/write-safety/” | relative_url }}).

Never enable blanket write access for agent or integrator automation without human approval per site.

141 Linux host hardening

142 Linux host hardening

Generic **Ubuntu edge VM** guidance for Open-FDD deployments scanned by IT tools (Tenable/Nessus, Qualys, etc.). This complements application-level LAN hardening({{ “/security/lan-hardening/” | relative_url }}) and [TLS]({{ “/security/tls-and-certs/” | relative_url }}).

Nessus-specific finding tables: [Tenable remediation]({{ “/security/tenable-remediation/” | relative_url }}).

142.1 Scope

Layer	Owner	Repo support
Ubuntu kernel, OpenSSH, apt packages	**Site IT / integrator**	`scripts/security/remediate_ubuntu_host.sh`
Docker engine, compose, Caddy	**Integrator**	Ansible + `check_openfdd_exposure.sh`
Open-FDD containers (pip/ pyOpenSSL)	**Open-FDD maintainers**	`docker/python-security-constraints.txt`, GHCR images
Self-signed TLS on OT LAN	**Expected lab finding**	Enterprise CA path in [TLS] ({{ “/security/tls-and-certs/”

142.2 Quick operator workflow

On the edge VM (SSH):

```
        cd ~/open-fdd
        ./scripts/security/check_host_security.sh
        ./scripts/security/check_openfdd_exposure.sh --edge-ip "$
(hostname -I | awk '{print $1}')"
```

Remediate host OS (maintenance window):

```
./scripts/security/remediate_ubuntu_host.sh          # dry-run
sudo ./scripts/security/remediate_ubuntu_host.sh --apply
# if /var/run/reboot-required exists:
sudo reboot
./scripts/security/check_host_security.sh
```

Pull new Open-FDD images after a release that pins container pip/pyOpenSSL:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit <inventory_host>
```

142.3 Ubuntu version and EOL

Supported targets: **Ubuntu 22.04 LTS** or **24.04 LTS** (64-bit). Rebuild hosts still on **16.04** or **18.04** — `apt upgrade` cannot clear an SEoL finding.

Validate a surprising 16.04 Nessus hit:

```
lsb_release -a
cat /etc/os-release
# Scan credential / stale DNS / container banner mismatch?
```

142.4 Kernel and package updates

```
sudo apt update
sudo apt full-upgrade -y
uname -r
test -f /var/run/reboot-required && echo "reboot required"
sudo reboot   # maintenance window
```

Optional status tools:

```
ubuntu-security-status    # or: pro status
apt list --upgradable | grep -i security
```

142.5 Host Python (pip, wheel, pyOpenSSL)

Tenable often reports **host** `pip`, `wheel`, and `pyOpenSSL` versions installed by Ubuntu packages or admin `pip install`. Prefer **apt** on the host:

```
python3 -m pip show pip setuptools wheel pyOpenSSL
sudo apt install --only-upgrade python3-pip python3-openssl python3-wheel
```

Open-FDD **containers** pin toolchain versions at image build — see `docker/python-security-constraints.txt`. Rescan after `docker compose pull`.

142.6 Ollama — no LAN exposure

Ollama has **no built-in authentication** in typical deployments. Treat LAN exposure as **Critical**.

Rule	Implementation
Bind loopback only	`OLLAMA_HOST=127.0.0.1:11434` in systemd (`ollama.service`) or dev bootstrap
Bridge calls server-side	`OFDD_OLLAMA_BASE_URL=http://127.0.0.1:11434` or `host.docker.internal:11434` from containers
Never publish `:11434` on `0.0.0.0`	Compose uses `127.0.0.1:11434:11434` when Docker Ollama is enabled

Validation:

```
        ss -tulpn | grep 11434
        curl -sf http://127.0.0.1:11434/api/tags | head
        curl -sf --max-time 3 http://<edge-lan-ip>:11434/api/tags
# must FAIL from another host
```

Firewall (if management host needs GPU Ollama — rare):

```
        # Example: block LAN → 11434; allow loopback only (adjust
interface names)
        sudo ufw deny 11434/tcp
        sudo ufw allow from 127.0.0.0/8 to any port 11434 proto tcp
```

142.7 OpenSSH Terrapin (CVE-2023-48795)

Apply Ubuntu security updates first (`openssh-server`, `openssh-client`). Verify:

```
        ssh -V
        sudo sshd -T | grep -Ei '^(ciphers|macs|kexalgorithms)'
```

Manual algorithm hardening is **optional** and can lock out clients — document rollback (`sshd_config.bak`) before changes. Do not automate cipher stripping in repo scripts.

142.8 ICMP timestamp (Low)

Nessus may report ICMP timestamp disclosure. Optional hardening at host firewall or upstream network — low priority for isolated OT VLANs.

Example (nftables — site-specific):

```
        # Block ICMP timestamp / type 13 — verify with IT before applying
        sudo nft add rule inet filter input icmp type timestamp-request
drop
```

142.9 TLS and certificate scans

Self-signed or internal CA certificates produce **Medium** Nessus SSL findings (untrusted, self-signed, expiry). That is **expected** for lab/OT encryption-in-transit. For clean SSL plugin results, install a **site enterprise CA** certificate on Caddy — TLS and certificates({{ “/security/tls-and-certs/” | relative_url }} #enterprise-ca-production).

142.10 Maintainer security audits (CI parity)

```
        ./scripts/security/run_maintainer_audits.sh
```

Matches GitHub Actions `operator-bridge-security` job where possible.

143 Tenable / Nessus remediation

144 Tenable / Nessus remediation

Remediation plan for **IT/Tenable.io Nessus** findings on an Open-FDD **OT/LAN edge** deployment (e.g. Acme GL36 lab VM). Open-FDD runs behind a building firewall on Ubuntu with Docker Compose, Caddy, and optional host Ollama.

Related: Linux host hardening({{ “/security/linux-host-hardening/” | relative_url }}), TLS and certificates({{ “/security/tls-and-certs/” | relative_url }}), LAN hardening({{ “/security/lan-hardening/” | relative_url }}), Security testing cycle({{ “/developer/security-testing/” | relative_url }}).

144.1 Prioritized action plan

Priority	Action	Owner
P0	Confirm Ubuntu is **not** 16.04/18.04; rebuild if EOL	IT / integrator
P0	Bind Ollama to **127.0.0.1** only; verify LAN cannot reach `:11434`	Integrator
P0	`apt full-upgrade` + **reboot** for kernel USNs (8254, 8278, 8373)	IT / integrator
P1	Upgrade host `pip`/`wheel`/`pyOpenSSL` via apt or documented pip path	IT / integrator
P1	Pull GHCR images built with `docker/python-security-constraints.txt`	Integrator
P2	OpenSSH packages current; review Terrapin after patch	IT
P2	TLS: accept self-signed findings **or** install enterprise CA cert	IT decision
P3	ICMP timestamp: optional firewall rule	Network / IT

144.2 Finding classification table

Nessus finding	Severity	Class	Remediation
	Critical		`lsb_release -a` — rebuild VM on 24.04 LTS if truly

Nessus finding	Severity	Class	Remediation
Canonical Ubuntu Linux SEoL (16.04.x)		Host OS / **false positive?**	16.04; if host is 22.04/24.04, check scan target IP, credentials, stale asset record
Ollama Unauthenticated Access	Critical	Deployment config	`OLLAMA_HOST=127.0.0.1:11434`; no `0.0.0.0:11434`; firewall deny; see Ollama
Ubuntu kernel USN-8254-1	Critical	Host OS	`sudo apt update && sudo apt full-upgrade -y && sudo reboot`
pyOpenSSL 22.0.x < 26.0.0 Buffer Overflow	Critical	Host pip **and** container	Host: `apt upgrade python3-openssl` or pip ≥26; Container: GHCR image with constraints file
Ubuntu pip USN-8344-1	High	Host OS + container build	`apt full-upgrade`; image build upgrades `pip>=26`
SSL Certificate Cannot Be Trusted	Medium	**Accepted risk** (lab) / deployment	Self-signed Caddy — use enterprise CA for clean scan
SSL Self-Signed Certificate	Medium	**Accepted risk** (lab)	Same as above
SSL Certificate Expiry	Medium	Deployment / monitoring	`openssl x509 -dates`; renew `setup_caddy_certs.sh` or site PKI
SSH Terrapin (CVE-2023-48795)	Medium	Host OS	Upgrade `openssh-server`; validate `ssh -V`
Ubuntu wheel USN-8221-1	Medium	Host OS	`apt full-upgrade`
Ubuntu kernel USN-8278-1, USN-8373-1	Medium	Host OS	`apt full-upgrade` + reboot
pyOpenSSL 0.14.x Security Bypass	Medium	Host + container	Same as pyOpenSSL Critical — upgrade to ≥26
ICMP Timestamp Request	Low	Network / optional host	Firewall drop; often accepted on isolated OT VLAN

Class legend: (1) repo-owned fix · (2) Docker base/image · (3) host OS · (4) deployment/config · (5) accepted risk · (6) validate false positive

144.3 Operator commands (edge VM)

144.3.1 Validate (read-only)

```
        cd ~/open-fdd
        ./scripts/security/check_host_security.sh
        ./scripts/security/check_openfdd_exposure.sh --edge-ip "$
(hostname -I | awk '{print $1}')"
        lsb_release -a
        uname -r
        test -f /var/run/reboot-required && cat /var/run/reboot-
required.pkgs
        python3 -m pip show pip setuptools wheel pyOpenSSL 2>/dev/null |
grep -E '^(Name|Version):'
        ss -tulpn | grep -E ':80|:443|:8765|:11434'
        curl -sf http://127.0.0.1:11434/api/tags | head -3
```

From a **different LAN workstation** (must fail):

```
        curl -sf --max-time 3 http://<edge-ip>:11434/api/tags
```

144.3.2 Remediate host (maintenance window)

```
        ./scripts/security/remediate_ubuntu_host.sh              # dry-run
        ./scripts/security/remediate_ubuntu_host.sh --apply
        sudo reboot    # if reboot-required
        ./scripts/security/check_host_security.sh
```

144.3.3 Remediate Open-FDD containers

```
        OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit
<inventory_host>
        docker compose pull && docker compose up -d
```

144.3.4 Rescan

After reboot and image pull, request IT **Nessus rescan** of the edge IP. Compare plugin output to the table above.

144.4 Ollama unauthenticated access (Critical)

Not an Open-FDD code bug — Ollama's API is unauthenticated by design. Risk is **network exposure**.

Repo defaults (3.0.5+):

- Systemd: `Environment=OLLAMA_HOST=127.0.0.1:11434` in `infra/ansible/templates/ollama.service.j2`
- Compose Ollama (optional profile): `127.0.0.1:11434:11434` only
- Acme-style edge: host Ollama + bridge via `host.docker.internal:11434` (container → host loopback path)
- `post_deploy_check.sh` warns if `:11434` listens on `0.0.0.0`

Re-apply Ansible Ollama unit after upgrade:

```
        ansible-playbook -i inventory.yml ollama_bootstrap.yml --limit
<host> -e enable_ollama=true
```

144.5 pyOpenSSL and pip in containers

Docker build installs `docker/python-security-constraints.txt` before application requirements:

- `pip>=26.0`, `setuptools>=75.0`, `wheel>=0.45`
- `pyOpenSSL>=26.0.0`, `cryptography>=44.0.0`

CI `pip-audit` runs against bridge + bacnet requirements with the same constraints.

144.6 TLS — expected vs production

Mode	Nessus SSL plugins	When to use
Caddy HTTP `:80`	No TLS plugins	Dev / HTTP-only VLAN
Caddy self-signed : `443`	Medium: untrusted, self-signed, expiry	OT LAN encryption, lab
Enterprise CA cert on Caddy	Clean SSL plugins (if chain trusted)	IT mandates clean Nessus

See [TLS — enterprise CA]({{ “/security/tls-and-certs/” | relative_url }}#enterprise-ca-on-caddy-edge).

144.7 Residual findings after remediation

Even with a patched host and loopback-only Ollama, IT may still see:

- **SSL** Medium on self-signed TLS (accepted for lab; document in risk register)
- **ICMP** Low (optional network block)
- **Container** plugins if Nessus scans Docker overlay networks — clarify scan scope with IT

144.8 Notes for IT / security team

1. Scan the **edge VM IP**, not individual container IPs, unless container CVE correlation is required.
2. Open-FDD **bridge :8765** should be **loopback-only**; LAN entry is **Caddy :80/:443**.
3. **BACnet UDP 47808** on commission is expected OT traffic — out of scope for web/plugin remediation.
4. Do not expose Ollama **11434** to the building LAN; agent chat is server-side only.
5. Kernel CVEs require **host reboot** — schedule with operations.
6. For authenticated web/API depth, see roadmap [Authenticated scanning]({{ “/security/authenticated-scanning/” | relative_url }}).

144.9 Repo-owned fixes (this document’s release)

Item	Location
Container pip/ pyOpenSSL pins	`docker/python-security-constraints.txt`, `docker/Dockerfile`
Ollama loopback systemd	`infra/ansible/templates/ollama.service.j2`
Host/exposure check scripts	`scripts/security/check_*.sh`, `remediate_ubuntu_host.sh`
Post-deploy Ollama bind warn	`infra/ansible/scripts/post_deploy_check.sh`
CI pip-audit + constraints	`.github/workflows/ci.yml`

145 AI agents

146 AI agents and MCP

- AI agent context({{ “/ai-agent-context/” | relative_url }}) — canonical map for deploy, drivers, model, rules, validation (**start here**)
- MCP server({{ “/ai/mcp-server/” | relative_url }}) — FastMCP tools over the operator bridge
- Skills under `skills/openfdd-*-agent/` and `skills/openfdd-mcp-server/`
- Operator memory: `workspace/MEMORY.md` (see `workspace/MEMORY.md.example`)

Internal runbooks (not primary nav): [agent-skills/]({{ “/agent-skills/bench-validation-agent/” | relative_url }}).

147 Security

148 Security

LAN deployment security for the Operator Bridge and BACnet write guard. Open-FDD is for **trusted edge / OT LAN** — not direct public internet exposure.

Page	Topic
LAN hardening({{ “/security/lan-hardening/”	relative_url }})
ZAP baseline (local bench)({{ “/security/zap-baseline/”	relative_url }})
TLS and certificates({{ “/security/tls-and-certs/”	relative_url }})
Linux host hardening({{ “/security/linux-host-hardening/”	relative_url }})
Tenable / Nessus remediation({{ “/security/tenable-remediation/”	relative_url }})
Authenticated scanning (roadmap)({{ “/security/authenticated-scanning/”	relative_url }})

Page	Topic
Secrets and auth({{ “/security/secrets-auth/”	relative_url }})
BACnet write allowlists({{ “/security/bacnet-writes/”	relative_url }})
[Release & LAN scan cycle]({{ “/developer/security-testing/”	relative_url }})
Logging and audit({{ “/ops/logging/”	relative_url }})

149 API routes

150 API routes

REST API served by **`openfdd-bridge`** (default `http://127.0.0.1:8765`). Production: Caddy on `:80` proxies to the bridge.

OpenAPI: `GET /docs` and `GET /redoc` when the bridge runs.

Auth: JWT via `POST /api/auth/login`. Most routes require `Authorization: Bearer <token>`. Roles: `viewer`, `operator`, `integrator`, `commission`, `agent`, `admin`. BACnet writes require commission role + write safety({{ “/bacnet/write-safety/” | relative_url }}). Dev: `OFDD_AUTH_DISABLED` on loopback only.

WebSocket: `WS /ws/dashboard` — `POST /api/auth/ws-ticket`, then pass the ticket via `Sec-WebSocket-Protocol: ofdd.ws, <ticket>`. Query `?ticket=` is dev-only when `OFDD_WS_ALLOW_QUERY_TICKET=1`. Tickets are short-lived and not interchangeable with the Bearer token.

150.1 Health & audit

Method	Path	Role	Description
GET	`/health`	public	Minimal liveness (`ok`, `service`, `version`, `auth_required`)
GET	`/health/stack`	auth	Stack traffic-light (verbose URLs/bind with `OFDD_DEBUG_DIAGNOSTICS`)

Method	Path	Role	Description
POST	`/api/auth/ws-ticket`	auth	Short-lived WebSocket ticket
GET	`/api/audit/summary`	auth	Audit counters
GET	`/api/audit/events`	auth	Audit log (query params)
GET	`/api/audit/errors`	auth	Recent API errors

*Exact public set depends on `OFDD_AUTH_DISABLED` and middleware.

150.2 Auth

Method	Path	Description
POST	`/api/auth/login`	Username/password → JWT
GET	`/api/auth/me`	Current user
GET	`/api/auth/status`	Auth mode / lockout hints

150.3 Rules & FDD

Method	Path	Role	Description
GET	`/api/rules/saved`	user	Rule index
POST	`/api/rules/save`	operator+	Create/update rule metadata
GET	`/api/rules/saved/{id}/source`	user	Python source
PUT	`/api/rules/saved/{id}/source`	operator+	Write `.py` file
GET	`/api/rules/assignments`	user	Point bindings
POST	`/api/rules/bind`	integrator+	Bind rule to point
POST	`/api/rules/bindings`	integrator+	Bulk bindings

Method	Path	Role	Description
POST	`/api/rules/batch`	operator+	Run all enabled rules → `fdd_results.json`
GET/POST	`/api/rules/drafts`	user	Draft rules

150.4 Rule Lab playground

Method	Path	Description
POST	`/api/playground/lint`	AST lint Python rule
POST	`/api/playground/test-rule`	Run `apply_faults_arrow()` on feather PyArrow window
POST	`/api/playground/run-script`	Execute script mode
GET	`/api/playground/sample-frame`	Sample DataFrame for editor

150.5 BRICK / data model

Method	Path	Role	Description
GET	`/api/model/sites`	user	Site list
GET	`/api/model/tree`	user	Equipment tree (SPARQL)
GET	`/api/model/graph`	user	RDF graph fragment
GET	`/api/model/scope`	user	Scoped query
GET	`/api/model/export`	user	Export TTL/JSON
GET	`/api/model/commissioning-export`	user	Full commissioning bundle (points + `fdd_rule_ids` + `fdd_rules`)
POST		integrator+	

Method	Path	Role	Description
	`/api/model/commissioning-import`		Bulk import; validate in UI dry-run first
GET	`/api/model/health`	user	Model point/ equipment counts
GET	`/api/model/ttl`	user	TTL metadata
POST	`/api/model/sync-ttl`	integrator+	Regenerate TTL from JSON
POST	`/api/model/import`	integrator+	Import model payload
POST	`/api/model/sites`	integrator+	Create site
GET/POST	`/api/model/bacnet-sync`	integrator+	BACnet ↔ model sync state
GET	`/api/model/fdd-query-presets`	user	Saved FDD query presets
GET	`/api/model/fdd-coverage`	user	Rule coverage summary
POST	`/api/model/sparql`	user	SPARQL query (scoped)

150.6 Rule Lab (dashboard)

Method	Path	Role	Description
POST	`/api/rules/lab/validate`	operator+	Validate rule source
POST	`/api/rules/lab/preview`	operator+	Preview rule on historian window
POST	`/api/rules/lab/compare`	operator+	Compare two rule versions
POST	`/api/rules/lab/lint-sql`	operator+	DataFusion SQL safety lint
POST	`/api/rules/lab/save`	operator+	Save Rule Lab draft

Mounted at `/api/rules/lab` — see OpenAPI `/docs`.

150.7 Bench validation (read-only)

Method	Path	Role	Description
GET	`/api/bench/health`	operator+	Bench harness health
GET	`/api/bench/mapping`	operator+	Semantic cross-source mapping
GET	`/api/bench/poll-status`	operator+	BACnet + Niagara poll heartbeat
POST	`/api/bench/validate/bacnet-vs-niagara`	operator+	Dual-source equivalence check
POST	`/api/bench/long-fdd/evaluate`	operator+	Long FDD smoke evaluate (PyArrow/ DataFusion)

See Bench 5007 long FDD smoke({{ “/operations/bench-5007-long-fdd-smoke/” | relative_url }}).

150.8 Niagara baskStream

Method	Path	Role	Description
GET	`/api/niagara/health`	read	Connector health
GET	`/api/niagara/stations`	read	Configured stations
POST	`/api/niagara/stations/{id}/discover`	read	Discover points
POST	`/api/niagara/stations/{id}/poll/once`	poll	One poll cycle

Method	Path	Role	Description
GET	`/api/niagara/stations/{id}/poll/status`	read	Poll worker status
GET	`/api/niagara/driver/tree`	read	Driver commissioning tree

Read-only by default. See Niagara baskStream connector({{ “/niagara-baskstream-connector/” | relative_url }}).

150.9 BACnet

Method	Path	Role	Description
GET	`/config/bacnet`	read	Commission URL / config
GET	`/api/bacnet/commission/status`	read	Commission agent health
GET	`/api/bacnet/server/points`	read	Server point table
GET	`/api/bacnet/inventory`	read	Discovered inventory
POST	`/api/bacnet/whois`	commission	Who-Is
POST	`/api/bacnet/discover`	commission	Discover devices
POST	`/api/bacnet/read`	commission	Read property
POST	`/api/bacnet/read-multiple`	commission	RPM read
POST	`/api/bacnet/write`	write	Supervisory write (gated)

Method	Path	Role	Description
POST	`/api/bacnet/import-to-model`	integrator	Inventory → BRICK
POST	`/api/bacnet/driver/sync-discovery`	commission	Sync driver tree
POST	`/api/bacnet/poll/once`	commission	Trigger poll cycle
GET	`/api/bacnet/poll/status`	read	Poll worker status
POST	`/ingest/bacnet`	read	Ingest samples into feather

Bind address: `BACNET_BIND` in commission env — see [BACnet network setup]({{ “/bacnet/network-setup/” | relative_url }}).
Operator guide: Discover and read({{ “/bacnet/discover-read/” | relative_url }}).

PATCH | `/api/bacnet/driver/point` | commission | Enable poll + interval on point |
PATCH | `/api/bacnet/driver/device` | commission | Enable poll for all points on device |
PATCH | `/api/bacnet/driver/device/remap` | commission | Change instance and/or address |
DELETE | `/api/bacnet/driver/point/{point_id}` | commission | Remove point from registry |
DELETE | `/api/bacnet/driver/device/{device_instance}` | commission | Remove device |
DELETE | `/api/bacnet/driver/registry` | commission | Clear CSVs + model sync |

150.10 Faults (check-engine)

Method	Path	Description
GET	`/api/faults/catalog`	Fixed fault codes
GET	`/api/faults/status`	GREEN/YELLOW/RED aggregate
GET	`/api/faults/tree`	Fault tree by equipment
GET		Code metadata

Method	Path	Description
	`/api/faults/code/{code}`	

150.11 Timeseries & sites

Method	Path	Description
GET	`/api/timeseries/sites`	Feather sites
GET	`/api/timeseries/series`	Series metadata
GET	`/api/timeseries/plot`	Plot payload
GET	`/api/timeseries/readings`	Raw readings
GET	`/api/sites`	Site list
GET	`/api/sites/{site_id}/frame`	Wide frame for site

150.12 Building & host

Method	Path	Description
GET	`/api/building/status`	Building summary
GET	`/api/building/alerts`	Active alerts
GET	`/api/host/stats`	CPU/mem/disk

150.13 Local agent (Ollama)

Prefix **`/openfdd-agent`**. Role **`agent`** for tool execution.

Method	Path	Description
GET	`/openfdd-agent/context`	Composed prompt context
GET	`/openfdd-agent/tools`	Tool manifest
POST	`/openfdd-agent/tool`	Execute tool (e.g. `rules.save`)
GET	`/openfdd-agent/ollama/health`	Ollama reachability
GET		

Method	Path	Description
	`/openfdd-agent/building-insight`	Check-engine narrative context
GET	`/openfdd-agent/zone-temps`	Zone temperature insight
POST	`/openfdd-agent/chat`	Chat completion (catalog-bound)

Optional host Ollama for building insight — not required for core FDD. Configure via compose profile or host service; see [Architecture overview]({{ “/architecture/overview/” | relative_url }}).

150.14 Modbus

Method	Path	Role	Description
GET	`/api/modbus/driver/tree`	operator+	Commissioning tree
GET	`/api/modbus/poll/status`	operator+	Poll worker status
POST	`/api/modbus/poll/once`	integrator+	Force poll cycle
POST	`/api/modbus/read_registers`	operator+	One-shot register read
POST	`/api/modbus/read_and_store`	operator+	Read + CSV + feather (`source=modbus`)
POST	`/api/modbus/refresh`	operator+	Re-read one register
PATCH	`/api/modbus/register/poll`	integrator+	Enable poll interval

See Driver framework({{ “/drivers/” | relative_url }}).

150.15 JSON API (HTTP/HTTPS)

Method	Path	Role	Description
GET	`/api/json-api/env/status`	operator+	

Method	Path	Role	Description
			Env var configured flags (no secret values)
POST	`/api/json-api/presets/openweather`	integrator+	Register 3-point OpenWeather bundle + optional poll
GET	`/api/json-api/driver/tree`	operator+	Endpoints grouped by host
GET	`/api/json-api/poll/status`	operator+	Poll worker status
POST	`/api/json-api/poll/once`	integrator+	Force poll cycle
POST	`/api/json-api/request`	operator+	One-shot GET/POST (no store)
POST	`/api/json-api/read_and_store`	operator+	Request + CSV + feather (`source=json_api`)
POST	`/api/json-api/refresh`	operator+	Re-fetch endpoint
PATCH	`/api/json-api/endpoint/poll`	integrator+	Enable poll interval
DELETE	`/api/json-api/endpoint/{point_id}`	integrator+	Remove endpoint

Outbound auth: `auth_type` = `none` | `bearer` | `basic`; optional `verify_tls` (default true). See JSON API driver({{ “/drivers/json-api/” | relative_url }}).

150.16 PyPI package (separate)

Library-only HTTP is **not** bundled. See Python package({{ “/appendix/python-package/” | relative_url }}) for `open_fdd.arrow_runtime` (Arrow rules offline).

150.17 Errors

JSON error bodies; stack traces hidden unless `OFDD_DEBUG_TRACEBACKS=1`. See LAN hardening({{ “/security/lan-hardening/” | relative_url }}).

151 Python package

152 Python package (`open-fdd`)

Install from PyPI (embeddable FDD runtime — not the Docker edge stack):

```
        pip install open-fdd
        pip install "open-fdd[analytics]"     # optional NumPy helpers
        pip install "open-fdd[ml]"            # optional sklearn for
offline experiments
```

152.1 Modules

Module	Purpose
`open_fdd.arrow_runtime`	**Default** — `apply_faults_arrow` on PyArrow Tables, cookbook masks, column maps
`open_fdd.playground`	Rule Lab lint/compile helpers for Arrow rules

Retired in 3.0.1+: `open_fdd.engine` (YAML/pandas `RuleRunner`) is **not** shipped on PyPI. Use Operator Bridge Rule Lab or import rules from `workspace/data/rules_py/`.

152.2 When to use package-only

Scenario	Use
Lint/test Arrow rules offline	`open_fdd.arrow_runtime` + `open_fdd.playground`
Graph ML experiments (Layer B)	`open-fdd[ml]` in `experiments/` (see issue #211)
Full building operator stack	**Docker** Operator Bridge

152.3 Versioning

Publish: git tag `vX.Y.Z` (or legacy `open-fdd-vX.Y.Z`) → GitHub Actions **Publish open-fdd to PyPI**. See Release process({{ “/developer/release-process/” | relative_url }}).

```
import open_fdd
print(open_fdd.__version__)
```

152.4 Offline Arrow example

```
import pyarrow as pa
import pyarrow.compute as pc
from open_fdd.arrow_runtime import run_arrow_rule

code = '''
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["SAT"], float(cfg["high"]))
'''
table = pa.table({"SAT": [70.0, 90.0, 88.0]})
result = run_arrow_rule(code, table, {"high": 85})
print(result.flagged_count)
```

API surface: `open_fdd/arrow_runtime` docstrings and [Arrow recipes]({{ “/rule-cookbook/arrow-recipes/” | relative_url }}).

153 Configuration reference

154 Configuration reference

154.1 Bridge / auth

Variable	Default	Notes
`OFDD_BRIDGE_HOST`	`127.0.0.1`	`0.0.0.0` for LAN
`OFDD_AUTH_SECRET`	—	Required for LAN
`OFDD_AUTH_DISABLED`	`0`	Localhost dev only
`OFDD_INSECURE_LAN_DEV`	`0`	Lab only with auth disabled

154.2 BACnet

Variable	Notes
`BACNET_BIND`	`ip/mask:47808` in `commission.env`
`OFDD_BACNET_WRITE_ENABLED`	

Variable	Notes
	Default off — site-specific name may vary

154.3 Docker edge

Variable	Notes
`OPENFDD_IMAGE_TAG`	GHCR tag for Ansible deploy
`openfdd_docker_pull_from_ghcr`	Ansible host_var

Rule Lab Python rules use `rules_store.json` + `workspace/data/rules_py/*.py`. Legacy YAML schema (`docs/config_schema.json`) is archived — not used by Operator Bridge 3.0.1+.

155 Glossary

156 Glossary

Term	Definition
Operator Bridge	FastAPI service serving API + static dashboard
Rule Lab	UI and APIs for Arrow `apply_faults_arrow()` rules
Feather store	On-disk columnar historian under `workspace/data/feather_store/`
fdd_input	Brick tag linking a point to a rule binding role
Check-engine	Building-level GREEN/YELLOW/RED fault summary
Commission	BACnet discovery/read (and optional write) agent container
Poll driver	Scheduled BACnet RPM reads into historian
GHCR	GitHub Container Registry — `ghcr.io/bbartling/openfdd-*`

157 Workspace env files

158 Workspace env files

All live under `workspace/`. Copy from `*.example` templates; committed `*.local` files are gitignored.

File	Loaded by	Purpose
`auth.env.local`	`run_local.sh`, Docker compose, Ansible	JWT secret (`OFDD_AUTH_SECRET`), integrator/operator/ agent passwords, `OFDD_AUTH_DISABLED` for localhost dev only
`data.env.local`	`run_local.sh`, bridge, feather maintain	Feather historian caps (`OFDD_FEATHER_*`), audit/ error log rotation (`OFDD_LOG_*`), BACnet discovery mutation toggle
`json_api.env.local`	bridge, Ansible edge sync	API keys for JSON API driver `${ENV:VAR}` substitution (e.g. `OPENWEATHER_API_KEY`)
`ollama.env.local`	`run_local.sh`, bridge	Ollama base URL, model name, RAM tier, GPU mode, timeouts — see Ollama and analytics({{ “/operator-bridge/ollama-analytics/”
`mcp.env.local`	`run_local.sh`, MCP sidecar	Enable MCP RAG REST (`OFDD_MCP_*`), index path
`caddy.env.local`	`run_local.sh`, Ansible edge	Reverse proxy mode (`http` / `tls` / `off`), TLS CN, port overrides
`pentest.env.local`	`pentest_production_stack.sh`	Production-like LAN scan stack: CORS, auth, passive BACnet defaults

158.1 Ansible / edge extras

Path	Purpose
`infra/ansible/secrets/<host>.env.local`	SSH deploy credentials, GHCR pull tokens, site-specific BACnet bind — maintainers only

158.2 Related docs

- Secrets and auth({{ “/security/secrets-auth/” | relative_url }}) — required production variables
- Logging and audit({{ “/ops/logging/” | relative_url }}) — `OFDD_LOG_*` detail
- JSON API driver({{ “/drivers/json-api/” | relative_url }}) — `${ENV:VAR}` pattern
- Configuration reference({{ “/appendix/configuration/” | relative_url }}) — bridge runtime flags not in env files

159 Appendix

160 Appendix

Reference material for integrators and library users.

Page	Content
API routes({{ “/appendix/bridge_api/”	relative_url }})
Python package({{ “/appendix/python-package/”	relative_url }})
Configuration reference({{ “/appendix/configuration/”	relative_url }})
Workspace env files({{ “/appendix/workspace-env-files/”	relative_url }})
Glossary({{ “/appendix/glossary/”	relative_url }})

161 GL36 lab site (Acme)

162 GL36 lab site (Acme)

> **Lab note only.** This documents a maintainer **example edge host** (site `acme`, building `vm-bbartling`) used for Guideline 36 supervisory FDD development. Replace inventory host names, IPs, and BACnet ranges for your site.

Reference for **ASHRAE Guideline 36** supervisory monitoring on a live lab edge. Rule code lives in `workspace/data/rules_py/`; commissioning uses **Arrow constants** (no `config.json`).

Trim & respond Niagara reference: README_TRIM_RESPOND.

162.1 BACnet poll scope

Device filter (`infra/ansible/scripts/acme_trim_devices.sh`):

Range	Equipment
1–100	JCI VMA/VAV
1100	RTU-01 (AHU)
1002	Hot water plant
11000–13000	Trane UC210 VAV

Poll interval: **60 s**. Commission from control machine:

```
cd infra/ansible
POLL_INTERVAL=60 ./scripts/acme_commission_gl36.sh
```

Points land in `edge_backup/local/acme/vm-bbartling/points.csv` for Ansible sync.

162.2 Example FDD rules (Arrow)

162.2.1 Duct static pressure high (AHU)

```
"""AHU duct static high — GL36 trim advisory."""

import pyarrow.compute as pc

VALUE_COLUMN = "duct-static"
HIGH_INWC = 1.20
LOOKBACK_HOURS = 3
```

```
        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin, tmax = pc.min(ts).as_py(), pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            return pc.greater(pc.cast(table[VALUE_COLUMN], "float64"),
HIGH_INWC)
```

162.2.2 SAT too cold vs setpoint

```
        """AHU SAT too cold vs setpoint."""

        import pyarrow.compute as pc

        SAT_COLUMN = "sa-t"
        SP_COLUMN = "sa-t-sp"
        COLD_DELTA_F = 5.0


        def apply_faults_arrow(table, cfg, context=None):
            sat = pc.cast(table[SAT_COLUMN], "float64")
            sp = pc.cast(table[SP_COLUMN], "float64")
            return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

162.2.3 Site rule stubs in the repo

Rule file	Use
`bench_stat_zn-t_flatline_1h.py`	Zone temp flatline — bind `ZN-T` per VAV
`acme_duct_static_flatline_1h_gl36_duct_t_r_input.py`	RTU duct static flatline
`oat_vs_web_spread_1h.py`	Local OAT vs OpenWeather JSON API

162.2.4 Local OA-T vs web weather (JSON API)

1. Control machine: `workspace/json_api.env.local` with `OPENWEATHER_API_KEY`
2. JSON API tab → **OpenWeatherMap** preset → **Register** (default poll **30 min**)
3. Bind BACnet local OAT in the model (`oa-t` historian column)
4. Enable `oat_vs_web_spread_1h.py` in Rule Lab — flags when `|oa-t − web-oat-t| > 8 °F`

162.3 Deploy / upgrade (Ansible, this lab only)

```
        cd infra/ansible
        OPENFDD_IMAGE_TAG=latest ./deploy.sh docker --limit
<inventory_host> -e enable_bacnet_poll_driver=true
```

Full UI + image upgrade from control machine:

```
        OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
        infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full
```

Do **not** rsync dev `auth.env.local` to the edge (`deploy_sync_auth: false` in private host_vars). Regenerate credentials on the edge host if login fails after upgrade.

162.4 AI-assisted data modeling (lab workflow)

1. BACnet poll running (~340 GL36 points @ 60 s on this lab host).
2. **Model & assignments** → export commissioning JSON for LLM-assisted binding.
3. Import returned JSON via commissioning import.
4. Pin FDD rules on `ZN-T`, `DA-T`, `duct-static`, `OAT`, etc. — export shows readable names in `fdd_rules_linked`.

Model seed: `workspace/data/acme_gl36_model.json`.

163 Examples & lab notes

164 Examples & lab notes

Site-specific commissioning notes, GL36 examples, and maintainer lab workflows. These pages are **not** required for a generic edge deployment.

Page	Description
GL36 lab site (Acme)({{ “/examples/acme-gl36-lab/” | relative_url }})	

For production IT operators, start with Quick Start({{ “/quick-start/” | relative_url }}) and Operations({{ “/ops/” | relative_url }}).

165 Arrow-native runtime

166 Arrow-native runtime

Open-FDD 3.0 executes Rule Lab rules on **PyArrow tables** end to end.

```
feather_store.read_site_table()
  → fdd_runner / playground.run_arrow_table()
  → apply_faults_arrow(table, cfg, context)
  → pyarrow.compute masks
```

166.1 Authoring contract

```python
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["zone_temp"], cfg["max_zone_temp"])
```

Cookbook helpers: `open_fdd.arrow_runtime.cookbook` (flatline, spread, OOB, after-hours fan).

Script-mode analytics rules receive `table` (PyArrow) and `cfg` in the sandbox — not pandas DataFrames.

166.2 PyArrow-only policy (edge + Rule Lab)

Allowed	Forbidden
`pyarrow` (`pa`), `pyarrow.compute` (`pc`)	`import pandas`, `pd.DataFrame`, `to_pandas()`
`open_fdd.arrow_runtime.cookbook` helpers	`import numpy` for rule logic
Script mode globals: `table`, `cfg`, `out`	Legacy `df` DataFrame scripts

Rule Lab **lint** (`POST /api/playground/lint`) and rule **save** reject pandas patterns with agent-facing errors, e.g. *use `table` (PyArrow), not `df`*.

Fault rules: `apply_faults_arrow(table, cfg, context=None)`. Analytics scripts: top-level code on `table`/`cfg`, set `out = {"events": [...], "metrics": {...}}`.

166.3 Package layout

Module	Role
`open_fdd.arrow_runtime.backend`	Execute rule code, batch chunks
`open_fdd.arrow_runtime.cookbook`	Shared fault masks
`open_fdd.arrow_runtime.windows`	Rolling min/max, consecutive-true
`open_fdd.playground.arrow_templates`	Rule Lab starter templates

167 Release process

168 Release process

Open-FDD has **two publish paths**:

Artifact	Trigger	Registry
PyPI `open-fdd`		

Artifact	Trigger	Registry
	Git tag `vX.Y.Z` (or legacy `open-fdd-vX.Y.Z`)	pypi.org/project/open-fdd
Docker edge stack	Same release tag (or manual workflow)	`ghcr.io/bbartling/openfdd-*`

Do **not** publish from ordinary branch pushes. Merge to `master`, bump version, tag, push tag.

168.1 1. Prepare version

```
git checkout master
git pull --ff-only
# Edit pyproject.toml + open_fdd/__init__.py → same X.Y.Z
python scripts/release/check_version.py
```

168.2 2. Merge and tag

```
git tag -a v3.0.30 -m "open-fdd 3.0.30"
git push origin v3.0.30
```

Tag push runs:

- `.github/workflows/publish-open-fdd.yml` — build, test, `twine check`, PyPI via **Trusted Publishing**
- `.github/workflows/docker-publish.yml` — GHCR images with tags `3.0.30`, `3.0`, `latest`

Legacy tag `open-fdd-v3.0.30` is still accepted during transition.

168.3 3. PyPI Trusted Publishing setup (one-time)

On pypi.org → project **open-fdd** → Publishing → **Add a new trusted publisher**:

Field	Value
PyPI project name	`open-fdd`
Owner	`bbartling`
Repository	`open-fdd`
Workflow name	`Publish open-fdd to PyPI`

Field	Value
Environment	`pypi` (optional; enables approval gate)

No long-lived API token in the repo.

168.4 4. Verify release

```
pip install open-fdd==3.0.30
python -c "import open_fdd; print(open_fdd.__version__)"

docker pull ghcr.io/bbartling/openfdd-bridge:3.0.30
docker manifest inspect ghcr.io/bbartling/openfdd-bridge:3.0.30
```

168.5 5. Manual workflow_dispatch

- **PyPI:** `Publish open-fdd to PyPI` with `dry_run=true` (default) — build/test only
- **Docker:** `Publish Docker images to GHCR` — publishes `pyproject.toml` version + minor alias + `latest` (optional `image_tag` override)

168.6 Package vs Docker

See PyPI package({{ “/developer/pypi-package/” | relative_url }}) and [Docker images]({{ “/ops/docker-images/” | relative_url }}).

169 PyPI package

170 PyPI package (`open-fdd`)

170.1 What it is

Embeddable **Arrow-native FDD compute** for consultants, cloud pipelines, and test benches:

- Lint and test `apply_faults_arrow(table, cfg, context)` rules
- Run rules on PyArrow Tables, Feather, or Parquet
- Optional NumPy analytics helpers (`pip install "open-fdd[analytics]"`)

170.2 What it is not

The PyPI wheel does **not** include:

- Operator Bridge / FastAPI app
- React dashboard
- BACnet commission/poll service
- MCP server
- Ansible / Tailscale deploy tooling

Use **GHCR Docker images** for the full edge stack.

170.3 Install

```
pip install open-fdd
pip install "open-fdd[analytics]"   # optional NumPy helpers
pip install "open-fdd[ml]"          # offline sklearn experiments
```

170.4 CLI

```
open-fdd version
open-fdd lint-rule path/to/rule.py
open-fdd test-rule path/to/rule.py --input sample.feather --
config cfg.json
open-fdd run-arrow path/to/rule.py --input in.feather --output
out.feather
open-fdd validate-rule-pack ./rules
```

170.5 Rule contract

```
def apply_faults_arrow(table, cfg, context=None):
    # return PyArrow bool mask, length == table.num_rows
    ...
```

Use `open_fdd.arrow_runtime.run_arrow_rule` for structured `ArrowRuleResult` (counts, errors, preview).

170.6 Modules shipped on PyPI

Module	Purpose
`open_fdd.arrow_runtime`	Rule execution, cookbook masks, column maps
`open_fdd.playground`	Lint/compile helpers
`open_fdd.faults`	Fault catalog YAML

Module	Purpose
`open_fdd.schema`	Result schemas

`open_fdd.engine` (pandas/YAML) is **not** in the wheel.

170.7 Examples

Repo `examples/` — arrow_minimal, Feather batch, generic IoT pipeline.

170.8 Local validation

```
pip install -e ".[test,dev,analytics]"
python scripts/release/check_version.py
pytest open_fdd/tests -q
python -m build && twine check dist/*
```

171 MCP server

172 Open-FDD MCP server

FastMCP agent interface over the **operator bridge** — not a parallel FDD stack.

172.1 Modes

Mode	`OFDD_MCP_MODE`	Site resolution
Edge	`edge`	`OPENFDD_BRIDGE_BASE_URL` (default `http://127.0.0.1:8765`)
Portfolio	`portfolio`	`portfolio/sites.json` + optional `site_id` per tool

RCx Central (`:8050` Dash, `:8060` API) is a **separate HTTP stack** — not the MCP server. Agents doing RCx overview, FDD preset tables, or DOCX reports should call Central API (`/api/central/*`) or use the Dash; see RCx Central agent guide.

Acme production edge runs **without** MCP (`enable_mcp: false`). Run MCP centrally and call `http://100.122.106.124` via Tailscale, or use RCx Central with Edge Connections registry.

172.2 Transport

Transport	Env	Use
Streamable HTTP	`OFDD_MCP_TRANSPORT=streamable-http` (default in Docker)	`http://127.0.0.1:8090/mcp`
Stdio	`OFDD_MCP_TRANSPORT=stdio`	Cursor / Claude Desktop

Legacy bridge doc search: `POST /tools/search_docs` on `:8090` (unchanged).

172.3 Human approval

These tools require `human_approved=true`:

- `save_rule`
- `apply_fdd_tuning`
- `run_fdd_batch`

MCP does not command BACnet writes — only Open-FDD bridge APIs.

172.4 Client examples

Cursor (`~/.cursor/mcp.json`):

```
        {
          "mcpServers": {
            "open-fdd": {
              "command": "python",
              "args": ["-m", "mcp_server.run"],
              "env": {
                "PYTHONPATH": "/path/to/open-fdd/workspace:/path/to/open-
fdd/workspace/api",
                "OPENFDD_REPO_ROOT": "/path/to/open-fdd",
                "OFDD_MCP_TRANSPORT": "stdio",
                "OFDD_MCP_MODE": "portfolio",
                "OFDD_PORTFOLIO_SITES_PATH": "/path/to/open-fdd/portfolio/
sites.json"
              }
            }
```

```
    }
  }
```

HTTP (bensserver dev stack):

```
docker compose -f docker/compose.dev.yml up -d mcp-rag
curl -fsS http://127.0.0.1:8090/health
```

172.5 Site registry example

```
{
  "sites": [
    {
      "site_id": "acme",
      "name": "Acme GL36 Lab",
      "base_url": "http://100.122.106.124",
      "username": "integrator"
    }
  ]
}
```

Store passwords in env or OS keychain — not in git.

172.6 RCx Central (not MCP)

Service	Default URL	Agent docs
Central API	`http://127.0.0.1:8060`	central-api.md
RCx Dash	`http://127.0.0.1:8050`	rcx-central-dash-agent.md

MCP `portfolio_rollup` hits Edge `/api/building/portfolio-rollup` per site. RCx `GET /api/central/overview/{site_id}` adds Dash KPIs, fault pie (with short fault descriptions), and mechanical narrative from FDD presets.

172.7 Fault code labels (agents)

Use fixed lookup — do not paraphrase codes in UI copy:

- Docs: fault-codes/short-lookup.md
- Python: `portfolio/central/fault_code_lookup.py`
- Edge API: `GET /api/faults/catalog`

172.8 Doc RAG refresh

After editing `docs/` consumed by MCP search:

```
./scripts/build_mcp_rag_index.sh
```

- `workspace/mcp_server/` — FastMCP tools, resources, prompts
- `workspace/mcp_rag/retrieval.py` — RAG index (rebuild separately)
- Docker image `openfdd-mcp-rag` — compatibility name; runs unified server

172.9 Module layout

173 Adr Datafusion Rust Ready Rule Backend

174 ADR: DataFusion as optional Rust-ready rule backend

174.1 Status

Accepted — Open-FDD 3.1.0

174.2 Context

Open-FDD v3 executes FDD rules over **Apache Arrow** tables. The primary contract is:

```
apply_faults_arrow(table, cfg, context=None) -> BooleanArray | ChunkedArray
```

Operators and integrators author rules in Python/PyArrow. BACnet polling, bridge APIs, dashboard, MCP, and Docker deployment all consume normalized `ArrowRuleResult` payloads.

We want a migration path toward Rust/DataFusion components **without** rewriting the project in Rust or weakening the PyArrow contract.

174.3 Decision

Add an **optional** second rule backend:

Backend	Entry
`arrow`	`apply_faults_arrow(...)` (default, first-class)
`datafusion_sql`	Restricted `SELECT` over registered `telemetry`

DataFusion is Apache Arrow-native and implemented in Rust. The Python `datafusion` package gives Open-FDD a clean seam where a future Rust service can replace or accelerate the wrapper while preserving the same result schema.

174.4 Why

- Same Arrow memory model end-to-end
- SQL expressions are easy to read for simple threshold/CASE rules
- Compare mode can prove SQL masks match PyArrow before migration
- Optional extra — normal `pip install open-fdd` unchanged

174.5 Non-goals (this milestone)

- No Rust rewrite
- No replacing PyArrow rules
- No pandas on edge runtime
- No arbitrary SQL console (injection-safe subset only)
- No changes to BACnet, bridge auth, or MCP unless required for tests

174.6 Future path

Phase A — PyArrow + optional DataFusion Python backend (this ADR)

Phase B — Shared Arrow rule result contract across batch, Rules Lab, and reporting

Phase C — Optional Rust DataFusion crate/service replaces Python wrapper

Phase D — ML/graph analytics consume Arrow/Parquet outputs without changing the FDD rule contract

174.7 Consequences

- CI adds a job with `pip install -e ".[test,dev,datafusion]"` for backend validation
- Rule store persists `sql` and `fault_column` separately from Python `code`
- Rules Lab gains backend selector; SQL executes server-side only

175 Adr Rust Ready Arrow Fdd Contract

176 ADR: Rust-ready Arrow FDD contract

176.1 Status

Accepted

176.2 Context

Open-FDD runs supervisory FDD over historian telemetry from BACnet, Niagara, JSON API, and future connectors. PyArrow is the primary rule path; DataFusion SQL is an optional backend for SQL-authored rules and Rust migration prep.

176.3 Decision

1. **PyArrow remains primary.** Rules implement `apply_faults_arrow(table, cfg, context=None)` and return Arrow boolean masks normalized to `ArrowRuleResult`.

2. **DataFusion SQL is optional.** Install via `open-fdd[datafusion]`. Rules declare `backend: datafusion_sql`, execute server-side only, read the registered `telemetry` Arrow table, and normalize to the same `ArrowRuleResult` shape.

3. **Common historian contract.** Drivers ingest `timestamp`, `site_id`, point columns, and `metadata.source` into Feather shards. Point metadata (`fdd_input`, `cross_source_semantic`, `equipment_id`) drives rule binding — not vendor-specific code in rule engines.

4. **Future Rust.** A Rust execution service may replace the Python Arrow/DataFusion wrappers behind the same table-in, boolean-mask-out contract. UI, commissioning, and driver orchestration stay in Python until migrated.

5. **Drivers must not bypass the contract.** No connector may emit a custom FDD result shape or row-loop rule path for production batch evaluation.

6. **No browser-side SQL.** DataFusion runs in the bridge process only.

7. **No pandas row-loop runtime on edge.** PyArrow columnar masks only for production FDD.

8. **Confirmation.** May use correctness-first Python streak loops today; future vectorized/window implementations must keep the same raw vs confirmed mask contract.

176.4 Consequences

- Rule Lab, batch FDD, and validation smokes share one mask contract.
- Confirmation window filters apply after raw masks for all backends.
- Vectorizing confirmation (window functions / Arrow kernels) is planned; correctness-first Python streak loops are acceptable for bench scale today.

177 Bench Validation Agent

178 Bench validation agent (local benserver)

Local bench only. Read-only.

178.1 Topology

Item	Value
Open-FDD host	benserver
BACnet device	instance **5007**, router `2000:7` (MS/TP)
Niagara station	`https://192.168.204.11`
Niagara BACnet folder	`slot:/Drivers/BacnetNetwork/BENS$20BENCHTEST$20BOX/points`
Password env	`OPENFDD_NIAGARA_ADMIN_PASSWORD` (never commit)

`/stream/health` returns **302** before login (reachability). **200** after SCRAM.

178.2 One-command validation

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='…'
python3 scripts/bootstrap_bench_dual_source.py
python3 scripts/bench_validate_bacnet_vs_niagara.py --write-report
```

178.3 API tools (bridge must be running)

Action	Endpoint
Bench health	`GET /api/bench/health`
Cross-source validate	`POST /api/bench/validate/bacnet-vs-niagara`
Poll status	`GET /api/bench/poll-status`
Poll cadence	`GET /api/bench/poll-cadence`
Niagara test	`POST /api/niagara/stations/bench9065/test`
Niagara discover	`POST /api/niagara/stations/bench9065/discover`
Start/stop Niagara poll	`POST …/poll/start` / `…/poll/stop`
BACnet driver tree	`GET /api/bacnet/driver/tree`

178.4 MCP / agent notes

- Browser → Open-FDD API only (no direct Niagara WebSocket from React).

- Preserve Niagara ORDs exactly (`$20`, `$2d`, `$23`). PowerShell: single-quote ORDs.
- Default Niagara discovery excludes `proxyExt` (10 points, not 20).
- Branch: `fix/3.0.34-bugs`; push and watch PR #300 CI + CodeRabbit.

178.5 Overnight

```
python3 scripts/run_overnight_bench_smoke.py
```

See overnight-bench-smoke.md and bench-bacnet-vs-niagara.md.

179 Rcx Central Dash Agent

180 RCx Central Dash — agent guide

For Cursor/Codex agents working on **OpenFDD RCx Central** (`portfolio/`) — analyst Dash + Central API over read-only Edge REST.

180.1 Runtime stack (benserver / analyst workstation)

Service	Script	Port	Bind
RCx Central API	`./scripts/run_central_api.sh`	8060	`0.0.0.0` (`OPENFDD_CENTRAL_API_HOST`)
RCx Central Dash	`./scripts/run_portfolio_dash.sh`	8050	`0.0.0.0` (`OPENFDD_RCX_DASH_HOST`)

Env: `OPENFDD_CENTRAL_API_URL=http://127.0.0.1:8060` (Dash → API). LAN access: `sudo ./scripts/open_rcx_central_lan_ports.sh` (UFW 8050+8060).

Not the operator bridge (`:8765`) — RCx Central polls Edge sites via Tailscale/LAN URLs in `portfolio/config/sites.json`.

180.2 Dash UI (2026-06)

Single **Dashboard** tab (faults, building summary, FDD rules, preset buttons, RCx report) + **Edge Connections** tab.

Area	Data source
Fault pie + legend	`fault_code_lookup.py` + live/CSV faults
Building summary	`build_mechanical_narrative()` → Edge FDD presets
FDD rules table	`GET /api/central/fdd-analytics/{site_id}`
FDD preset buttons	`GET /api/central/fdd-preset/{site_id}/{preset_id}` → same as Edge Data Model tab
P8 overrides	KPI count always; chart only when overrides present
RCx DOCX	`POST /api/central/rcx/report`

180.3 Edge parity (Data Model tab)

RCx Central must **not** scrape React. Use Edge REST:

- `GET /api/model/fdd-query-presets` + `GET /api/model/fdd-query-presets/{id}`
- `GET /api/model/sparql/predefined` + `POST /api/model/sparql`
- `GET /api/model/health`, `GET /api/model/tree`

Preset ids: `rules_to_equipment`, `rules_to_sensors`, `ahus_vavs_zones`, `missing_rule_bindings`, `orphan_points`, `rule_coverage_by_equipment_type`, etc. — see `portfolio/central/fdd_preset_catalog.py`.

180.4 BRICK equipment typing (common bug)

Acme and many sites store types in **`brick_type`** (`AHU`, `VAV`), not `equipment_type`.

- Count/classify with `portfolio/central/equipment_classify.py` (mirrors `workspace/api/openfdd_bridge/equipment_classify.py`).
- TTL sync maps short types → Brick classes (`AHU` → `Air_Handling_Unit`, `VAV` → `Variable_Air_Volume_Box`) in `ttl_service.py`.
- After model.json edits on Edge: **sync TTL** before SPARQL HVAC buttons return counts.
- Packaged rooftop serving VAVs: prefer **`brick_type: AHU`** and display name `AHU 01` (not `Rtu 01`) when it is a full air handler.

180.5 Fault codes (stable labels)

Layer	Path
Docs table	`docs/fault-codes/short-lookup.md`
Dash lookup	`portfolio/central/fault_code_lookup.py`
Edge catalog	`GET /api/faults/catalog` · `workspace/api/openfdd_bridge/fault_catalog.py`

Never invent codes. Letter suffix only (`VAV-C`, not `VAV-03`).

180.6 Central API routes (agent smoke)

```
curl -s http://127.0.0.1:8060/health
curl -s http://127.0.0.1:8060/api/central/overview/acme
curl -s "http://127.0.0.1:8060/api/central/fdd-analytics/acme?hours=24"
curl -s http://127.0.0.1:8060/api/central/fdd-preset/acme/ahus_vavs_zones
```

180.7 Tests

```
pytest tests/portfolio/ tests/scripts/test_rcx_overnight_patch_cycle.py
pytest tests/workspace_bridge/test_fdd_query_presets.py
```

180.8 PRs (typical split)

- **Edge bridge + TTL + model fixes** → `fix/edge-dashboard-bridge` (#299)
- **RCx Dash / portfolio only** → `feature/rcx-central-overview-ui` (#298)

180.9 Related

- Overnight patch cycle
- Model queries
- MCP server — portfolio mode vs RCx Central HTTP API

181 Rcx Central Overnight Patch Cycle

182 RCx Central overnight patch-cycle (agent runbook)

See also: RCx Central Dash agent guide for UI/API modules and BRICK typing.

Mission: validate **OpenFDD RCx Central** against local services and (optionally) live **ACME Edge** read-only, up to **10 patch cycles** with **2-hour** spacing.

182.1 Safety (non-negotiable)

- **Read-only** on BACnet and live BAS equipment.
- Never commit secrets, `.env`, tokens, or raw building dumps.
- Merge, publish, and deploy only when `RCX_ALLOW_MERGE=1`, `RCX_ALLOW_PUBLISH=1`, or `RCX_ALLOW_DEPLOY=1` and the owner authorizes.

182.2 One patch cycle

1. `git pull --ff-only` on `feature/rcx-central-overview-ui` (or follow-up branch).
2. `gh pr checks 298` and `gh run view <id> --log-failed` for failures.
3. Classify CodeRabbit comments: must-fix / nice-to-have / false positive.
4. Run tests: `python3 -m pytest tests/portfolio tests/workspace_bridge -q`.
5. Start RCx Central: `./scripts/run_central_api.sh`, `./scripts/run_portfolio_dash.sh` (bind `0.0.0.0`; LAN: `sudo ./scripts/open_rcx_central_lan_ports.sh`).
6. Run `python3 scripts/rcx_central_overnight_patch_cycle.py` (try-out or overnight env).
7. Patch bugs, commit, push, confirm CI green.
8. Write `reports/rcx_central_overnight_logs/cycle_NN.md`.

182.3 GH Actions

```
gh pr checks 298
gh run list --branch feature/rcx-central-overview-ui --limit 10
gh run view <RUN_ID> --log-failed
```

Fix order: unit tests → operator-bridge → docs → Docker supervisor.

182.4 RCx Central local

```
./scripts/run_central_api.sh          # :8060
OPENFDD_CENTRAL_API_URL=http://127.0.0.1:8060 ./scripts/
run_portfolio_dash.sh  # :8050
curl -s http://127.0.0.1:8060/health
curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:8050/
```

Edge connections: save URL + credentials on **Edge Connections** tab (local `portfolio/config/`).

182.5 Live ACME (gated)

```
export OPENFDD_LIVE_ACME=1
export RCX_EDGE_SITE=acme
export RCX_CENTRAL_API_URL=http://127.0.0.1:8060
```

Requires ACME credentials in local registry — never log passwords.

182.6 Try-out vs overnight

Mode	`RCX_PATCH_CYCLES`	`RCX_CYCLE_SLEEP_MINUTES`
Try-out	2	0
Overnight	10	120

182.7 Rev bump / deploy

Only when tests and CI pass and owner sets `RCX_ALLOW_PUBLISH=1` / `RCX_ALLOW_DEPLOY=1`. Use existing `scripts/upgrade_edge_full.sh` — do not invent new deploy paths.

182.8 Branch cleanup after merge

```
git checkout master && git pull --ff-only
git branch -d feature/rcx-central-overview-ui
git push origin --delete feature/rcx-central-overview-ui
```

183 Bench Bacnet Vs Niagara

184 BACnet direct vs Niagara bench validation

Local **benserver** read-only cross-check: same physical BACnet device via two paths.

184.1 Topology

```
BENS BENCHTEST BOX (BACnet device 5007, MS/TP 2000:7)
  ├─ Open-FDD native BACnet poll (commission agent → samples.csv → feather source=bacnet)
  └─ Niagara Windows station → baskStream WebSocket → Open-FDD Niagara driver (source=niagara_baskstream)
```

184.2 Quick start

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='…'   # never commit
python3 scripts/bootstrap_bench_dual_source.py
python3 scripts/bench_validate_bacnet_vs_niagara.py --write-report
```

Or via API (bridge running):

```
curl -X POST http://127.0.0.1:8765/api/bench/validate/bacnet-vs-niagara \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"write_report": true}'
```

184.3 Mapping config

Edit `workspace/data/bench_bacnet_vs_niagara.yaml` for semantic point mapping, tolerances, and Niagara ORDs (`$20` / `$2d` preserved).

184.4 Brick model

Import dual-source semantics:

```
        # Included in bootstrap; or manually:
        python3 -c "
        import json, os, sys
        from pathlib import Path
        REPO = Path('.').resolve()
        sys.path.insert(0, str(REPO/'workspace/api'))
        os.environ['OFDD_DESKTOP_DATA_DIR'] = str(REPO/'workspace/data')
        from openfdd_bridge.model_service import ModelService
        from openfdd_bridge.ttl_service import TtlService
        ModelService().import_json(json.loads((REPO/'workspace/data/
bench_dual_source_model.json').read_text()), replace=False)
        TtlService().sync()
        "
```

184.5 Normalized data model

All drivers normalize through `driver_point_contract.py`:

- **bacnet** → canonical `bacnet_direct`
- **niagara_baskstream**, **modbus**, **json_api**

Semantic identity (`fdd_input` / `cross_source_semantic`) links both sources to one Brick point role.

184.6 Poll intervals

- BACnet: `poll_interval_s` in `points.csv` (standard: 60, 300, …)
- Niagara: `poll_interval_seconds` on station config; worker re-reads each cycle (no restart required)
- Lab minimum Niagara interval: 15s (API validates ≥15)

Validate cadence: `GET /api/bench/poll-cadence?source=bacnet_direct&expected_interval_s=60`

184.7 Overnight smoke

```
        python3 scripts/run_overnight_bench_smoke.py --dry-run   # one
checkpoint
        python3 scripts/run_overnight_bench_smoke.py             # 12h,
checkpoint every 2h
```

Reports: `reports/overnight_bench/<timestamp>/`

184.8 Read-only rules

No BACnet writes, no Niagara overrides, no alarm ack. Relay booleans are read-only observations only.

185 Dashboard Analytics

186 Dashboard analytics

Engineering analytics pages live under **Analytics** in the operator dashboard (`/analytics`).

186.1 Pages

Route	Purpose
`/analytics`	Overview KPIs, fault-hour charts, top faults table
`/analytics/faults`	Time-range fault analytics
`/analytics/equipment`	AHU / VAV fault-hour summary
`/analytics/health`	BACnet / BRICK model health
`/analytics/rcx`	RCx Report Builder (preview + DOCX download)

186.2 API (read-only)

- `GET /api/analytics/overview`
- `GET /api/analytics/faults?hours=24`
- `GET /api/analytics/model-health`
- `POST /api/reports/rcx/preview`

- `POST /api/reports/rcx/generate` → DOCX download

186.3 Fault hours

Elapsed fault hours are estimated from FDD batch run `analytics.estimated_fault_duration_sec` when present, otherwise from flagged/total sample ratio. See `open_fdd/reports/fault_hours.py`.

186.4 Theme

All Plotly charts use `getPlotlyThemeLayout(theme)` from `workspace/dashboard/src/lib/plotly-theme.ts`.

186.5 Tests

```
        pytest tests/workspace_bridge/test_dashboard_analytics.py
open_fdd/tests/test_fault_hours.py
        cd workspace/dashboard && npm test
```

Live BACnet is not required. Gate live ACME validation with `OPENFDD_LIVE_ACME=1`.

187 Datafusion Sql Rules

188 DataFusion SQL rules (optional)

Open-FDD v3 remains **PyArrow-first**. Rules implement `apply_faults_arrow(table, cfg, context=None)` and return a boolean Arrow mask.

The optional **DataFusion SQL** backend (`backend: datafusion_sql`) is a future-Rust-prep seam:

- Input: PyArrow `Table` registered as `telemetry`
- Rule body: a restricted `SELECT` that adds a boolean `fault` column
- Output: the same `ArrowRuleResult` shape as PyArrow rules

Install:

```
pip install 'open-fdd[datafusion]'
```

188.1 When to use SQL vs PyArrow

Use DataFusion SQL	Use PyArrow
Simple thresholds, `CASE WHEN`, boolean logic	Custom HVAC helpers, rolling windows, ML prep
SQL-readable expressions for operators	Complex graph/feature work
Proving parity before Rust migration	Primary supported authoring path

188.2 Rule metadata

```yaml
id: zone_temp_high_sql
name: Zone temperature high — SQL
backend: datafusion_sql
fault_column: fault
sql: |
  SELECT
    *,
    zone_temp > 75.0 AS fault
  FROM telemetry
```

188.3 Safety

SQL rules are **not** a database console. The linter rejects:

- Multiple statements, DDL/DML
- Reads other than `FROM telemetry`
- File paths and external URLs
- Missing or non-boolean `fault` column
- Row counts that differ from input telemetry
- Wrong fault column alias (e.g. `not_fault` when `fault` is required)

Server-side execution only — the Rules Lab browser editor never runs SQL locally.

188.4 Rules Lab

In **Rule Lab**, choose **PyArrow** or **DataFusion SQL**:

- **Validate SQL** — lint + optional server preview on latest historian sample

- **Run Preview** — bounded sample against site telemetry
- **Compare backends** — prove an equivalent SQL rule matches a PyArrow rule mask

See ADR: DataFusion Rust-ready backend.

189 Dashboard Rcx Makeover Inventory

190 Dashboard + RCx Makeover — Inventory

Branch: `feature/dashboard-rcx-mega-makeover`
Date: 2026-05-30

190.1 What exists

Area	Location	Notes
React operator dashboard	`workspace/dashboard/`	Vite + React 19, theme via `contexts/theme-context.tsx`, CSS vars
Plotly charts	`lib/plot-chart.ts`, `PlotPage.tsx`, `HostStatsPage.tsx`	Partial theme handling; not centralized
Building check-engine	`building_status.py`, `FaultsPage`, `BuildingInsightDashboard`	Single source for alerts
FDD batch + analytics	`fdd_results.py`, `fdd_fault_analytics.py`	Per-run sample analytics, equipment enrichment
Poll / model health	`poll_throughput.py`, `model_health.py`, `device_poll_health.py`	Used by agent, not dashboard analytics pages
Zone temp analytics	`zone_temp_analytics.py`	VAV comfort metrics via agent routes
Central RCx DOCX		Central-only (`POST /api/`

Area	Location	Notes
	`portfolio/central/rcx_report.py`, `fault_hours.py`	`central/rcx/report`)
Generic open_fdd reports	`open_fdd/reports/docx_generator.py`, `fault_viz.py`	Equipment FDD reports, optional deps
Portfolio desk	`portfolio/dash/app.py`, `portfolio/central/api.py`	Separate Dash + FastAPI stack
Frontend unit tests	14 Vitest files under `workspace/dashboard/src/lib/`	Not in CI
Bridge Python tests	`tests/workspace_bridge/`, `tests/portfolio/`	Strong coverage for building insight, central

190.2 What partially exists

- **Theme toggle** — app shell updates; Plotly uses inline colors in `plot-chart.ts` / `HostStatsPage.tsx` (inconsistent).
- **Fault display** — equipment names in check-engine alerts when model maps columns; catalog page is not analytics-focused.
- **RCx DOCX** — portfolio central builder with basic sections; not exposed on edge bridge; no chart/section checkboxes.
- **Fault hours** — `portfolio/central/fault_hours.py` estimates from rollups; FDD run analytics has duration estimates but no REST aggregation.
- **Demo data** — `workspace/data/acme_gl36_model.json`, `fdd_results.json`, `samples/demo_site.csv`; no unified analytics fixture.

190.3 What is missing (this branch targets)

- Shared `getPlotlyThemeLayout()` helper + all charts using it
- Analytics pages: Overview, Fault Analytics, Equipment, BACnet/Model Health, RCx Report Builder
- Bridge REST: `/api/analytics/overview`, `/faults`, `/equipment/{id}`, `/model-health`, `/reports/rcx/preview`, `/reports/rcx/generate`
- Edge RCx DOCX with selectable sections/charts (`open_fdd/reports/rcx_docx.py`)
- Fault-hour aggregation module with tests
- Demo analytics fixture for CI
- Docs: `docs/dashboard-analytics.md`, `docs/rcx-report-builder.md`

190.4 Files inspected

```
AGENTS.md, README.md, pyproject.toml
workspace/dashboard/src/App.tsx, AppLayout.tsx, theme-context.tsx,
plot-chart.ts
workspace/api/openfdd_bridge/main.py, building_status.py,
fdd_results.py, fdd_fault_analytics.py
workspace/api/openfdd_bridge/routes/analytics_routes.py
portfolio/central/rcx_report.py, fault_hours.py, api.py
open_fdd/reports/
tests/workspace_bridge/, tests/portfolio/test_central.py
```

190.5 Recommended implementation path

1. **Foundation** — `open_fdd/reports/fault_hours.py`, `charts.py`, `rcx_docx.py`; bridge `dashboard_analytics.py` + routes; deps in bridge requirements.
2. **Theme** — `plotly-theme.ts`; refactor `plot-chart.ts` + `HostStatsPage.tsx`.
3. **UI** — Analytics nav section + pages wired to new APIs; RCx Report Builder with preview/generate.
4. **Fixtures + tests** — demo JSON, backend pytest, Vitest for theme + report builder.
5. **Docs** — operator-facing how-to.

190.6 Known risks

- **Scope** — full prompt is multi-PR; this branch delivers edge dashboard + bridge RCx; Central Dash remains separate.
- **Live BACnet** — analytics read persisted FDD/model/poll data only (read-only).
- **Trend charts in RCx** — need timeseries API + point roles; may be partial when historian empty.
- **ACME coupling** — use demo fixture + model-driven roles, not hard-coded point names.

191 Rcx Central Current State

192 OpenFDD RCx Central — current state

Updated: 2026-05-30
Branches: `feature/rcx-central-overview-ui` (#298)
Status: Local TTL mirror + SPARQL on Central; benserver sign-off before Docker GHCR publish.

192.1 Product names

User-facing	Internal path
OpenFDD Edge	`workspace/`, `docker/`, GHCR images
OpenFDD RCx Central	`portfolio/` (Dash + Central API)

192.2 Entrypoints

Service	Command	Port	Bind default
RCx Central Dash	`./scripts/run_portfolio_dash.sh`	8050	`0.0.0.0`
RCx Central API	`./scripts/run_central_api.sh`	8060	`0.0.0.0`
Docker (both)	`docker/rcx-central/docker-compose.yml`	8050, 8060	

LAN firewall (benserver): `sudo ./scripts/open_rcx_central_lan_ports.sh`

192.3 Central API

- `GET/POST/PUT/DELETE /api/central/edges`, `POST /api/central/edges/test`
- `GET /api/central/overview/{site_id}` — KPIs, fault pie, mechanical narrative, P8
- `GET /api/central/fdd-analytics/{site_id}` — rules table + descriptions

- `GET /api/central/fdd-preset/{site_id}/{preset_id}` — Edge FDD preset proxy + Central enrichment
- `POST /api/central/model/remediate/{site_id}` — equipment types, BACnet metadata, Edge TTL sync, Central mirror
- `POST /api/central/model/sync-ttl/{site_id}` — pull `data_model.ttl` to `portfolio/data/sites/{site_id}/model/`
- `GET /api/central/model/sparql/validate/{site_id}` — local AHU/VAV/site SPARQL counts
- `POST /api/central/model/sparql/{site_id}` — read-only SPARQL on mirrored TTL
- `GET /api/central/mechanical-summary/{site_id}`
- `GET /api/central/model-tree/{site_id}` — on-demand BRICK tree (lazy; not on auto-refresh)
- `GET /api/central/rcx/points/{site_id}` — BACnet point catalog for Report Builder
- `POST /api/central/rcx/preview`, `/charts/preview`, `/report`

192.4 Edge REST (read-only, shared with Data Model tab)

Model: `/api/model/tree`, `/api/model/health`, SPARQL, `/api/model/fdd-query-presets/{id}`
Trends: `/api/timeseries/readings`
Analytics: `/api/analytics/faults`
Faults: `/api/faults/status`, `/api/faults/catalog`

192.5 Dash UI tabs

Dashboard · Report Builder · Edge Connections

Dashboard: fault mix + legend, building summary, P8 KPI, FDD rules (lazy load), FDD/BRICK preset buttons, **local SPARQL** (TTL mirror).
Report Builder: 3-step wizard — building/time → charts & sections → custom points → preview DOCX.
Edge data fetched on demand (not with dashboard load).

192.6 Pre-Docker owner checklist (benserver :8050)

1. Dashboard loads in under 5s without hanging on model tree.
2. **Load full BRICK model** and **Load FDD rules** buttons work when clicked.
3. Report Builder: load catalog → preview charts → generate DOCX for Acme.
4. Fault overlays visible on trend previews when enabled.

5. Sign off here before `docker/rcx-central` image rebuild or `RCX_ALLOW_PUBLISH=1`.

192.7 BRICK model notes (agents)

- Classify equipment with `brick_type` **and** `equipment_type` — see `equipment_classify.py`
- TTL maps `AHU`/`VAV` → Brick schema classes for SPARQL HVAC queries
- Acme roof unit: `brick_type: AHU`, name `AHU 01` (not `Rtu 01` in new models)

192.8 Fault codes

- Stable short labels: `docs/fault-codes/short-lookup.md`, `portfolio/central/fault_code_lookup.py`
- Full catalog: `GET /api/faults/catalog` on Edge

192.9 Agent docs

- RCx Central Dash agent guide
- AI agent workflow
- MCP server

192.10 Tests

`pytest tests/portfolio/` · `tests/workspace_bridge/test_fdd_query_presets.py`

192.11 Remaining gaps

- **Owner Dash sign-off** on benserver before Docker GHCR publish
- GHCR publish for `openfdd-rcx-central` image (gated: `RCX_ALLOW_PUBLISH=1`)
- Optional OpenAI insights hook (templates used today)
- True Edge `start`/`end` bounded timeseries (hours lookback used for now)
- Summarize HVAC SPARQL buttons on Dash — **done** (local TTL mirror on Central)

193 Short Lookup

194 Fault code short descriptions

Stable lookup for pie charts, tables, and RCx reports. Titles match `workspace/api/openfdd_bridge/fault_catalog.py`.

Code	Short description
AHU-A	Supply fan performance degradation
AHU-B	Simultaneous heating and cooling
AHU-C	Supply air temperature sensor fault
AHU-D	Mixed-air temperature inconsistent with OAT/RAT
AHU-E	Economizer not economizing
AHU-F	Damper/valve command vs feedback mismatch
AHU-G	PID hunting / excessive control oscillation
VAV-A	Reheat active during cooling demand
VAV-B	Airflow not meeting setpoint
VAV-C	Zone temperature sensor fault
VAV-D	Damper command vs airflow mismatch
VAV-E	Rogue zone (chronic reheat/overcooling)
VAV-F	VAV actuator PID hunting
BLD-A	Whole-building energy deviation
BLD-B	Outdoor-air temperature sensor fault
BLD-C	Equipment running outside occupancy schedule
BLD-D	Point data dropout (stale points)
RTU-A	Supply fan performance degradation
RTU-B	Simultaneous heating and cooling
RTU-C	Discharge air temperature sensor fault
RTU-D	Damper/valve command vs feedback mismatch
RTU-E	Cooling capacity PID hunting
CH-G	Pump / variable-speed PID hunting

Full catalog: index.md and live `GET /api/faults/catalog` on Edge.

Python mirror: `portfolio/central/fault_code_lookup.py` (RCx Central Dash).

195 Niagara Baskstream Connector

196 Niagara baskStream connector (read-only)

Open-FDD integrates with **Niagara Framework** stations through the vendor **baskStream** module. The bridge API performs SCRAM login, opens a persistent WebSocket, and reads point values via MessagePack — the browser never connects directly to Niagara.

196.1 Read-only posture

This connector is **read-only**. It does not implement writes, overrides, alarm acknowledge, or emergency actions. Use Niagara Workbench for OT commands.

196.2 Niagara station setup

1. Install the **baskStream** module on the Niagara station (same module validated by `niagara-baskstream-python-tools` on Linux).
2. Add **BASkStreamService** (or equivalent baskStream service) on the station.
3. Ensure HTTPS (port **443**) is reachable from the Open-FDD bridge host.
4. Firewall: allow the bridge server → station on 443.

196.2.1 Health check behavior

- **Before login:** `GET /stream/health` returns **302** redirect to `/login`.
- **After SCRAM login:** `GET /stream/health` returns **200** with JSON including `authenticatedUser`.

From Linux bench:

```
curl -k -i https://192.168.204.11/stream/health
# Expect 302 before login
```

196.3 Open-FDD configuration

1. Copy `workspace/niagara.env.example` → `workspace/niagara.env.local` (gitignored).
2. Set `OPENFDD_NIAGARA_ADMIN_PASSWORD` (or your custom env name).
3. In the dashboard **Niagara** tab, add a station:
   - URL: `https://<station-ip>`
   - Username: Niagara web user
   - **password_env:** `OPENFDD_NIAGARA_ADMIN_PASSWORD`
   - **verify_tls:** off for self-signed bench certs
   - **default_points_root:** e.g. `slot:/Drivers/BacnetNetwork/BENS$20BENCHTEST$20BOX/points`

196.3.1 ORD encoding

Niagara ORDs use encoded characters (`$20` = space, `$2d` = hyphen, `$23` = `#`). **Preserve these strings exactly** in config, API, and UI. Do not URL-decode or shell-expand them.

PowerShell users: use **single-quoted** strings for ORDs so `$20` is not treated as a variable.

196.3.2 Allowed Origins

The Open-FDD **backend** connector does not require Niagara "Allowed Origins" for browser CORS — traffic is **Browser → Open-FDD `/api/niagara` → Niagara**. Direct browser-to-Niagara WebSocket is discouraged.

196.4 Dependencies

Bridge API requires `aiohttp` and `msgpack` (see `workspace/api/requirements.txt`). If missing, the app still boots; `/api/niagara/health` reports `dependencies_ok: false`.

196.5 Polling

Enable poll per station in the UI. The worker uses one persistent WebSocket per active station, batched reads, cached discovery metadata, and historian ingest with `source=niagara_baskstream`.

Disable background worker: `OFDD_DISABLE_NIAGARA_POLL_WORKER=1`.

196.6 Bench reference (benserver)

Item	Value
Station URL	`https://192.168.204.11`
BACnet device folder	`slot:/Drivers/BacnetNetwork/BENS$20BENCHTEST$20BOX/points`
Expected points	~10 BACnet components (proxy extensions excluded by default)

Validate: **Test connection → Discover → Read selected / Poll once**. Values should match the standalone `baskstream_cli.py values` output (OA-T, DUCT-T, STAT ZN-T, etc.).

197 Data Model

198 Data Model — sync and FDD queries

198.1 Explorer (BACnet poll sync)

The **Explorer** tab focuses on **BACnet poll CSV ↔ model.json** sync status and **Sync poll → model**. Write TTL lives under **Advanced** (or SPARQL RDF tools).

198.2 FDD query presets

The **SPARQL** tab includes **FDD / BRICK query presets** — composed queries across `model.json`, rule bindings, and BACnet metadata:

- Rules → Equipment / Sensors / BACnet devices
- Equipment → Points, AHUs/VAVs/Zones
- Missing rule bindings, orphan points, sensor classes, coverage by equipment type

API: `GET /api/model/fdd-query-presets`, `GET /api/model/fdd-query-presets/{id}`.

199 Host Resources

200 Host Resources — Easy Pooge reset

Integrators see a **Danger Zone** on the Host Resources tab for job-site redeploys.

- Typed confirmation: `POOGE THIS EDGE`
- Dry-run preview by default
- Allowlisted path cleanup (historian, BACnet scratch, exports)
- Optional Linux apt upgrade and GHCR image pull

API: `POST /api/host/pooge/preview`, `POST /api/host/pooge/run` (integrator only).

201 Trend Plots

202 Trend plots

The **Trend plot** tab reads feather historian data via `GET /api/timeseries/readings`.

202.1 HTTP / Tailscale

Charts work over plain HTTP on localhost, LAN, and Tailscale — COOP/CORP security headers are omitted on non-HTTPS origins so Plotly is not blocked.

202.2 Troubleshooting

If the chart is blank, open **Trend plot debug** at the bottom of the tab:

- Endpoint called
- Site, equipment, selected keys
- Series and timestamp row counts
- Last error message

Common fixes: enable BACnet polling, pick points with feather columns, reduce FDD overlay scope.

203 Cloud Exporter

204 Cloud exporter sidecar

Optional `openfdd-cloud-exporter` container polls the bridge API and POSTs JSON to a webhook-compatible URL (PostBin, webhook.site, etc.).

204.1 Enable (edge compose)

```
export OPENFDD_EXPORT_ENDPOINT=https://webhook.site/your-uuid
export OPENFDD_EXPORT_DRY_RUN=1
docker compose --profile cloud-export up -d cloud-exporter
```

Set `OPENFDD_EXPORT_DRY_RUN=0` only when you intend live export.

204.2 Security

- Disabled by default (`OPENFDD_EXPORT_DRY_RUN=1`).
- Does not export operator passwords or auth env files.
- Tokens are redacted in logs.

See `workspace/cloud_exporter/README.md` for environment variables.

205 Overnight Bench Smoke

206 Overnight bench smoke test

12-hour local validation on **benserver** only. Read-only.

206.1 Command

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='…'
python3 scripts/run_overnight_bench_smoke.py
```

Options:

Flag	Default	Purpose
`--duration-hours`	12	Total run length
`--checkpoint-hours`	2	Wake interval
`--dry-run`	off	Single checkpoint then exit
`--skip-bootstrap`	off	Skip BACnet/Niagara setup
`--test-poll-freq`	off	Run poll interval change test at t=0

206.2 Each checkpoint

1. Cross-source validate (`bench_validate_bacnet_vs_niagara`)
2. Poll cadence report (BACnet + Niagara)
3. Targeted pytest (contract, validator, Niagara)
4. JSON + Markdown report under `reports/overnight_bench/<run_id>/`

206.3 Final deliverable

`final_report.md` / `final_report.json` with PASS/FAIL and next actions.

206.4 GH Actions loop

After code fixes during a wake cycle:

```
git add … && git commit -m "…" && git push origin fix/3.0.34-bugs
gh pr checks 300 --watch
```

No empty commits when validation-only cycles pass.

207 Ai Agent Workflow

208 AI agent workflow vs runtime use

208.1 Three agent surfaces

Surface	Port	Use when
OpenFDD Edge (bridge)	8765 / :80 Caddy	Rule Lab, BACnet, Data Model SPARQL, live FDD
OpenFDD RCx Central	8050 Dash, 8060 API	Multi-site analyst dashboard, RCx DOCX, Edge registry
OpenFDD MCP	8090	Cursor/Codex tools over bridge or `portfolio/sites.json`

Acme production edge runs **without** MCP (`enable_mcp: false`). Use **RCx Central API** or **portfolio MCP** against `http://100.122.106.124` (Tailscale).

208.2 Runtime analyst (no git clone)

1. `docker compose -f docker/rcx-central/docker-compose.yml up` **or** `./scripts/run_central_api.sh` + `./scripts/run_portfolio_dash.sh`
2. Open http://localhost:8050 (LAN: `sudo ./scripts/open_rcx_central_lan_ports.sh` on benserver)
3. **Edge Connections** → add Acme URL → Test → Save
4. **Dashboard** → select building → verify building summary (AHU/VAV counts), FDD preset buttons, fault pie descriptions
5. **RCx report** → Preview → Generate DOCX

Kill containers when done. Registry in Docker volume `rcx-central-config`.

208.3 Developer / coding agent (git working tree)

Use Cursor or Codex against `open-fdd`. Read:

- RCx Central Dash agent guide
- Overnight patch cycle

Key modules: `portfolio/dash/overview_tab.py`, `portfolio/central/mechanical_narrative.py`, `portfolio/central/fault_code_lookup.py`, `workspace/api/openfdd_bridge/fdd_query_presets.py`, `ttl_service.py`.

After doc changes consumed by MCP RAG: `./scripts/build_mcp_rag_index.sh` (optional on benserver).

208.4 Model / SPARQL validation (Acme)

1. Confirm `model.json` equipment uses `brick_type` (`AHU`, `VAV`) — not only legacy `equipment_type`
2. Sync TTL on Edge (Data Model tab or bridge restart)
3. Edge UI: **Summarize HVAC → AHUs / VAV boxes** must return non-zero for Acme lab
4. RCx Dash: **AHUs / VAVs / Zones** preset → same row counts as Edge

208.5 Secrets

- Never commit `portfolio/config/credentials.json` or passwords in `sites.json`
- Env: `OFDD_AGENT_PASSWORD`, per-site credentials in config store
- Live ACME tests: `OPENFDD_LIVE_ACME=1` only when explicitly requested

208.6 Model routing (test triage)

See `AGENTS.md` — classify CI failures as SIMPLE vs COMPLEX before processing.

209 Docker Desktop Windows

210 RCx Central on Docker Desktop (Windows)

210.1 Prerequisites

- Docker Desktop for Windows
- Edge reachable from the host (e.g. Tailscale)

210.2 Run

```
cd open-fdd
docker compose -f docker/rcx-central/docker-compose.yml up --build
```

Open:

- Dash: http://localhost:8050
- API health: http://localhost:8060/health

No git clone required at runtime if you pull `ghcr.io/bbartling/openfdd-rcx-central` (when published). Build locally with `OPENFDD_IMAGE_TAG=local`.

210.3 Volumes

Volume	Purpose
`rcx-central-data`	Rollups, reports (`portfolio/data/reports/`)
`rcx-central-config`	`sites.json`, masked credentials

210.4 Tailscale networking

If Edge is reachable on the Windows host but not from inside the Linux container:

1. Test from host: `curl https://<tailscale-ip>:8765/health`
2. Test from container: `docker compose exec rcx-central-api curl -s http://<tailscale-ip>:8765/health`

3. If container cannot route, use the host's Tailscale IP (Docker Desktop often routes LAN correctly) or document `host.docker.internal` proxy for dev.

210.5 Smoke test

```
./scripts/test_rcx_central_docker.sh
```

Windows PowerShell variant: run compose up, then curl health URLs manually per steps above.

211 Edge Auth

212 Edge authentication (RCx Central)

212.1 Browser flow

1. Open RCx Central → **Edge Connections**
2. Enter site_id, name, Edge base URL (`https://<tailscale-ip>:8765`)
3. Enter username/password (or bearer token via API)
4. **Test Connection** — probes `/health`, `/api/model/health`, `/api/faults/status`
5. **Save Edge** — writes to `portfolio/config/sites.json` + `credentials.json` (Docker volume)

212.2 API

```
GET  /api/central/edges
POST /api/central/edges
POST /api/central/edges/test
PUT  /api/central/edges/{site_id}
DELETE /api/central/edges/{site_id}
```

Secrets are **never** returned in API responses or committed to git. Passwords are masked in UI after save.

212.3 Auth types

- `password` — `POST /api/auth/login` on Edge

- `bearer` — static token in credentials store
- `none` — public Edge endpoints only (dev)

212.4 Rejected URLs

`file://`, `ftp://`, and hosts without scheme are rejected.

213 Index

214 OpenFDD RCx Central

OpenFDD RCx Central is the local analyst / RCx engineer dashboard. It runs on an analyst workstation or **Docker Desktop (Windows/Linux)** — not on the OT edge by default.

Product	Role
OpenFDD Edge	Arrow-native, no-pandas OT-LAN FDD machine at the building
OpenFDD RCx Central	Read-only multi-edge analytics, chart preview, DOCX reports

Internal source path: `portfolio/` (package name unchanged).

214.1 Quick start (host Python)

```
pip install -r portfolio/requirements.txt
./scripts/run_central_api.sh   # :8060
./scripts/run_portfolio_dash.sh # :8050
```

214.2 Docker Desktop

```
./scripts/run_rcx_central_docker.sh
# Dash http://localhost:8050  API http://localhost:8060/health
```

See docker-desktop-windows.md.

214.3 Pages

- Overview — collected rollup trends (Plotly, light/dark theme)
- Edge Connections — add/test/save Edge URLs (secrets in local volume)
- Mechanical Summary — BRICK equipment counts from Edge APIs
- FDD Rules & Analytics — configured rules, chart packs
- Trend Explorer — pointer to Overview Plotly charts (local collect CSVs)
- RCx Report Builder — matplotlib preview, fault overlays, DOCX download
- Validation Runs — one-off read-only Edge probes
- Settings — API URL and volume paths

Model query details: model-queries.md

214.4 Safety

RCx Central is **read-only** toward Edge and BACnet. No writes, commands, or setpoint changes.

215 Model Queries

216 Model queries (RCx Central → OpenFDD Edge + local TTL mirror)

OpenFDD RCx Central reuses the **same read-only Edge REST APIs** as the Edge operator React **Data model** tab. RCx Central does not scrape the React UI.

Local BRICK SPARQL: `POST /api/central/model/sync-ttl/{site_id}` fetches Edge `model.json`, builds TTL with the same `ttl_service` as Edge, and stores it under `portfolio/data/sites/{site_id}/model/data_model.ttl`. Central runs SPARQL locally (rdflib). Optionally calls Edge `POST /api/model/sync-ttl` first so on-site TTL stays aligned when the Edge image is current.

216.1 Primary endpoints

Method	Edge path	Purpose
GET	`/api/model/tree`	Equipment + points (BRICK roles, timeseries columns)
GET	`/api/model/health`	Point/equipment counts, model issues
GET	`/api/model/sparql/predefined`	Prebuilt SPARQL query list
POST	`/api/model/sparql`	Run SPARQL (`{"query": "..."}`)
GET	`/api/model/fdd-query-presets`	FDD commissioning query presets
GET	`/api/model/fdd-query-presets/{id}`	Run preset by id
GET	`/api/model/ttl?save=false`	Turtle text (used by Central TTL mirror)
POST	`/api/model/sync-ttl`	Regenerate TTL on Edge from `model.json`

216.2 Trends (matplotlib preview + fault overlays)

Method	Edge path	Purpose
GET	`/api/timeseries/series?site_id=`	Plottable columns per site
GET	`/api/timeseries/readings?site_id=&columns=&hours=`	Trend samples + optional FDD fault flags

RCx Central maps BRICK **point roles** (e.g. `supply_air_temperature`) to historian columns via `/api/model/tree`, then calls `/api/timeseries/readings` with `include_faults=true` for overlay bands.

216.3 Analytics

Method	Edge path	Purpose
GET	`/api/analytics/overview`	KPIs, fault hours by severity/ equipment
GET	`/api/analytics/faults?hours=`	Fault rows for selected window
GET	`/api/analytics/model-health`	Stale/missing roles

216.4 RCx Central API wrappers

Method	Central path
GET	`/api/central/overview/{site_id}`
GET	`/api/central/mechanical-summary/{site_id}`
GET	`/api/central/fdd-analytics/{site_id}`
GET	`/api/central/fdd-preset/{site_id}/{preset_id}`
POST	`/api/central/model/remediate/{site_id}`
POST	`/api/central/model/sync-ttl/{site_id}`
GET	`/api/central/model/ttl/{site_id}`
GET	`/api/central/model/sparql/predefined/{site_id}`
POST	`/api/central/model/sparql/{site_id}`
GET	`/api/central/model/sparql/validate/{site_id}`
POST	`/api/central/rcx/preview`

Edge model mutations use integrator auth via remediate; SPARQL on Central is read-only.

216.5 FDD preset ids (Edge parity)

Same as React `DataModelSparqlPanel` FDD buttons:

`rules_to_equipment`, `rules_to_sensors`, `rules_to_bacnet_devices`, `equipment_to_points`, `ahus_vavs_zones`, `missing_rule_bindings`, `points_by_bacnet_device`, `sensor_classes_used_by_fdd`, `rule_coverage_by_equipment_type`, `orphan_points`

Dash **Local BRICK SPARQL** buttons call Central paths above (mirrored TTL).

216.6 BRICK typing (do not mis-count AHUs/VAVs)

Many sites (including Acme) set **`brick_type`** (`AHU`, `VAV`) without `equipment_type`. Presets and mechanical narrative use `equipment_classify` — not `equipment_type` alone.

After `model.json` changes on Edge, **sync TTL** (`POST /api/central/model/sync-ttl/{site_id}` or remediate) so SPARQL predefined queries (`Air_Handling_Unit`, `Variable_Air_Volume_Box`) match equipment counts.

Packaged rooftop air handlers feeding VAVs should be modeled as **`brick_type: AHU`** even when the legacy BACnet name contains "RTU".

217 Overnight Acme Validation

218 Overnight ACME validation via RCx Central

RCx Central drives read-only validation of the ACME OpenFDD Edge over Tailscale.

218.1 Prerequisites

- Local RCx Central API (`:8060`) and Dash (`:8050`).
- ACME Edge registered under **Edge Connections** with credentials in `portfolio/config/` (gitignored).
- `gh` CLI authenticated for PR/CI inspection.

218.2 Try-out (2 cycles, no sleep)

```
OPENFDD_LIVE_ACME=1 \
RCX_CENTRAL_API_URL=http://127.0.0.1:8060 \
RCX_CENTRAL_DASH_URL=http://127.0.0.1:8050 \
RCX_EDGE_SITE=acme \
RCX_PATCH_CYCLES=2 \
```

```
RCX_CYCLE_SLEEP_MINUTES=0 \
python3 scripts/rcx_central_overnight_patch_cycle.py
```

218.3 Overnight (10 cycles, 2 h spacing)

Set `RCX_PATCH_CYCLES=10` and `RCX_CYCLE_SLEEP_MINUTES=120`.

Reports land in `reports/rcx_central_overnight_logs/cycle_NN.md`.

218.4 Read-only scope

Collects health, model tree counts, fault status, and overview data. No BACnet writes.

219 Patch Cycle Release Process

220 Patch-cycle release process

A **patch cycle** completes when CI is green, local smoke passes, optional ACME read-only validation passes, changes are committed, and the cycle log is written.

220.1 Authorization flags

Flag	Action
`RCX_ALLOW_MERGE=0` (default)	Do not merge PRs
`RCX_ALLOW_PUBLISH=0`	Do not push GHCR images
`RCX_ALLOW_DEPLOY=0`	Do not deploy to ACME

Set to `1` only when the repo owner authorizes.

220.2 Image bump (when authorized)

1. Tests and `gh pr checks` green.
2. Bump patch tag only.

3. `./scripts/docker_build.sh` and publish via existing GHCR workflow.
4. `./scripts/upgrade_edge_full.sh --limit <host>` when deploy authorized.
5. Re-run overnight validator with `OPENFDD_LIVE_ACME=1`.

221 Report Builder

222 RCx Report Builder (RCx Central)

222.1 Flow

1. Select saved Edge site
2. Choose date range (2h / 24h / 7d)
3. **Collect Data / Preview** — `POST /api/central/rcx/preview`
4. **Preview Charts** — matplotlib PNG as base64 in browser
5. Select chart/section checkboxes
6. **Generate DOCX** — `POST /api/central/rcx/report`

Reports save to `portfolio/data/reports/` when `save_to_volume=true`.

222.2 Fault overlays

Preview charts include severity-colored bands when fault rows exist and overlay toggle is on.

222.3 DOCX

Uses `open_fdd/reports/rcx_docx.py` when available (selectable sections/charts). Falls back to `portfolio/central/rcx_report.py`.

Read-only safety note is included in every report.

223 Rcx Report Builder

224 RCx Report Builder

The **RCx Report Builder** (`/analytics/rcx`) collects a read-only snapshot from the edge bridge and generates a Word document.

224.1 Workflow

1. Choose report window (2 h / 24 h / 7 d).
2. Click **Collect Data / Preview** — calls `POST /api/reports/rcx/preview`.
3. Review available vs disabled charts (disabled charts show missing point roles or no fault data).
4. Select sections and charts.
5. Click **Generate DOCX** — `POST /api/reports/rcx/generate` returns `application/vnd.openxmlformats-officedocument.wordprocessingml.document`.

224.2 Implementation

- Preview/readiness: `workspace/api/openfdd_bridge/dashboard_analytics.py`
- DOCX builder: `open_fdd/reports/rcx_docx.py`
- Charts: `open_fdd/reports/charts.py` (matplotlib → BytesIO, no temp PNGs)

224.3 Safety

Report generation is **read-only**. No BACnet writes, commands, overrides, or setpoint changes.

224.4 Dependencies

Bridge API requires `python-docx` and `matplotlib` (see `workspace/api/requirements.txt`).

224.5 Central vs edge

Portfolio Central (`portfolio/central/rcx_report.py`, port 8060) supports multi-site rollups. The edge builder uses local FDD results and building status for single-site RCx exports from the operator UI.

225 3.1.2

226 Open-FDD 3.1.2

Release date: 2026-06-15

226.1 Highlights

- **DataFusion / Rust-prep validation** — Bench long FDD smoke exercises PyArrow and DataFusion SQL backends with execution evidence and backend equivalence checks.
- **Bench 5007 long FDD smoke** — Model-driven dual-source validation (BACnet MS/TP device 5007 + Niagara bench9065) with 10-minute fault confirmation window.
- **Fault confirmation window** — Confirmed faults require consecutive true samples; reports include raw vs confirmed timing.
- **GHCR tags** — `ghcr.io/bbartling/openfdd-bridge:3.1.2`, `openfdd-commission:3.1.2`, `openfdd-mcp-rag:3.1.2`.

226.2 Known warnings

- `confirmation_engine=python_loop_over_arrow_mask` — confirmation still uses a Python loop over the Arrow fault mask; vectorized confirmation is planned.

226.3 Validation

```
        python scripts/smoke_bench_5007_long_fdd.py --synthetic --dry-run
        python scripts/smoke_bench_5007_long_fdd.py --live --duration-
minutes 35 --baseline-minutes 10 \
          --confirmation-rows 10 --confirmation-minutes 10 --poll-seconds
60 --strict-live-freshness
```

```
        python scripts/inspect_bench_5007_long_fdd_report.py workspace/
reports/bench_5007_long_fdd_*.json
```

Reports now classify **historian_mode** (`live_recent`, `historical_replay`, `synthetic`, `dry_run`) and refuse to claim live polling was validated when sample timestamps are outside the wall-clock run window. Use `--allow-historical-replay` only for debugging replay data.

For live bench sign-off, the inspector must show `data_is_fresh: true` and `historian_mode: live_recent`, plus non-zero BACnet/ Niagara rows.

227 3.1.3

228 Open-FDD 3.1.3

Release date: 2026-06-15

228.1 Patch fixes

- **Bench long FDD evaluate** — when wall-clock `run_started_at` / lookback filtering would empty a non-empty historian table (demo CSV fallback or stale feather replay), evaluate now uses the full loaded table and sets `time_filter_relaxed: true`. Strict smoke freshness gates still FAIL/WARN honestly on non-fresh samples.
- **Live smoke lookback** — final matrix evaluate uses the same 24h minimum lookback as cycle evaluates.

228.2 GHCR tags

- `ghcr.io/bbartling/openfdd-bridge:3.1.3`
- `ghcr.io/bbartling/openfdd-commission:3.1.3`
- `ghcr.io/bbartling/openfdd-mcp-rag:3.1.3`

228.3 Validation

```
        python scripts/smoke_bench_5007_long_fdd.py --synthetic --dry-run
        python scripts/smoke_bench_5007_long_fdd.py --live --duration-
minutes 35 \
          --baseline-minutes 10 --confirmation-rows 10 --confirmation-
minutes 10 \
```

```
          --poll-seconds 60 --strict-live-freshness
        python scripts/inspect_bench_5007_long_fdd_report.py workspace/
reports/bench_5007_long_fdd_*.json
```